

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-30001A-Bottom- RP.pvdb

Date of Analysis : Aug 12,2026 9:50am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User

FileName : 401-E-30001A-Bottom- RP

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Vessel Design Summary:

Step: 31 9:50am Aug 12,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 700.000 mm ID
Vessel Design Length, Tangent to Tangent 4970.70 mm
Specified Datum Line Distance 0.00 mm
Shell Side Design Temperature 155 °C
Channel Side Design Temperature 155 °C
Shell Side Design Pressure 1.000 MPa
Channel Side Design Pressure 4.600 MPa
Shop Shell Side Test Pressure 1.704 MPa
Field Shell Side Test Pressure 1.300 MPa
Shop Channel Side Test Pressure 6.156 MPa
Field Channel Side Test Pressure 5.980 MPa

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Shell	SA-240 304L	...	...	S30403	
Head	SA-240 304L	...	...	S30403	
Flange	SA-266 4	...	...	K03017	
Flange	SA-965 F304L	...	...	S30403	
Nozzle	SA-266 4	...	...	K03017	
Nozzle	SA-312 TP304L	...	...	S30403	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F304L	...	> 130	S30403	
Tubes	SA-213 TP304L	...	...	S30403	
Tubesheet	SA-965 F304L	...	...	S30403	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Chnl rear cover	4.600	0.00	5.000	1.5000	N/A	No
Chnl rear flang	4.600	0.00	5.000	1.5000	N/A	No
Chnl. rear	4.600	0.00	6.000	1.5000	N/A	No
Shell flange	1.000	0.00	1.000	0.0000	N/A	No
Shell	1.007	0.00	3.000	0.0000	N/A	No
shell head	1.007	0.00	2.000	0.0000	No	No

Liquid Level: 700.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.000



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Vessel Design Summary:

Step: 31 9:50am Aug 12,2026

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	99.5	95.0	73.0	95.0	90.5	43.4	1.00	1.00
Body Flg	235.0	131.0	90.0	108.0	101.9	79.5	1.00	1.00
Cylinder	1235.0	1000.0	18.0	18.0	13.5	...	1.00	1.00
Body Flg	1425.5	81.0	6.0	72.0	67.6	67.6	1.00	1.00
Cylinder	4894.5	3469.0	10.0	10.0	3.1	...	1.00	1.00
Ellipse	4970.7	76.2	6.0	5.1	3.1	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1- Channel in	WRC-297	0.938	Shell	Passed
N2B- channel o	WRC-297	0.938	Shell	Passed
N4B- Shell out	WRC-297	0.868	Nozzle	Passed
K1- Pressure r	WRC-297	0.868	Nozzle	Passed
N3B- Shell inl	WRC-297	0.868	Nozzle	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	150.000 mm
Saddle Bearing Angle	150.000 deg.
Centerline Dimension	720.000 mm
Wear Pad Width	200.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	162.000 deg.
Distance from Saddle to Tangent	620.000 mm

Baseplate Length	930.000 mm
Baseplate Thickness	22.000 mm
Baseplate Width	180.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	15.000 mm
Web Thickness	15.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	2

Baseplate Sketch

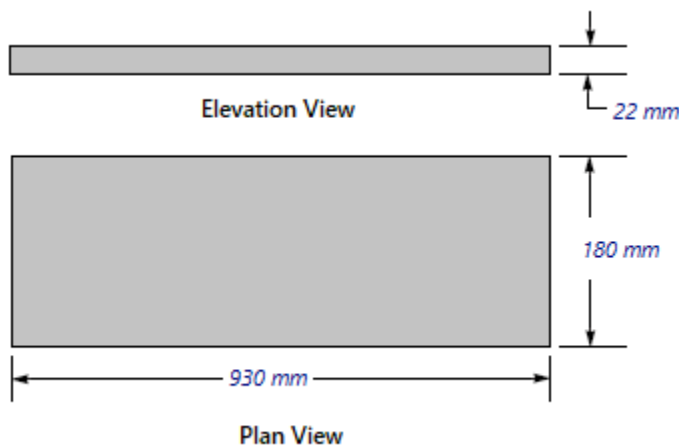
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Vessel Design Summary:

Step: 31 9:50am Aug 12, 2026



Maximum Tensile Bolt Load	23553. N
---------------------------	----------

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	222729.33	N
Maximum Transverse Saddle Shear Load	44432.64	N
Maximum Longitudinal Saddle Shear Load	41996.34	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	132551.58	N
Maximum Transverse Saddle Shear Load	3254.90	N
Maximum Longitudinal Saddle Shear Load	422.29	N

Weights:

Fabricated	- Bare w/o removable internals	5224.3	kg
Shop Test	- Fabricated + water (full)	13705.9	kg
Shipping	- Fab. + rem. intls.+ shipping app.	5224.3	kg
Erected	- Fab. + rem. intls.+ insul. (etc)	10324.3	kg
Empty	- Fab. + intls. + details + wghts.	10364.2	kg
Operating	- Empty + operating liquid (no ca)	12122.9	kg
Field Test	- Empty weight + test medium (full)	13717.4	kg

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Nozzle Summary: Step: 29 9:50am Aug 12,2026

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1- Channel inl	6	...	...	OK	9.60	OK	Passed
N2B- channel ou	6	...	...	OK	9.60	OK	Passed
N4B- Shell outl	3	...	...	OK	3.07	OK	Passed
K1- Pressure re	3	...	...	OK	3.07	OK	Passed
N3B- Shell inle	3	...	...	OK	3.09	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 3 Nozzle : N3B- Shell inle [Shellside]
 Minimum MAWP Nozzles : 6 Nozzle : N2B- channel ou [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1- Channel inle	735.000	270.000	491.750
30	N2B- channel out	735.000	90.000	491.750
50	N4B- Shell outle	1725.499	270.000	285.451
50	K1- Pressure rel	4425.499	270.000	285.451
50	N3B- Shell inlet	1725.499	90.000	285.451

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 28 9:50am Aug 12,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N4B- Shell outle	6.000 in	120	WNF	6.625	14.275	368.27	8.00	325.26	600
K1- Pressure rel	6.000 in	120	WNF	6.625	14.275	368.27	8.00	325.26	300
N3B- Shell inlet	6.000 in	120	WNF	6.625	14.275	368.27	8.00	325.26	300
N1- Channel inle	10.000 in	80	WNF	10.750	15.088	...	...	350.73	600
N2B- channel out	10.000 in	80	WNF	10.750	15.088	...	...	350.73	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N4B- Shell	SA-312 TP304L	6.000	10.000	8.000	8.000	...
K1- Pressur	SA-312 TP304L	6.000	10.000	8.000	8.000	...
N3B- Shell	SA-312 TP304L	6.000	10.000	8.000	8.000	...
N1- Channel	SA-266 4	15.000	30.000	...	...	...
N2B- channe	SA-266 4	15.000	30.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N4B- Shell outle	1725.499	270.0	305.00	0.00	Shell
K1- Pressure rel	4425.499	270.0	305.00	0.00	Shell
N3B- Shell inlet	1725.499	90.0	305.00	0.00	Shell
N1- Channel inle	735.000	270.0	305.00	0.00	Chnl. rear
N2B- channel out	735.000	90.0	305.00	0.00	Chnl. rear

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Bill of Material:

Step: 27 9:50am Aug 12,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 95.0mm THK X 940.0mm OD	SA-266 4
1	WELD NECK FLANGE: 108.0mm THK X 940.0mm OD	SA-266 4
1	WELD NECK FLANGE: 72.0mm THK X 860.0mm OD	SA-965 F304L
1	ELLIPTICAL HEAD: 2.0 X 1, 6.0mm THK X 700.0mm ID X 76.2mm	SA-240 304L
1	INSULATION: 1000mm X 45mm THK	Hot
9	BAFFLES: 680.0mm DIA X 6.0mm THK	SA-240 304L
1	TUBESHEET: 860.0mm X 105.0mm THK	SA-965 F304L
545	TUBES: 3300.0mm X 19.0mm DIA X 2.0mm THK	SA-213 TP304L
2	SADDLE: 150.0mm X 150 DEG	SA-516 70
1	INSULATION: 3400mm X 25mm THK	Hot
1	INSULATION: 251mm X 25mm THK	Hot
2	GASKET: 755.0mm OD X 729.0mm ID	...
24	BODY FLANGE BOLTS: 31.8mm DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 31.8mm DIA	...
20	BODY FLANGE BOLTS: 25.4mm DIA	SA-193 B7
20	NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA	...
2	CYLINDRICAL SEGMENT 18.0mm THK X 873.5mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 452.0mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 10.0mm THK X 1765.4mm X 3469.0mm	SA-240 304L
1	CYLINDRICAL SEGMENT 10.0mm THK X 433.7mm X 3469.0mm	SA-240 304L
1	NAMEPLATE	...



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MDMT Summary:

Step: 30 9:50am Aug 12,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Bolting [21]		-48						
Shell [16]		-196						
shell head [16]		-196						
N4B- Shell out[1]	B	-29						No
K1- Pressure r[1]	B	-29						No
N3B- Shell inl[1]	B	-29						No
Warmest MDMT:								
		-29						
Chnl rear cov[11]	B	-2	-2	-29	1.000	23.750	1.00	No
Chnl rear fla[11]	B	-16	-30	-29	0.752	15.000	1.00	No
Chnl. rear [8]	B	-11	-25	-29	0.746	18.000	1.00	No
N1- Channel in[1]	B	-11	-29	-29	0.725	18.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
N2B- channel o[1]	B	-11	-29	-29	0.725	18.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
Bolting [21]		-48						
Warmest MDMT:								
		-2	-2					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	11.4	Pass
Channel/Tube	-29.0	11.4	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).



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MDMT Summary:

Step: 30 9:50am Aug 12,2026

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:50am Aug 12,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:50am Aug 12,2026

Units used in this Analysis (Newtons):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kg
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	kg/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	1	MPa
Channel Side Design Pressure	4.6003	MPa
Shell Side Design Temperature	155.0	°C
Channel Side Design Temperature	155.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	11.4	°C
MDMT (CET), Tube Side	11.4	°C
User defined MAWP, Shell Side	0	MPa
User defined MAWP, Channel Side	0	MPa
User defined MAPnc, Shell Side	0	MPa
User defined MAPnc, Channel Side	0	MPa
User defined Test Pres., Shell Side	0	MPa
User defined Test Pres., Channel Side	0	MPa

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070



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Input Echo:

Step: 1 9:50am Aug 12,2026

0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	



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Input Echo: Step: 1 9:50am Aug 12,2026

Damping Factor (Beta) for Wind (Empty) 0.0000
Damping Factor (Beta) for Wind (Filled) 0.0000

Seismic Design Code IS 1893-RSM
Importance Factor for IS-1893 2.000
Soil Factor 1.000
Zone Number 2.000
Percent Seismic for Hydrotest 0.000
Period of Vibration (User defined) Sec 0.000
Damping Factor 2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y
Consider MAP New and Cold in Noz. Design N
Consider External Loads for Nozzle Des. Y
Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No
Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
Use Table G instead of exact equation for "A" Yes
Shell Head Joints are Tapered Yes
Compute "K" in corroded condition Yes
Use Code Case 2286 No
Use the MAWP to compute the MDMT Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes
Calculate B31.3 type stress for Nozzles with Loads Yes
Reduce the MDMT due to lower membrane stress Yes
Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:



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Input Echo:

Step: 1 9:50am Aug 12,2026

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Chnl rear cover
Distance "FROM" to "TO"	95 mm
Flange Outside Diameter	940 mm
Element Thickness	95 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	73 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.6003 MPa
Design Temperature Internal Pressure	155 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 4
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	218.4 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03017
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Chnl rear flang
Distance "FROM" to "TO"	131 mm
Flange Inside Diameter	700 mm
Element Thickness	108 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	90 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.6003 MPa
Design Temperature Internal Pressure	155 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 4
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Chnl. rear
Distance "FROM" to "TO"	1000 mm
Inside Diameter	700 mm
Element Thickness	18 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	18 mm



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Input Echo:

Step: 1 9:50am Aug 12,2026

External Corrosion Allowance	0	mm
Design Internal Pressure	4.6003	MPa
Design Temperature Internal Pressure	155	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	231.2	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Gas	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	700	mm
Liquid Density	0.14976E-04	kg/cm ³
Element From Node	30	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1000	mm
Thickness of Insulation	45	mm
Density	0.00012	kg/cm ³
Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1- Channel inle	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1331.2	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	
Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2B- channel out	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1331.2	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	



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Input Echo:

Step: 1 9:50am Aug 12,2026

Element From Node	40
Element To Node	50
Element Type	Flange
Description	Shell flange
Distance "FROM" to "TO"	81 mm
Flange Inside Diameter	700 mm
Element Thickness	72 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	6 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 MPa
Design Temperature Internal Pressure	155 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304L
Allowable Stress, Ambient	115 MPa
Allowable Stress, Operating	87.41 MPa
Allowable Stress, Hydrotest	153 MPa
Material Density	0.008027 kg/cm³
P Number Thickness	0 mm
Yield Stress, Operating	130.8 MPa
External Pressure Chart Name	HA-3
UNS Number	S30403
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	3469 mm
Inside Diameter	700 mm
Element Thickness	10 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 MPa
Design Temperature Internal Pressure	155 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304L
Allowable Stress, Ambient	115 MPa
Allowable Stress, Operating	114.5 MPa
Allowable Stress, Hydrotest	153 MPa
Material Density	0.008027 kg/cm³
P Number Thickness	0 mm
Yield Stress, Operating	130.8 MPa
External Pressure Chart Name	HA-3
UNS Number	S30403
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	620 mm
Width of Saddle	150 mm



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Input Echo:

Step: 1 9:50am Aug 12,2026

Height of Saddle at Bottom	720	mm
Saddle Contact Angle	150.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	200	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	162.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	3110	mm
Width of Saddle	150	mm
Height of Saddle at Bottom	720	mm
Saddle Contact Angle	150.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	200	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	162.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Water	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	700	mm
Liquid Density	0.0009984	kg/cm³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3400	mm
Thickness of Insulation	25	mm
Density	0.00012	kg/cm³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N4B- Shell outle	
Dist. from "FROM" Node / Offset dist	300	mm
Nozzle Diameter	6	in.
Nozzle Schedule	120	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	604.36	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-312 TP304L	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K1- Pressure rel	
Dist. from "FROM" Node / Offset dist	3000	mm
Nozzle Diameter	6	in.
Nozzle Schedule	120	



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Input Echo:

Step: 1 9:50am Aug 12,2026

Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	446.64	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-312 TP304L	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	N3B- Shell inlet	
Dist. from "FROM" Node / Offset dist	300	mm
Nozzle Diameter	6	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	446.64	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-312 TP304L	
Element From Node	50	
Detail Type	Weight	
Detail ID	Top HE Empty-L	
Dist. from "FROM" Node / Offset dist	620	mm
Miscellaneous Weight	25005	N
Offset from Element Centerline	2000	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Top HE empty-R	
Dist. from "FROM" Node / Offset dist	3110	mm
Miscellaneous Weight	25005	N
Offset from Element Centerline	2000	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Lft sadl.-oper.0	
Dist. from "FROM" Node / Offset dist	620	mm
Miscellaneous Weight	29418	N
Offset from Element Centerline	2000	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Op. wt-R	
Dist. from "FROM" Node / Offset dist	3110	mm
Miscellaneous Weight	29418	N
Offset from Element Centerline	2000	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Hydro-top HE	
Dist. from "FROM" Node / Offset dist	620	mm
Miscellaneous Weight	33340	N
Offset from Element Centerline	2000	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Hydro- Top HE	
Dist. from "FROM" Node / Offset dist	3110	mm
Miscellaneous Weight	33340	N
Offset from Element Centerline	2000	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Log. baffle	



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Input Echo:

Step: 1 9:50am Aug 12,2026

Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	0	mm
Element From Node	50	
Detail Type	For./Mom.	
Detail ID	Wind shear-left	
Dist. from "FROM" Node / Offset dist	620	mm
Force in X Direction	2418.2	N
Force in Y Direction	0	N
Force in Z Direction	15989	N
Moment about X Axis	0	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	0	N-mm
Force/Moment Combination Method	SRSS	
Element From Node	50	
Detail Type	For./Mom.	
Detail ID	Right saddle- w	
Dist. from "FROM" Node / Offset dist	3110	mm
Force in X Direction	2418.2	N
Force in Y Direction	0	N
Force in Z Direction	15989	N
Moment about X Axis	0	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	0	N-mm
Force/Moment Combination Method	SRSS	
Element From Node	50	
Detail Type	For./Mom.	
Detail ID	Left saddle- Se	
Dist. from "FROM" Node / Offset dist	620	mm
Force in X Direction	18580	N
Force in Y Direction	0	N
Force in Z Direction	18580	N
Moment about X Axis	0	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	0	N-mm
Force/Moment Combination Method	SRSS	
Element From Node	50	
Detail Type	For./Mom.	
Detail ID	Right saddle- S	
Dist. from "FROM" Node / Offset dist	3110	mm
Force in X Direction	18580	N
Force in Y Direction	0	N
Force in Z Direction	18580	N
Moment about X Axis	0	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	0	N-mm
Force/Moment Combination Method	SRSS	

Element#	6/6
Element From Node	60
Element To Node	70
Element Type	Elliptical
Description	shell head
Distance "FROM" to "TO"	76.2 mm
Inside Diameter	700 mm
Element Thickness	5.1 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	6 mm
External Corrosion Allowance	0 mm



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Input Echo:

Step: 1 9:50am Aug 12,2026

Design Internal Pressure	1	MPa
Design Temperature Internal Pressure	155	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-240 304L	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Liquid	
Detail ID	Water	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	700	mm
Liquid Density	0.0009984	kg/cm ³
Element From Node	60	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	251.2	mm
Thickness of Insulation	25	mm
Density	0.00012	kg/cm ³

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XY Coordinate Calculations: Step: 2 9:50am Aug 12,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Chnl rear cover		99.5	...	95	...
Chnl rear flang		235	...	131	...
Chnl. rear		1235	...	1000	...
Shell flange		1425.5	...	81	...
Shell		4894.5	...	3469	...
shell head		4970.7	...	76.2	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

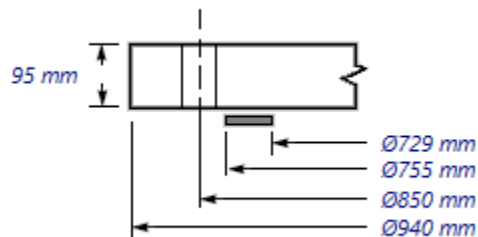
Flng: 7 9:50am Aug 12,2026

Flange Input Data Values

Description: New Flange :

Chnl rear cover

Flange Type		Blind	
Design Pressure	P	4.60	MPa
Design Temperature		155	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	940.000	mm
Flange Thickness	t	95.0000	mm
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Perimeter along the Center of the Bolts	L	2670.354	mm
Diameter of Bolt Circle	C	850.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	766.000	mm
Flange Face Inside Diameter	Fid	700.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	755.000	mm
Gasket Inside Diameter	Gi	729.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	742.151 mm

Basic Flange and Bolt Loads:



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Flg Calc [Int P]: New Flange Flng: 7 9:50am Aug 12,2026

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 742.1509^2 \cdot 4.6$$

$$= 1989850.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 742.1509 \cdot 3.0 \cdot 4.6$$

$$= 413411.594 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1989850 + 413412 + 0, 0)$$

$$= 2403261.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 742.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1032720.062 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$$

$$= \max((2403262 + 0 + 4 \cdot 0 / 742.15) / 172, 1032720 / 172)$$

$$= 13973.892 \text{ mm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	13973.892	14384.487	
Radial Distance between Bolts and Edge:	31.750	45.000	
Circ. Spacing between the Bolts:	71.450	110.947	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 14384.487 \cdot 172.0 / (68.95 \cdot \pi (755.0 + 729.0))$$

$$= 7.697 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172.0(13973.8916 + 14384.4873) / 2$$

$$= 2438569.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 2403261.50 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (850.0 - 742.1509) / 2$$

$$= 53.9246 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 2403262 \cdot 53.9246 \cdot 1.0 / (93.5^2 \cdot 742.1509) + 0.3 \cdot 1.0 \cdot 4.6 \cdot 742.1509^2 / 93.5^2$$

$$= 124.90 \text{ MPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 2438569 \cdot 53.9246 \cdot 1.0 / (93.5^2 \cdot 742.1509)$$



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Flg Calc [Int P]: New Flange

Flng: 7 9:50am Aug 12,2026

= 38.51 MPa

Bolt Stress, Operating [BSO]:

= $Wm1 / Ab$

= 2403262/14384.4873

= 167.09 MPa

Bolt Stress, Seating [BSa]:

= $(Wm2 / Ab)$

= (1032720/14384.4873)

= 71.80 MPa

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	124.90	138.00	38.51	138.00
Bolting	167.09	172.00	71.80	172.00

Reqd. Blind Flange Thickness at Center 90.454 mm

Reqd. Blind Flange Thickness at Gasket 50.894 mm

Estimated M.A.W.P. (Operating) 5 MPa

Estimated Finished Weight of Flange at given Thk. 511.0 kg

Estimated Unfinished Weight of Forging at given Thk 511.0 kg

Minimum Design Metal Temperature Results:

Thickness Ratio = 1., Temperature Reduction per Fig. UCS 66.1 = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	120.99	138.00	37.31	138.00
Bolting	167.09	172.00	71.80	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 88.976 mm

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Flg Calc [Int P]: FLANGE

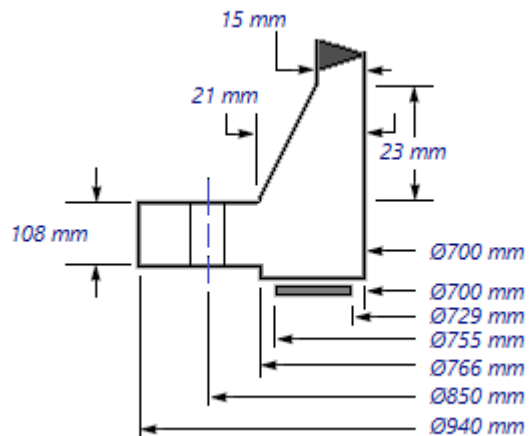
Fling: 8 9:50am Aug 12, 2026

Flange Input Data Values

Description: FLANGE :

Chnl rear flang

Flange Type	Integral Weld Neck		
Design Pressure	P	4.60	MPa
Design Temperature		155	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	700.000	mm
Flange Outside Diameter	A	940.000	mm
Flange Thickness	t	108.0000	mm
Thickness of Hub at Small End	go	15.0000	mm
Thickness of Hub at Large End	g1	21.0000	mm
Length of Hub	h	23.0000	mm
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	850.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	766.000	mm
Flange Face Inside Diameter	Fid	700.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	755.000	mm
Gasket Inside Diameter	Gi	729.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: FLANGE

Flng: 8 9:50am Aug 12, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4.6 \cdot (700.0/2 + 1.5)) / (138.0 \cdot 1.0 - 0.6 \cdot 4.6) + Ca$$

$$= 13.4567 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.5) / (351.5 + 0.6 \cdot 13.5)$$

$$= 5.181 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (350.0 + 0.6 \cdot 15.0)$$

$$= 5.766 \text{ MPa}$$

Corroded Flange Thickness, $t_c = T - c_i$	106.500 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	703.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	19.500 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.500 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	54.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	742.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 742.1509^2 \cdot 4.6$$

$$= 1989850.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 742.1509 \cdot 3.0 \cdot 4.6$$

$$= 413411.594 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 703.0^2 \cdot 4.6003$$

$$= 1785445.625 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 1989850 - 1785446$$

$$= 204404.359 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1989850 + 413412 + 0, 0)$$

$$= 2403261.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 742.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1032720.062 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/S_b, Wg/S_a)$$

$$= \max((2403262 + 0 + 4 \cdot 0/742.15)/172, 1032720/172)$$

$$= 13973.892 \text{ mm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 106.5/(3.0 + 0.5)$$

$$= 246.071 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 850.0 \cdot \sin(\pi/24)$$

$$= 110.947 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{110.947/(2 \cdot 31.75 + 106.5)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	13973.892	14384.487	
Radial Distance between Hub and Bolts:	44.450	54.000	
Radial Distance between Bolts and Edge:	31.750	45.000	
Circ. Spacing between the Bolts:	71.450	110.947	246.071

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 14384.487 \cdot 172.0 / (68.95 \cdot \pi(755.0 + 729.0))$$

$$= 7.697 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172.0(13973.8916 + 14384.4873) / 2$$

$$= 2438569.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 2403262 - 1989850$$

$$= 413411.56 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (850.0 - 742.1509) / 2$$

$$= 53.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5(C - B) / 2 + hg$$

$$= 0.5(850.0 - 703.0) / 2 + 53.9246$$

$$= 63.7123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (850.0 - 703.0 - 19.5) / 2$$

$$= 63.7500 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	1785446.	63.7500	1.0000	113868352.
Face Pressure, Mt	204404.	63.7123	1.0000	13028353.
Gasket Load, Mg	413412.	53.9246	1.0000	22302090.
Gasket Seating, Matm	2438569.	53.9246	1.0000	131552160.

Total Moment for Operation, Mo 149198784. N-mm
Total Moment for Gasket seating, Mg 131552160. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 97.419 mm
Hub Ratio, h/h0 = HL / H0 0.236
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.444

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.895
Factor V per 4.16.5 0.406
Factor f per 4.16.5 1.280

Stress factor from Table 4.16.4 [K]:

= A / B
= 940.0/703.0
= 1.337

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.337² • log10(1.337)))/(1.337² - 1))/(1.337 - 1)
= 6.838

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.337²(1 + 8.55246 • log10(1.337)) - 1)/((1.0472 + 1.9448 • 1.337²) • (1.337 - 1))
= 1.781

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.337²(1 + 8.55246 • log10(1.337)) - 1)/(1.36136(1.337² - 1) • (1.337 - 1))
= 7.514

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
= (1.337² + 1)/(1.337² - 1)
= 3.538

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 7.514 • 13.5² • 97.419/0.406
= 328955.469 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.895/97.419
= 0.009 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (106.5 • 0.009 + 1)/1.781 + 106.5³/328955.469
= 4.783

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.2797•14920E+09/703.0)/(4.7828•19.5²)
= 149.29 MPa

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{1^2})$$

$$= (1.2797 \cdot 0.13155\text{E}+09 / 703.0) / (4.7828 \cdot 19.5^2)$$

$$= 131.63 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.305 \cdot 1.4920\text{E}+09 / 703.0) / (4.7828 \cdot 106.5^2)$$

$$= 9.01 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.305 \cdot 1.3155\text{E}+09 / 703.0) / (4.7828 \cdot 106.5^2)$$

$$= 7.95 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (6.8379 \cdot 1.4920\text{E}+09 / (106.5^2 \cdot 703.0)) - 3.5384 \cdot 9$$

$$= 96.01 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (6.8379 \cdot 1.3155\text{E}+09 / (106.5^2 \cdot 703.0)) - 3.5384 \cdot 8$$

$$= 84.65 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (149 + \max(9, 96)) / 2$$

$$= 122.65 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (132 + \max(8, 85)) / 2$$

$$= 108.14 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 2403262 / 14384.4873$$

$$= 167.09 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (1032720 / 14384.4873)$$

$$= 71.80 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	149.29	207.00	131.63	207.00
Radial Flange	9.01	138.00	7.95	138.00
Tangential Flange	96.01	138.00	84.65	138.00
Maximum Average	122.65	138.00	108.14	138.00
Bolting	167.09	172.00	71.80	172.00

Minimum Required Flange Thickness 101.879 mm

Estimated M.A.W.P. (Operating) 5 MPa
 Estimated Finished Weight of Flange at given Thk. 266.0 kg
 Estimated Unfinished Weight of Forging at given Thk 313.9 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot \text{Ma} / \text{Bsc} \cdot \text{Cnvfac} \cdot V / (\lambda \cdot \text{Eamb} \cdot \text{go}^2 \cdot \text{ho} \cdot \text{Ki})$$



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Flg Calc [Int P]: FLANGE

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$$= 52.14 \cdot 0.13155E+09 / 1.0 \cdot 1.0 \cdot 0.406 / (4.152 \cdot 202713 \cdot 13.5^2 \cdot 97.419 \cdot 0.3)$$

$$= 0.620 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 0.14920E+09 / 1.0 \cdot 1.0 \cdot 0.406 / (4.152 \cdot 194826 \cdot 13.5^2 \cdot 97.419 \cdot 0.3)$$

$$= 0.732 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 0.13155E+09 / 1.0 \cdot 1.0 \cdot 0.406 / (4.783 \cdot 202713 \cdot 13.5^2 \cdot 97.419 \cdot 0.3)$$

$$= 0.538 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 0.14920E+09 / 1.0 \cdot 1.0 \cdot 0.406 / (4.783 \cdot 194826 \cdot 13.5^2 \cdot 97.419 \cdot 0.3)$$

$$= 0.635 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.752, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating	
			Actual	Allowed
Longitudinal Hub	144.51	207.00	126.16	207.00
Radial Flange	9.98	138.00	8.71	138.00
Tangential Flange	89.56	138.00	78.18	138.00
Maximum Average	117.03	138.00	102.17	138.00
Bolting	167.09	172.00	71.80	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 99.035 mm

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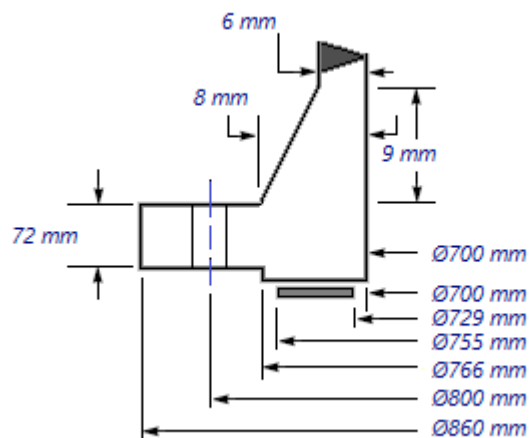
Flg Calc [Int P]: New Flange

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Flange Input Data Values**Description: New Flange :**

Shell flange

Flange Type	Integral Weld Neck		
Design Pressure	P	1.00	MPa
Design Temperature		155	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	700.000	mm
Flange Outside Diameter	A	860.000	mm
Flange Thickness	t	72.0000	mm
Thickness of Hub at Small End	go	6.0000	mm
Thickness of Hub at Large End	g1	8.0000	mm
Length of Hub	h	9.0000	mm
Flange Material	SA-965 F304L		
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	87.41	MPa
Flange Allowable Stress At Ambient	Sfa	115.00	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	800.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		20	
Flange Face Outside Diameter	Fod	766.000	mm
Flange Face Inside Diameter	Fid	700.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	755.000	mm
Gasket Inside Diameter	Gi	729.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.0 \cdot (700.0/2 + 0.0)) / (87.41 \cdot 1.0 - 0.6 \cdot 1.0) + Ca$$

$$= 4.0320 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (87.41 \cdot 1.0 \cdot 6.0) / (350.0 + 0.6 \cdot 6.0)$$

$$= 1.483 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.0 \cdot 1.0 \cdot 6.0) / (350.0 + 0.6 \cdot 6.0)$$

$$= 1.951 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	700.000 mm
Corroded Large Hub,	g1Cor = g1-ci	8.000 mm
Corroded Small Hub,	g0Cor = go-ci	6.000 mm
Code R Dimension,	R = (C-B)/2 - g1	42.000 mm
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm
Basic Gasket Width,	bo = N/2	6.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm
Gasket Reaction Diameter,	G = Go - 2 * b	742.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 742.1509^2 \cdot 1.0$$

$$= 432576.125 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 742.1509 \cdot 3.0 \cdot 1.0$$

$$= 89872.094 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 700.0^2 \cdot 1.0$$

$$= 384834.750 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 432576 - 384835$$

$$= 47741.391 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(432576 + 89872 + 0, 0)$$

$$= 522448.219 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 742.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1032720.062 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((522448 + 0 + 4 \cdot 0 / 742.15) / 172, 1032720 / 172)$$

$$= 6004.806 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$



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$$= 2 \cdot 25.4 + 6 \cdot 72.0 / (3.0 + 0.5)$$

$$= 174.229 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 800.0 \cdot \sin(\pi / 20)$$

$$= 125.148 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{(Bs / (2a + t))}, 1)$$

$$= \max(\sqrt{(125.148 / (2 \cdot 25.4 + 72.0))}, 1)$$

$$= 1.0095$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	6004.806	7109.663	
Radial Distance between Hub and Bolts:	34.925	42.000	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	125.148	174.229

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 7109.663 \cdot 172.0 / (68.95 \cdot \pi(755.0 + 729.0))$$

$$= 3.804 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172.0(6004.8057 + 7109.6631) / 2$$

$$= 1127728.12 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 522448 - 432576$$

$$= 89872.09 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (800.0 - 742.1509) / 2$$

$$= 28.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((800.0 - 700.0) / 2 + 28.9246)$$

$$= 39.4623 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (800.0 - 700.0 - 8.0) / 2$$

$$= 46.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	384835.	46.0000	1.0095	17878052.
Face Pressure, Mt	47741.	39.4623	1.0095	1902678.



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Gasket Load, Mg	89872.	28.9246	1.0095	2625305.
Gasket Seating, Matm	1127728.	28.9246	1.0095	32942714.

Total Moment for Operation, Mo 22406034. N-mm
Total Moment for Gasket seating, Mg 32942714. N-mm

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 64.807 mm
Hub Ratio, $h/h_o = H_L / H_o$ 0.139
Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 1.333

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.907
Factor V per 4.16.5 0.468
Factor f per 4.16.5 1.369

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 860.0/700.0$
 $= 1.229$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169(1.229^2 \cdot \log_{10}(1.229))) / (1.229^2 - 1)) / (1.229 - 1)$
 $= 9.550$

Stress factor from Table 4.16.4 [T]:

$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.229^2(1 + 8.55246 \cdot \log_{10}(1.229)) - 1) / ((1.0472 + 1.9448 \cdot 1.229^2) \cdot (1.229 - 1))$
 $= 1.827$

Stress factor from Table 4.16.4 [U]:

$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 $= (1.229^2(1 + 8.55246 \cdot \log_{10}(1.229)) - 1) / (1.36136(1.229^2 - 1) \cdot (1.229 - 1))$
 $= 10.495$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.229^2 + 1) / (1.229^2 - 1)$
 $= 4.926$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$
 $= 10.495 \cdot 6.0^2 \cdot 64.807 / 0.468$
 $= 52263.027 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_o$
 $= 0.907 / 64.807$
 $= 0.014 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3/d$
 $= (72.0 \cdot 0.014 + 1) / 1.827 + 72.0^3 / 52263.027$
 $= 8.240$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$
 $= (1.3691 \cdot 22406034 / 700.0) / (8.2404 \cdot 8.0^2)$
 $= 83.07 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:

$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$
 $= (1.3691 \cdot 32942714 / 700.0) / (8.2404 \cdot 8.0^2)$



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Flg Calc [Int P]: New Flange

Flng: 9 9:50am Aug 12, 2026

= 122.13 MPa

Radial Flange Stress, Operating [SRo]:

= ($\beta \cdot Mop / Bcor$) / ($L \cdot t^2$)

= (2.3436•22406034/700.0)/(8.2404•72.0²)

= 1.76 MPa

Radial Flange Stress, Seating [SRa]:

= ($\beta \cdot Matm/Bcor$) / ($L \cdot t^2$)

= (2.3436•32942714/700.0)/(8.2404•72.0²)

= 2.58 MPa

Tangential Flange Stress, Operating [STo]:

= ($Y \cdot Mo / (t^2 \cdot Bcor)$) - $Z \cdot SRO$

= (9.5501•22406034/(72.0²•700.0))-4.9263•2

= 50.30 MPa

Tangential Flange Stress, Seating [STa]:

= ($y \cdot Matm / (t^2 \cdot Bcor)$) - $Z \cdot SRA$

= (9.5501•32942714/(72.0²•700.0))-4.9263•3

= 73.95 MPa

Average Flange Stress, Operating [SAo]:

= ($SHo + \max(SRO, STo)$) / 2

= (83+max(2,50))/2

= 66.69 MPa

Average Flange Stress, Seating [SAa]:

= ($SHA + \max(SRA, STa)$) / 2

= (122+max(3,74))/2

= 98.04 MPa

Bolt Stress, Operating [BSo]:

= $Wm1 / Ab$

= 522448/7109.6631

= 73.49 MPa

Bolt Stress, Seating [BSa]:

= ($Wm2 / Ab$)

= (1032720/7109.6631)

= 145.27 MPa

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	83.07	131.12	122.13	172.50
Radial Flange	1.76	87.41	2.58	115.00
Tangential Flange	50.30	87.41	73.95	115.00
Maximum Average	66.69	87.41	98.04	115.00
Bolting	73.49	172.00	145.27	172.00

Minimum Required Flange Thickness 67.640 mm

Estimated M.A.W.P. (Operating) 1 MPa

Estimated Finished Weight of Flange at given Thk. 114.4 kg

Estimated Unfinished Weight of Forging at given Thk 127.5 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$

= 52.14•33543550/1.0279•1.0•0.468/(6.986•195129•6.0²•64.807•0.3)

= 0.835 (should be ≤ 1)



Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 22814696 / 1.0279 \cdot 1 \cdot 0.468 / (6.986 \cdot 185711 \cdot 6.0^2 \cdot 64.807 \cdot 0.3)$
 $= 0.596$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 32942712 / 1.0095 \cdot 1 \cdot 0.468 / (8.24 \cdot 195129 \cdot 6.0^2 \cdot 64.807 \cdot 0.3)$
 $= 0.708$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 22406034 / 1.0095 \cdot 1 \cdot 0.468 / (8.24 \cdot 185711 \cdot 6.0^2 \cdot 64.807 \cdot 0.3)$
 $= 0.506$ (should be ≤ 1)

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	83.07	172.50	122.13	172.50
Radial Flange	1.76	115.00	2.58	115.00
Tangential Flange	50.30	115.00	73.95	115.00
Maximum Average	66.69	115.00	98.04	115.00
Bolting	73.49	172.00	145.27	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 67.640 mm



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Internal Pressure Calculations: Step: 6 9:50am Aug 12,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Chnl rear cover		4.6003	73	1.5	940	138	1
Chnl rear flang		4.6003	90	1.5	700	138	2
Chnl. rear		4.6004	18	1.5	700	138	3
Shell flange	1	1	6	...	700	87.41	4
Shell		1.0069	10	...	700	114.5	5
shell head		1.0069	6	...	700	114.5	6

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Chnl rear cover		4.60027	4.73542	4.73544	95	90.4536	1
Chnl rear flang		4.60027	4.73544	4.73544	108	101.879	2
Chnl. rear		4.60027	6.30028	6.88458	18	13.457	3
Shell flange		1.00006	1.31087	1.72459	72	67.6402	4
Shell		1.00006	3.20941	3.23028	10	3.09427	5
shell head		1.00006	1.65913	1.67325	5.1	3.08065	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1 MPa
 Shell Side MAPnc = 2 MPa
 Channel Side MAWP = 5 MPa
 Channel Side MAPnc = 5 MPa

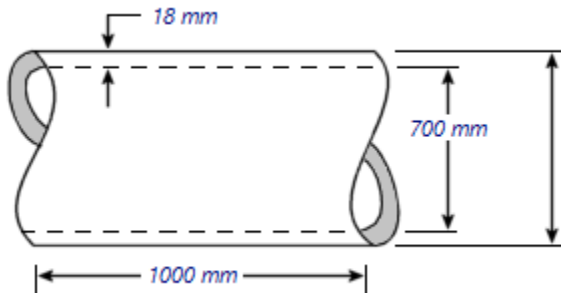
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Chnl. rear

Cylindrical Shell Design Information:			
Design Pressure P	4.600 MPa	Int. Design Temperature	155 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	1000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.5000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	700.000 mm	Actual Stress	100.8 MPa
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	2312212.0 mm ²	SG of Contents	0.015
Element Volume	384.9 ltr	Weight of Contents	56.6 N
Empty Weight	3085.8 N	Operating Weight	3142.4 N
		Comp. All. Ope. Stress	143.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4.6 \cdot 351.5) / (138.0 \cdot 1.0 - 0.6 \cdot 4.6) \\
 &= 11.9570 + 1.5000 = 13.4570 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 16.5) / (351.5 + 0.6 \cdot 16.5) \\
 &= 6 - 0 = 6 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 18.0) / (350.0 + 0.6 \cdot 18.0) \\
 &= 7 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4.6 \cdot (351.5 + 0.6 \cdot 16.5)) / (1.0 \cdot 16.5) \\
 &= 100.762 \text{ MPa}
 \end{aligned}$$

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Element: Chn1. rear

Step: 7 9:50am Aug 12, 2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.507 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18.0$, $t_r = 12.315$, $c = 1.5$ mm, $E^* = 1.0$

Thickness Ratio = $tr * E^* / (tg - c) = 0.746$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -25 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.735 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.746	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-25 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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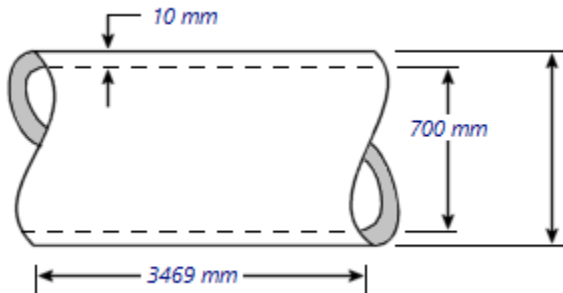
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Element: Shell

Step: 8 9:50am Aug 12,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	1.007 MPa	Int. Design Temperature	155 °C
Material	SA-240 304L	Ext. Design Temperature	40 °C
Length	3469.000 mm	External Pressure Chart	HA-3
		UNS Number	S30403
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	114.5 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	115.0 MPa
Inside Diameter Di	700.000 mm	Actual Stress	35.8 MPa
Spec. Min. Thk t,e, or sa	10.0000 mm	Spec. Nominal Thk.	10.000 mm
Surface Area	7846694.0 mm ²	SG of Contents	1.000
Element Volume	773.7 ltr	Weight of Contents	7581.8 N
Empty Weight	6090.8 N	Operating Weight	13672.6 N
		Comp. All. Ope. Stress	67.5 MPa
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1.007 \cdot 350.0) / (114.5 \cdot 1.0 - 0.6 \cdot 1.007) \\
 &= 3.0943 + 0.0000 = 3.0943 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (114.5 \cdot 1.0 \cdot 10.0) / (350.0 + 0.6 \cdot 10.0) \\
 &= 3 - 0.01 = 3 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (115.0 \cdot 1.0 \cdot 10.0) / (350.0 + 0.6 \cdot 10.0) \\
 &= 3 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1.007 \cdot (350.0 + 0.6 \cdot 10.0)) / (1.0 \cdot 10.0) \\
 &= 35.846 \text{ MPa}
 \end{aligned}$$



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Element: Shell

Step: 8 9:50am Aug 12,2026

% Elongation per Table UG-79-1 $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.408 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

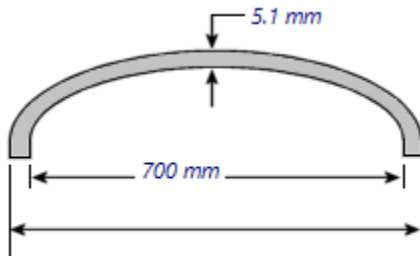
SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell head

Elliptical Head Design Information:							
Design Pressure		P	1.007 MPa	Int. Design Temperature		155 °C	
Material		SA-240 304L		Ext. Design Temperature		40 °C	
Length		76.200 mm		External Pressure Chart		HA-3	
				UNS Number		S30403	
Int. Corr. All.		c	0.0000 mm	Allowable Stress (ope)		S	114.5 MPa
Ext. Corr. All.		ce	0.0000 mm	Allowable Stress (amb)		Sa	115.0 MPa
Inside Diameter		Di	700.000 mm	Actual Stress		69.2 MPa	
Spec. Min. Thk t,e, or sa		5.1000 mm		Spec. Nominal Thk.		6.000 mm	
Surface Area		719965.1 mm²		SG of Contents		1.000	
Element Volume		74.2 ltr		Weight of Contents		727.5 N	
Empty Weight		361.6 N		Operating Weight		1089.1 N	
				Comp. All. Ope. Stress		67.5 MPa	
				Aspect Ratio		2.00	
Radiography Information:							
User Defined				User Defined			
Circ. Joint Efficiency		100 %		Long. Joint Eff.		E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.007 \cdot 700.0 \cdot 1.0) / (2 \cdot 114.5 \cdot 1.0 - 0.2 \cdot 1.007)$$

$$= 3.0806 + 0.0000 = 3.0806 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 114.5 \cdot 1.0 \cdot 5.1) / (1.0 \cdot 700.0 + 0.2 \cdot 5.1)$$

$$= 2 - 0.01 = 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 115.0 \cdot 1.0 \cdot 5.1) / (1.0 \cdot 700.0 + 0.2 \cdot 5.1)$$

$$= 2 \text{ MPa}$$



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Element: shell head

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Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (1.007 \cdot (1.0 \cdot 700.0 + 0.2 \cdot 5.1)) / (2 \cdot 1.0 \cdot 5.1) \\ &= 69.203 \text{ MPa} \end{aligned}$$

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (1.007 \cdot 350.0) / (114.5 \cdot 1.0 - 0.6 \cdot 1.007) + 0.0 \\ &= 3.094 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.007 MPa](#)

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (114.5 \cdot 1.0 \cdot 6.0) / (350.0 + 0.6 \cdot 6.0) \\ &= 2 - 0.01 = 2 \text{ MPa} \end{aligned}$$

% Elongation per Table UG-79-1 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.689 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

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Shop Pressure Test Results: Step: 10 9:50am Aug 12,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.704 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.300 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2.166 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.442 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.430 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.430 MPa
App 27-4	= M.A.W.P.	1.311 MPa

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.156 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5.980 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.147 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	5.209 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	6.578 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6.578 MPa
App 27-4	= M.A.W.P.	4.735 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Chnl. rear	135.0	234.0	0.577	6.16
Shell	60.9	153.0	0.398	1.71
shell head	117.6	153.0	0.769	1.71

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1- Channel inle	Nozzle	138.00	138.00	1.000
N2B- channel out	Nozzle	138.00	138.00	1.000
N4B- Shell outle	Nozzle	115.00	87.41	1.316
N4B- Shell outle	Pad	138.00	138.00	1.000
K1- Pressure rel	Nozzle	115.00	87.41	1.316
K1- Pressure rel	Pad	138.00	138.00	1.000
N3B- Shell inlet	Nozzle	115.00	87.41	1.316
N3B- Shell inlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Chnl rear cover	138.00	138.00	1.000
Chnl rear flang	138.00	138.00	1.000
Chnl. rear	138.00	138.00	1.000
Shell flange	115.00	87.41	1.316
Shell	115.00	114.50	1.004
shell head	115.00	114.50	1.004

Minimum 1.000



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Shop Pressure Test Results:

Step: 10

9:50am

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Stress ratios for Exchanger Materials (MPa):

Description	Ambient	Operating	Ratio
Tube Material	115.00	87.41	1.316
Tubesheet Material	115.00	87.41	1.316

Minimum

1.316

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1- Channel inle	59.46	225.00	0.264
N2B- channel out	59.46	225.00	0.264
N4B- Shell outle	10.58	153.00	0.069
K1- Pressure rel	10.58	153.00	0.069
N3B- Shell inlet	10.58	153.00	0.069

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 11 9:50am Aug 12,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.704 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.300 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2.166 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.442 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.430 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.430 MPa
App 27-4	= M.A.W.P.	1.311 MPa

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.156 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5.980 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.147 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	5.209 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	6.578 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6.578 MPa
App 27-4	= M.A.W.P.	4.735 MPa

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1.0 • 1.0

= 1.300 MPa

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4.6 • 1.0

= 5.980 MPa

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Chnl. rear	131.1	234.0	0.560	5.99
Shell	46.5	153.0	0.304	1.31
shell head	89.8	153.0	0.587	1.31

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1- Channel inle	Nozzle	138.00	138.00	1.000
N2B- channel out	Nozzle	138.00	138.00	1.000
N4B- Shell outle	Nozzle	115.00	87.41	1.316
N4B- Shell outle	Pad	138.00	138.00	1.000
K1- Pressure rel	Nozzle	115.00	87.41	1.316
K1- Pressure rel	Pad	138.00	138.00	1.000
N3B- Shell inlet	Nozzle	115.00	87.41	1.316
N3B- Shell inlet	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):



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Field Pressure Test Results: Step: 11 9:50am Aug 12,2026

Description	Ambient	Operating	Ratio
Chnl rear cover	138.00	138.00	1.000
Chnl rear flang	138.00	138.00	1.000
Chnl. rear	138.00	138.00	1.000
Shell flange	115.00	87.41	1.316
Shell	115.00	114.50	1.004
shell head	115.00	114.50	1.004

Minimum 1.000

Stress ratios for Exchanger Materials (MPa):

Description	Ambient	Operating	Ratio
Tube Material	115.00	87.41	1.316
Tubesheet Material	115.00	87.41	1.316

Minimum 1.316

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1- Channel inle	57.76	225.00	0.257
N2B- channel out	57.76	225.00	0.257
N4B- Shell outle	8.08	153.00	0.053
K1- Pressure rel	8.08	153.00	0.053
N3B- Shell inlet	8.08	153.00	0.053

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		1.30 MPa
Shell Thickness	ts	10.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	700.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-240 304L

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30403
Shell Allowable Stress at Temperature	Ss	115.00 MPa
Shell Allowable Stress at Ambient		115.00 MPa

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Test Pressure		5.98 MPa
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	700.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 MPa
Channel Allowable Stress at Ambient		138.00 MPa

Tube Data:

Number of Tube Holes	Nt	545
Tube Wall Thickness	et	2.0000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	3300.00 mm
Straight Tube Length (bet. inner tubsht faces) L		3195.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		115.00 MPa
Tube Allowable Stress At Ambient		115.00 MPa
Tube Yield Stress At design Temperature	Syt	170.00 MPa
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	330.948 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	312161.0 mm ²
Perimeter of Tube Layout	Cp	2058.4 mm



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ASME TS Calc: Pressure Test

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Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material	SA-965 F304L	
Tubesheet Material UNS Number	S30403	
Tubesheet Allowable Stress at Temperature	S	115.00 MPa
Tubesheet Allowable Stress at Ambient	Tt	115.00 MPa
Thickness of Tubesheet	h	105.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	860.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?	Yes
Is the Tubesheet Flat on Tubeside?	Yes
Is the Tubesheet Flat on Shellside?	No
Step 1 Diameter on the Shellside	ds1 763.00 mm
Step 1 Depth on the Shellside	hs1 8.00 mm
Step 2 Diameter on the Shellside	ds2 697.00 mm
Step 2 Depth on the Shellside	hs2 8.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4	860.00 mm

Note: Tubesheet diameter is now: 860.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 mm ²
Junction Stress Reduction option	Perform Plastic Calculation	
Perform Differential Pressure Design	No	
Run Multiple Load Cases	Yes	

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side	Shell
Flange Outside Diameter	A 860.000 mm
Flange Inside Diameter	B 700.000 mm
Flange Face Outside Diameter	Fod 766.000 mm
Flange Face Inside Diameter	Fid 700.000 mm
Gasket Outside Diameter	Go 755.000 mm
Gasket Inside Diameter	Gi 729.000 mm
Small end Hub thk.	g0 6.0000 mm
Large end Hub thk.	g1 8.0000 mm
Gasket Factor,	m 3.00
Gasket Design Seating Stress	y 68.95 MPa
Flange Facing Sketch	Code Sketch 1a
Gasket Thickness	tg 4.5000 mm
Full face Gasket Flange Option	Program Selects

Bolting Information:

Diameter of Bolt Circle	C	800.000 mm
Nominal Bolt Diameter	dB	25.4000 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	20
Bolt Material	SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00 MPa
Bolt Allowable Stress At Ambient	Sa	172.00 MPa
Weld between Flange and Shell/Channel		0.0000 mm

Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 89.0000 mm
Is Bolt Load Transferred to the Tubesheet	No



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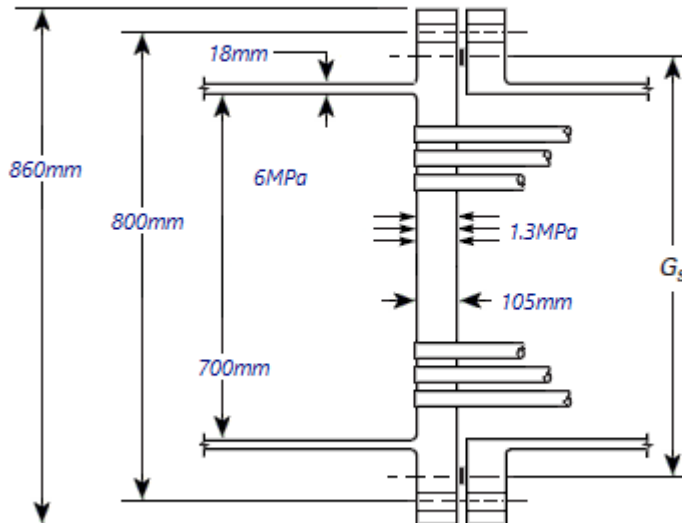
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Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Bolt load is not transferred to tubesheet which is extended for bolting. Req. Tubesheet extension thk. can't be computed. Please check input.

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group G

Elastic Mod. at Design Temperature	21.1 °C	0.19513E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 °C	0.19513E+06 MPa

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20271E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 MPa

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesheet Design Temp.	21.1 °C	0.19513E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 °C	0.19513E+06 MPa

TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature	21.1 °C	0.19513E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 °C	0.19513E+06 MPa

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(5.98 \cdot (19.0/2 - 0)) / (115.0 \cdot 1.0 + 0.4 \cdot 5.98)$
= $0.4840 + 0.0000 = 0.4840$ mm



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Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 21.11 °C
Elastic Modulus for Material 193060.00 MPa

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.0000	19.00	3525.95	9.50	50.0000	0.0121884	92.80

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 13.4381$ MPa

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3979	19.00	3525.95	47.75	50.0000	0.0004824	46.56

EMAWP = $(4 * B) / (3 * (D/T)) = (4 * 46.5638) / (3 * 47.7533) = 1.3001$ MPa

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4840 mm
Allowable Internal Pressure at Corroded thickness	26.44 MPa
Required Internal Design Pressure	5.98 MPa
Allowable External Pressure at Corroded thickness	13.44 MPa
Required External Design Pressure	1.30 MPa
Required Thickness due to Shell Side pressure	0.3979 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	742.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 * ro + dt$
 $= 2 * 330.948 + 19.0$
 $= 680.896$ mm

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (25.0 - 19.0) / 25.0$
 $= 0.240$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Gs / Do$ (Configurations d, e, f)
 $= 742.1556 / 680.896$
 $= 1.0900$

Ratio [pc]:

$= Dc / Do$ (Configurations a, e, f)
 $= 700.0 / 680.896$
 $= 1.0281$



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ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/105.0$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(19.0 - 2 \cdot 2.0 (195128/195128) \cdot$$

$$(114/114) \cdot (0); 19.0 - 2 \cdot 2.0)$$

$$= 19.0000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 25.0 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 680.896 \cdot 25.0) / (\pi \cdot 680.896^2)) }$$

$$= 25.0000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.0 - 19.0) / 25.0$$

$$= 0.2400$$

$$h/p = 4.200000 ; \mu^* = 0.240000 ; E^* = 39811 \text{ MPa}$$

$$E^*/E = 0.204029 ; \nu^* = 0.407183$$

Note: As h/p (4.200) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s and ω_s are all = 0

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((700.0 + 18.0) 18.0) }$$

$$= 0.0160 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.016 \cdot 202713.0 \cdot 18.0^3 / (6(1 - 0.3^2))$$

$$= 3462291.0000 \text{ MPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta +$$

$$h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(700.0 + 18.0 / \cos(0)) \cdot 3462291 / 105.0^3 \cdot (1.0 +$$

$$105.0 \cdot 0.016 \cdot (1.0 + 1.0) / 2 + (105.0 \cdot 0.016)^2 / 2 \cdot 1.0)$$

$$= 52678.4 \text{ MPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 700.0^2 / (4 \cdot 202713 \cdot 18.0) \cdot (1 - 700.0 \cdot 0.3 / (2(700.0 + 18.0)))$$

$$= 0.28663E-01 \text{ mm/MPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) -$$

$$\zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2))$$



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$$= 1.028 \cdot 3462291 [0.028663 (1.0 + 0.016 \cdot 1.0 \cdot 105.0) - 0 (1.0/105.0 + 1.0 \cdot 0.016/2)]$$

$$= 4370.4482 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 860.0/680.896$$

$$= 1.2630$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 39811 \cdot (52678.44 + 195128 \cdot \ln(1.26))$$

$$= 1.4629$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2/16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$$

$$= 680.896^2/16 \cdot [(1.09 - 1) \cdot (1.09^2 + 1) \cdot 1.3 - (1.028 - 1) \cdot (1.028^2 + 1) \cdot 5.98]$$

$$= -2584.7737 \text{ MPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + (\omega c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o)$$

$$= -2584 + (26136) + 679147 \cdot (800.0 - 742.156) / (2 \cdot \pi \cdot 680.896)$$

$$= 32735.3926 \text{ MPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((32735.39) - 680.896^2/32 \cdot 1.463 (1.3 - 5.98)) / (1 + 1.46)$$

$$= 53567.8945 \text{ MPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= 53567.89 + 680.896^2/64 \cdot (3 + 0.407) \cdot (1.3 - 5.98)$$

$$= -61949.6797 \text{ MPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p |; | M_o |)$$

$$= \max(| 53567.895 |; | -61949.68 |)$$

$$= 61949.6797 \text{ MPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 61949.68 / (0.24 (105.0 - 0)^2)$$

$$= 140.4755 \text{ MPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta_T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 170.0$$

$$= 322.99 \text{ MPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 59114.145 / (0.24 (67.7657 - 0)^2)$$

$$= 321.8191 \text{ MPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$



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$$= 67.7657 + 0 + 0$$

$$= 67.7657 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(\text{HreqB}; \text{HreqS})$$

$$= \max(67.7657; 22.891)$$

$$= 67.7657 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |1.3 - 5.98|$$

$$= 4.680 \text{ MPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y$$

$$= 4 \cdot 0.24 \cdot 105.0 / 680.9 \cdot 0.8 \cdot 0.95 \cdot 169.997$$

$$= 19.126 \text{ MPa}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.24 \cdot 129.198) \cdot (4 \cdot 312161.0 / 2058.35) \cdot |1.3 - 5.98| + 0$$

$$= 22.8910 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 18.000 mm

Min. Channel len. adjacent to the tubesheet is: 202.049mm

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19.0 - 2.0) 2.0$$

$$= 106.814 \text{ mm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.0 (19.0 - 2.0) 161.497$$

$$= 17248.707 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.0 (19.0 + 0.67 \cdot 2.0) 161.5$$

$$= 17248.7070 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 161.497 / 161.497$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.0 (19.0 + 0.67 \cdot 2.0) 161.5$$

$$= 11350.6641 \text{ N}$$

Minimum Length of Weld [ar]:



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$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19.0)^2 + 1.07 \cdot 2.0 \cdot (19.0 - 2.0) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 19.0 \right)$$

$$= 2.448 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2239 mm

Minimum Required Groove Weld Leg [agr]: 1.2239 mm

Tube-Tubesheet Jt allowable, 17248.71 is >= tube strength 17248.71 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	140.5 <=	323.0 MPa	Ok
Tubesheet shear stress	28.2 <=	129.2 MPa	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	67.766	105.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.224	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.224	2.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	81.730	...	203	323	36	129	Ps+Pt D1	Ok
D2uc	36.006	...	63	323	8	129	Ps+Pt D2	Ok
D3uc	67.766	...	140	323	28	129	Ps+Pt D3	Ok
Max:	81.7300	... mm		0.629		0.279 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	...	...	370	Ok
D2uc	...	...	...	...	...	370	Ok
D3uc	...	...	...	...	...	370	Ok



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Max

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	81.730	105.000	Ok
Tubesheet thickness flanged extension :	0.000	89.000	Ok
Tube thickness :	0.484	2.000	Ok
Tube-tubesheet fillet weld leg :	1.224	2.000	Ok
Tube-tubesheet groove weld leg :	1.224	2.000	Ok

Given Shell Thickness: 10.0000 mm

Given Channel Thickness: 18.0000 mm

Min Channel length of thk, (18.000) adj. to tubesheet: 202.049 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculations: Step: 12 9:50am Aug 12,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	...	93.5	No Calc	No Calc
20	30	No Calc	...	106.5	No Calc	No Calc
30	40	1000	736	16.5	0.0033062	113.569
40	50	No Calc	...	72	No Calc	No Calc
50	60	3603.53	720	10	0.00039112	37.7171
60	70	No Calc	710.2	5.1	0.00099737	60.5997

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	95	43.3827	...	No Calc
20	30	108	79.4512	...	No Calc
30	40	18	No Calc	...	3.39473
40	50	72	67.6402	...	No Calc
50	60	10	No Calc	...	0.69847
60	70	5.1	No Calc	...	0.48352

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	1000	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	3603.53	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 13

9:50am Aug 12,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume ltr	Corroded Metal Wgt. kg	Corroded ID Volume ltr	Extra due Misc % kg
10	20	510.974	...	510.974	...	51.0974
20	30	265.999	50.7808	262.642	50.8569	26.6
30	40	314.685	384.914	289.064	388.22	31.4685
40	50	114.425	31.178	114.425	31.178	11.4425
50	60	621.13	773.652	621.13	773.652	62.113
60	70	36.8725	74.2371	36.8725	74.2371	3.68725
Total		1864	1314.76	1835	1318.14	186



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
30	Liqd	5.77019	500	...	...	Gas
30	Ins1	13.2649	500	...	...	Hot insulation
30	Noz1	149.334	500	486.525	...	N1- Channel inle
30	Noz1	149.334	500	-486.525	...	N2B- channel out
50	Sad1	138.649	620	525	...	Left Saddle
50	Sad1	138.649	3110	525	...	Right saddle
50	Liqd	773.18	1734.5	...	...	Water
50	Ins1	23.9009	1700	...	...	Hot insulation
50	Noz1	67.795	300	434.137	...	N4B- Shell outle
50	Noz1	50.1028	3000	434.137	...	K1- Pressure rel
50	Noz1	50.1028	300	-434.137	...	N3B- Shell inlet
50	Wght	2550	620	2000	...	Top HE Empty-L
50	Wght	2550	3110	2000	...	Top HE empty-R
50	Wght	3000	620	2000	...	Lft sadl.-oper.
50	Wght	3000	3110	2000	...	Op. wt-R
50	Wght	3400	620	2000	...	Hydro-top HE
50	Wght	3400	3110	2000	...	Hydro- Top HE
50	Wght	100	1000	...	...	Log. baffle
50	Forc	...	620	...	...	Wind shear-left
50	Forc	...	3110	...	...	Right saddle- w
50	Forc	...	620	...	...	Left saddle- Se
50	Forc	...	3110	...	...	Right saddle- S
60	Liqd	74.1918	134.533	...	...	Water
60	Ins1	2.82063	125.6	...	...	Hot insulation
30	FTsh	395.299	1052.5	...	...	Fixed tubesheet
30	Tube	1883.88	2650	...	...	Tubes
40	Baf1	5.62461	350	...	...	Baffles
40	Baf1	5.62461	700	...	...	Baffles
40	Baf1	5.62461	1050	...	...	Baffles
40	Baf1	5.62461	1400	...	...	Baffles
40	Baf1	5.62461	1750	...	...	Baffles
40	Baf1	5.62461	2100	...	...	Baffles
40	Baf1	5.62461	2450	...	...	Baffles
40	Baf1	5.62461	2800	...	...	Baffles
40	Baf1	5.62461	3150	...	...	Baffles

Total Weight of Each Detail Type:



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Element and Detail Weights: Step: 13 9:50am Aug 12,2026

Saddles	277.3
Liquid	853.1
Insulation	40.0
Nozzles	466.7
Weights	18000.0
Exchanger Components	2329.8
Liquid in Tubes	5.5

Sum of the Detail Weights 21972.4 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	2050.5	2050.5	2050.5	2050.5	2050.5	2050.5
Saddles	277.3	277.3	277.3	277.3	277.3	277.3
Nozzles	466.7	466.7	466.7	466.7	466.7	466.7
Wld Weights	100.0	100.0	100.0	100.0	100.0	100.0
Exchanger	2329.8	2329.8	2329.8	2329.8	2329.8	2329.8
Insulation	...	...	...	...	40.0	40.0
Empty Weights	...	...	...	5100.0	5100.0	...
Test Weights	...	6800.0	...	...	...	...
Ope Weights	...	...	...	...	...	6000.0
Ope. Liquid	...	...	...	...	...	853.1
Tube Ope Lqd	...	...	...	...	...	5.5
Test Liquid	...	1314.0	...	...	...	...
Tube Tst Lqd	...	367.7	...	...	...	...
Totals	5224.3	13705.9	5224.3	10324.3	10364.2	12122.9

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	5224.3 kg
Shop Test Wt.	- Fabricated weight + water (full)	13705.9 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	5224.3 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	10324.3 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	10364.2 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	12122.9 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	12091.0 kg
Field Test Wt.	- Empty weight + test medium (full)	13717.4 kg

Exchanger Tube Data

Volume of Exchanger tubes :	368.0 ltr
Weight of Ope Liq in tubes :	5.5 kg
Weight of Water in tubes :	367.7 kg

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	974522	...



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Element and Detail Weights: Step: 13 9:50am Aug 12,2026

20	30	681682	288084
30	40	2312212	2199114
40	50	410807	178128
50	60	7846694	7628730
60	70	719965	698725

Total	12945882.0	10992782.0 mm ²
Total	12.9	11.0 m ²

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Nozzle Flange MAWP:

Step: 14 9:50am Aug 12,2026

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1- Channel in	8.99	10.20	155	600	GR 1.1	3.796	8.986	8.986	8.986
N2B- channel o	8.99	10.20	155	600	GR 1.1	3.796	8.986	8.986	8.986
N4B- Shell out	6.24	8.23	155	600	GR 2.3	5.033	4.331	6.242	4.331
K1- Pressure r	3.12	4.11	155	300	GR 2.3	...	...	...	...
N3B- Shell inl	3.12	4.11	155	300	GR 2.3	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 3.116 MPa

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 4.110 MPa

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 8.986 MPa

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 10.201 MPa

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1- Channel inle	296.037	0.50	17677670	22500	4.735	8.986	Pass
N2B- channel out	296.037	0.50	17677670	22500	4.735	8.986	Pass
N4B- Shell outle	192.151	0.50	6363961	13500	1.311	4.331	Pass

Evaluating Flange on Nozzle [N1- Channel inle]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(17677670) + 4 \cdot \text{abs}(22500)296.037 \leq \pi \cdot 296.037^3 [(8.986 - 4.735) + .500 \cdot 8.986]$$

$$= 0.30950E+09 \leq 0.71283E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1- Channel inle [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.986, 0.500 \cdot 8.986 + 8.986 - 309496832.0 / (\pi \cdot 296.037^3))$$

$$= \min(8.986, 9.682)$$

$$= 8.986 \text{ MPa}$$

Evaluating Flange on Nozzle [N2B- channel out]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(17677670) + 4 \cdot \text{abs}(22500)296.037 \leq \pi \cdot 296.037^3 [(8.986 - 4.735) + .500 \cdot 8.986]$$

$$= 0.30950E+09 \leq 0.71283E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2B- channel out [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.986, 0.500 \cdot 8.986 + 8.986 - 309496832.0 / (\pi \cdot 296.037^3))$$

$$= \min(8.986, 9.682)$$

$$= 8.986 \text{ MPa}$$

Evaluating Flange on Nozzle [N4B- Shell outle]:



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Nozzle Flange MAWP: Step: 14 9:50am Aug 12, 2026

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)] \\
 &= 16(6363961) + 4 \cdot \text{abs}(13500) 192.151 \leq \pi \cdot 192.151^3 [(6.242 - 1.311) + .500 \cdot 6.242] \\
 &= 0.11220E+09 \leq 0.17954E+09 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4B- Shell outle [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(6.242, 0.500 \cdot 6.242 + 6.242 - 112203768.0 / (\pi \cdot 192.151^3)) \\
 &= \min(6.242, 4.331) \\
 &= 4.331 \text{ MPa}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

Step: 15 9:50am Aug 12,2026

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Use Gust Response Factor (Dynamic Analysis)	No	

Calculated Values:

Vessel Length to Diameter ratio: 6.789

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 1.0 • 1.0 • 0.6 • 57.5²)

= 1983.5 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.1 • 0.836 • 1983.503

= 177.6 N

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	1020.00	1.0000	1.0000	1.0000	57.50	0.8358
20	1020.00	1.0000	1.0000	1.0000	57.50	0.8358
30	1020.00	1.0000	1.0000	1.0000	57.50	0.8358
40	1020.00	1.0000	1.0000	1.0000	57.50	0.8358
50	1020.00	1.0000	1.0000	1.0000	57.50	0.8358
60	1020.00	1.0000	1.0000	1.0000	57.50	0.8358

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm²	Elevation mm	Pressure kg/m²	Force N
Top HE Empty-	0.00	3020.00	202.35	0.00
Top HE empty-	0.00	3020.00	202.35	0.00
Lft sadl.-ope	2568290.00	3020.00	202.35	5096.05
Op. wt-R	2568290.00	3020.00	202.35	5096.05
Hydro-top HE	0.00	3020.00	202.35	0.00
Hydro- Top HE	0.00	3020.00	202.35	0.00
Log. baffle	0.00	300.00	202.35	0.00



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Wind Load Calculation:

Step: 15 9:50am Aug 12,2026

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kg/m ²	Element Wind Load N
10	20	1020	1128	107160	202.347	195.414
20	30	1020	840	110040	202.347	200.665
30	40	1020	991.2	991200	202.347	1807.52
40	50	1020	840	68040	202.347	124.076
50	60	1020	922.807	3201216	202.347	17048.9
60	70	1020	912.24	191970	202.347	350.07

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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 Earthquake Load Calculation: Step: 16 9:50am Aug 12,2026

Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 2.0000
 Time Period of Vibration (program or user) 0.0303 Seconds
 Value of Sa/g from user Spectrum Data 0.0939
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.0939 \cdot 1.0) \cdot 1.0$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
 Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
 The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 14519.75 \cdot 0.0939 \cdot 1.0$
 $= 1362.8 \text{ N}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.0939 \cdot 116158 \cdot 1.0$



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Earthquake Load Calculation:

Step: 16 9:50am Aug 12,2026

= 7268.1 N

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N
10	20	470	14519.7	1362.77
20	30	350	14519.7	1362.77
30	40	350	14519.7	1362.77
40	50	350	14519.7	1362.77
50	Sad1	350	14519.7	1362.77
Sad1	60	350	14519.7	1362.77
50	60	350	14519.7	1362.77
60	70	350	14519.7	1362.77

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 17 9:50am Aug 12,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3290.499 mm
Center of Gravity of the Liquid	3306.135 mm
Center of Gravity of the Insulation	2466.133 mm
Center of Gravity of the Nozzles	1381.459 mm
Center of Gravity of the Added Weights (Operating)	3276.319 mm
Center of Gravity of the Added Weights (Empty)	3273.865 mm
Center of Gravity of the Tubesheet(s)	1287.500 mm
Center of Gravity of the Tubes	2885.000 mm
Center of Gravity of the Baffles	3094.500 mm
Center of Gravity of Bare Shell New and Cold	1397.605 mm
Center of Gravity of Bare Shell Corroded	1409.111 mm
Vessel CG in the Operating Condition	2764.268 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	2668.239 mm
Vessel CG in the Test Condition	2645.209 mm

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Saddle Calcs: Bundle Pull New: Step: 18 9:50am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

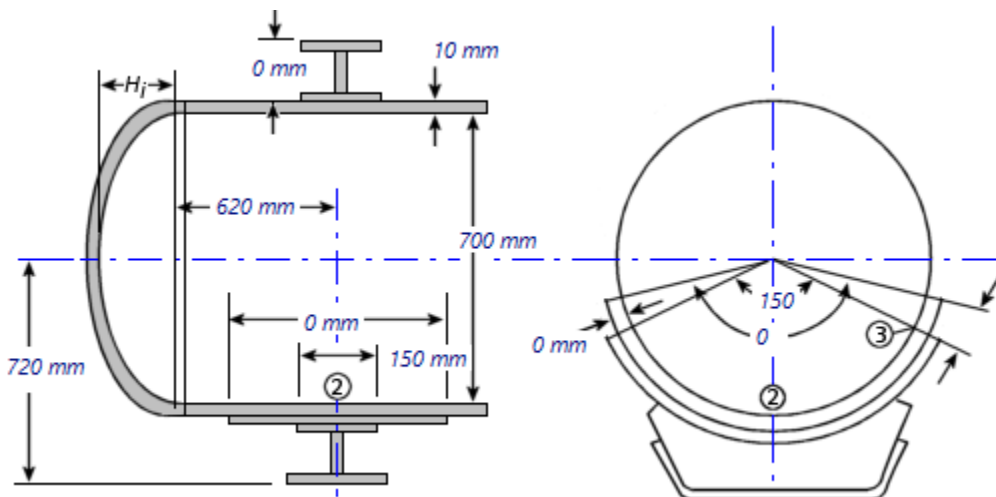
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.9477 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	3626.20	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Shell Allowable Stress used in Calculation		114.50	MPa
Head Allowable Stress used in Calculation		87.41	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		24514.99	N
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		86703.95	N
Pressure used in Analysis	P	0.000	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
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Saddle Calcs: Bundle Pull New: Step: 18 9:50am Aug 12,2026

Long. Stress at Top of Midspan	-6.66	-153.00
Long. Stress at Top of Midspan	6.66	114.50
Long. Stress at Bottom of Midspan	6.66	114.50
Long. Stress at Top of Saddles	12.08	114.50
Long. Stress at Bottom of Saddles	-6.95	-153.00
Long. Stress at Bottom of Saddles	6.95	114.50
Tangential Shear in Shell	12.84	69.72
Circ. Stress at Horn of Saddle	50.12	171.75
Circ. Compressive Stress in Shell	2.40	114.50

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

= Bundle Pull Out Force • Impact Factor • B / Ls

= 24515 • 1.5 • 720.0/2490.0002

= 10633.0 N

Saddle Reaction Force due to Earthquake FI [Fsl]:

= max(FI,eq; Sum of X Forces) • B / Ls

= max(0.0, 36772) • 720.0/2490.0002

= 12143.5 N

Saddle Reaction Force due to Earthquake Ft [Fst]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3.0(0/2 + 34569) • 720.0/714.7852

= 0.0 N

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 74560 + max(10633, 0, 12144, 0)

= 86703.9 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 88063.54 N

Transverse Shear Load 0.00 N

Longitudinal Shear Load 41996.34 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(700.0/2 + 10.0 + 10.0)sin(150.0/2), 2•355.0)

= 714.785 mm

The Computed K values from Table 4.15.1:

K1 = 0.1607 K2 = 0.7988 K3 = 0.4851 K4 = 0.2952

K5 = 0.6733 K6 = 0.0317 K7 = 0.0317 K8 = 0.3021

K9 = 0.2177 K10 = 0.0355 K1* = 0.2792

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -86704•620.0[1-(1-620.0/3626.2+(355.0²-0.0²)/

(2•620.0•3626.2))/(1+(4•0.0)/(3•3626.2))]

= -7687625.0 N-mm

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 86704•3626/4((1+2(355²-0²)/(3626²))/(1+(4•0)/

(3•3626))-4•620/3626)

= 26362354.0 N-mm



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Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) - 26362354 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= -6.66 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) + 26362354 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 6.66 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) - 7687625.0 / (0.1607 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 12.08 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) + 7687625.0 / (0.2792 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= -6.95 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 86704(3626.2 - 2 \cdot 620.0) / (3626.2 + (4 \cdot 0.0 / 3))$$

$$= 57055.0 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 57055.03 / (355.0 \cdot 10.0)$$

$$= 12.84 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{355.0 \cdot 10.0}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.6733 \cdot 86704 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -2.40 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -86704 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) - 3 \cdot 0.0317 \cdot 86704 / (2 \cdot 10.0^2)$$

$$= -50.12 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.95 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0)} + 200.0)10.0^3)/12$$

$$= 24357.543 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 24357.543 + 2922.908(171.814 - 5.0)^2$$

$$= 81359896.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	292.3	10.0	5.0	2922.9	14614.5	0.814E+08
Wearplate	200.0	10.0	15.0	2000.0	30000.0	0.492E+08
Web	15.0	328.0	184.0	4920.0	905279.8	0.448E+08
BasePlate	180.0	22.0	359.0	3960.0	1421639.8	0.139E+09
Totals	...	...	...	13802.9	2371534.2	0.314E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2371534/13802.907$$

$$= 171.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 150.0/2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0)\cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 86703.945$$

$$= 22488.6055 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.0671 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(150.0/2) / 1.309$$

$$= 461.730 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 10387884.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.6766 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(86704 + 1360)180.0 / (2 \cdot 930.0 \cdot 173.33))}$$

$$= 12.146 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (350.0 + 10.0 + 10.0)$$

$$= 714.785 \text{ mm}$$

Distance between Ribs [e]:



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= Web Length / (Nribs - 1)
 = 714.7851/(5 - 1)
 = 178.696 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 178.6963 • 180.0/2
 = 16082.664 mm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 88063.539/(930.0 • 180.0)
 = 0.526 N/mm²

Axial Load [P]:

= Ap • Bp
 = 16082.7 • 0.53
 = 8460.552 N

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 2025.0 + 1340.222
 = 3365.222 mm²

Compressive Stress [Sc]:

= P/Ar
 = 8460.6/3365.2219
 = 2.514 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.488E+07
Web	89.3	15.0	7.5	1340.2	10051.7	0.275E+07
Totals	...	...	...	3365.2	177114.1	0.764E+07

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 150.0 - 15.0
 = 135.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 177114.141/3365.222
 = 52.631 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(52.631, 150.0 - 52.631)
 = 97.369 mm

Radius of Gyration [r]:

= √(Total Inertia / Total Area)
 = √(7636983/3365.222)
 = 47.638 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 720.0 - cos(150.0/2)(350.0 + 10.0 + 10.0) - 22.0
 = 602.237 mm

Intermediate Term [Cc]:



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$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 602.237 / 47.638$$

$$= 12.642$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 36772.5 / (2 \cdot 930.0) \cdot 178.696 \cdot 602.24 / 2$$

$$= 1064238.500 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($12.6419 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.64)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (12.64) / (8 \cdot 123.21) - (12.64)^3 / (8 \cdot 123.21^3))$$

$$= 151.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2.51 / 151.69 + (1064239 \cdot 97.369 / 7636990) / 173.33$$

$$= 0.095$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.677E+07
Web	178.7	15.0	7.5	2680.4	20103.3	0.284E+07
Totals	...	...	...	4705.4	187165.8	0.961E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 187165.828 / 4705.444$$

$$= 39.776 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(39.776, \text{abs}(150.0 - 39.776))$$

$$= 110.224 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 720.0 - \cos(370.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 328.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(9614358 / 4705.444)}$$

$$= 45.202 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 328.0 / 45.202$$

$$= 7.256$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 36772.484 / (2 \cdot 930.0)$$

$$= 19.770 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:



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$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 19.77 \cdot 178.696 \cdot 328.0 \\
 &= 1159245.500 \text{ N-mm}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 1159246 / 87226.062 \\
 &= 13.286 \text{ MPa}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($7.2563 < 123.2107$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r)^2 / (8 \cdot C_c^2) - (KL/r)^4 / (8 \cdot C_c^4)) \\
 &= (1 - (7.26)^2 / (2 \cdot 123.21^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (7.26)^2 / (8 \cdot 123.21^2) - (7.26^4) / (8 \cdot 123.21^4)) \\
 &= 153.7 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba} \\
 &= 3.54 / 153.69 + (1159246 \cdot 110.224 / 9614358) / 173.33 \\
 &= 0.100
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	600.6192 mm ²
Saddle Load Q0 (Weight)	Q0	75920.0 N
Saddle Load QL (Wind/Seismic contribution)	QL	36772.5 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	F1	36772.5 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$\begin{aligned}
 &= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 36772.5 / (600.619 \cdot 2) \\
 &= 30.6 \text{ MPa. Must be less than } 103.2 \text{ MPa.}
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$\begin{aligned}
 &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 0.0 / (600.619 \cdot 2) \\
 &= 0.0 \text{ MPa. Must be less than } 103.2 \text{ MPa.}
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned}
 &= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\
 &= 36772.48 / (103.2 \cdot 2.0) \\
 &= 178.1796 \text{ mm}^2
 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned}
 &= B \cdot Ft + \text{Sum of X Moments} \\
 &= 720.0 \cdot 0.0 + 0.0 \\
 &= 0.00 \text{ N-mm}
 \end{aligned}$$



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Eccentricity (e):
= Rmom / QO
= 0.0/75920.02
= 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 178.1796, 0.0]
= 178.1796 mm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

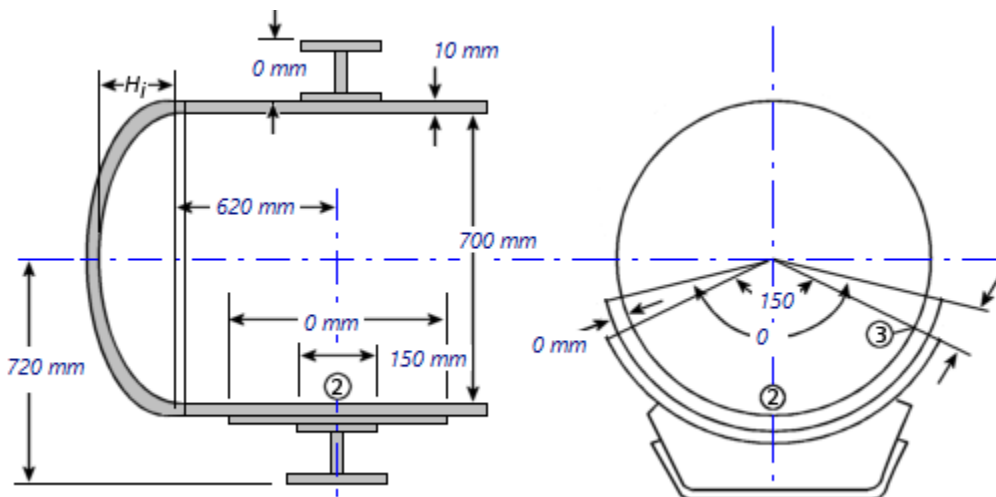
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.9477 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	3626.20	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Shell Allowable Stress used in Calculation		114.50	MPa
Head Allowable Stress used in Calculation		87.41	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		24514.99	N
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		86547.66	N
Pressure used in Analysis	P	0.000	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
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Long. Stress at Top of Midspan	-6.64	-153.00
Long. Stress at Top of Midspan	6.64	114.50
Long. Stress at Bottom of Midspan	6.64	114.50
Long. Stress at Top of Saddles	12.06	114.50
Long. Stress at Bottom of Saddles	-6.94	-153.00
Long. Stress at Bottom of Saddles	6.94	114.50
Tangential Shear in Shell	12.82	69.72
Circ. Stress at Horn of Saddle	50.03	171.75
Circ. Compressive Stress in Shell	2.40	114.50

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

= Bundle Pull Out Force • Impact Factor • B / Ls

= 24515 • 1.5 • 720.0/2490.0002

= 10633.0 N

Saddle Reaction Force due to Earthquake Fl [Fsl]:

= max(Fl,eq; Sum of X Forces) • B / Ls

= max(0.0, 36772) • 720.0/2490.0002

= 12143.5 N

Saddle Reaction Force due to Earthquake Ft [Fst]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3.0(0/2 + 34569) • 720.0/714.7852

= 0.0 N

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 74404 + max(10633, 0, 12144, 0)

= 86547.7 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 87907.25 N

Transverse Shear Load 0.00 N

Longitudinal Shear Load 41996.34 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(700.0/2 + 10.0 + 10.0)sin(150.0/2), 2•355.0)

= 714.785 mm

The Computed K values from Table 4.15.1:

K1 = 0.1607 K2 = 0.7988 K3 = 0.4851 K4 = 0.2952

K5 = 0.6733 K6 = 0.0317 K7 = 0.0317 K8 = 0.3021

K9 = 0.2177 K10 = 0.0355 K1* = 0.2792

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -86548•620.0[1-(1-620.0/3626.2+(355.0²-0.0²)/

(2•620.0•3626.2))/(1+(4•0.0)/(3•3626.2))]]

= -7673767.5 N-mm

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 86548•3626/4((1+2(355²-0²)/(3626²))/(1+(4•0)/

(3•3626))-4•620/3626)

= 26314836.0 N-mm



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Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) - 26314836 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= -6.64 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) + 26314836 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 6.64 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) - 7673767.5 / (0.1607 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 12.06 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) + 7673767.5 / (0.2792 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= -6.94 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 86548(3626.2 - 2 \cdot 620.0) / (3626.2 + (4 \cdot 0.0 / 3))$$

$$= 56952.2 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 56952.18 / (355.0 \cdot 10.0)$$

$$= 12.82 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{355.0 \cdot 10.0}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.6733 \cdot 86548 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -2.40 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -86548 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) - 3 \cdot 0.0317 \cdot 86548 / (2 \cdot 10.0^2)$$

$$= -50.03 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.95 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0)} + 200.0)10.0^3)/12$$

$$= 24357.543 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 24357.543 + 2922.908(171.814 - 5.0)^2$$

$$= 81359896.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	292.3	10.0	5.0	2922.9	14614.5	0.814E+08
Wearplate	200.0	10.0	15.0	2000.0	30000.0	0.492E+08
Web	15.0	328.0	184.0	4920.0	905279.8	0.448E+08
BasePlate	180.0	22.0	359.0	3960.0	1421639.8	0.139E+09
Totals	...	...	...	13802.9	2371534.2	0.314E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2371534/13802.907$$

$$= 171.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 150.0/2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0)\cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 86547.656$$

$$= 22448.0703 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.0634 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(150.0/2) / 1.309$$

$$= 461.730 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 10369160.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.6664 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt})\text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(86548 + 1360)180.0 / (2 \cdot 930.0 \cdot 173.33)}$$

$$= 12.135 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (350.0 + 10.0 + 10.0)$$

$$= 714.785 \text{ mm}$$

Distance between Ribs [e]:



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= Web Length / (Nribs - 1)
 = 714.7851/(5 - 1)
 = 178.696 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 178.6963 • 180.0/2
 = 16082.664 mm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 87907.25/(930.0 • 180.0)
 = 0.525 N/mm²

Axial Load [P]:

= Ap • Bp
 = 16082.7 • 0.53
 = 8445.537 N

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 2025.0 + 1340.222
 = 3365.222 mm²

Compressive Stress [Sc]:

= P/Ar
 = 8445.5/3365.2219
 = 2.510 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.488E+07
Web	89.3	15.0	7.5	1340.2	10051.7	0.275E+07
Totals	...	...	...	3365.2	177114.1	0.764E+07

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 150.0 - 15.0
 = 135.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 177114.141/3365.222
 = 52.631 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(52.631, 150.0 - 52.631)
 = 97.369 mm

Radius of Gyration [r]:

= √(Total Inertia / Total Area)
 = √(7636983/3365.222)
 = 47.638 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 720.0 - cos(150.0/2)(350.0 + 10.0 + 10.0) - 22.0
 = 602.237 mm

Intermediate Term [Cc]:



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$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 602.237 / 47.638$$

$$= 12.642$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 36772.5 / (2 \cdot 930.0) \cdot 178.696 \cdot 602.24 / 2$$

$$= 1064238.500 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($12.6419 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.64)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (12.64) / (8 \cdot 123.21) - (12.64)^3 / (8 \cdot 123.21^3))$$

$$= 151.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2.51 / 151.69 + (1064239 \cdot 97.369 / 7636990) / 173.33$$

$$= 0.095$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.677E+07
Web	178.7	15.0	7.5	2680.4	20103.3	0.284E+07
Totals	...	...	...	4705.4	187165.8	0.961E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 187165.828 / 4705.444$$

$$= 39.776 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(39.776, \text{abs}(150.0 - 39.776))$$

$$= 110.224 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 720.0 - \cos(370.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 328.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(9614358 / 4705.444)}$$

$$= 45.202 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 328.0 / 45.202$$

$$= 7.256$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 36772.484 / (2 \cdot 930.0)$$

$$= 19.770 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:



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$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 19.77 \cdot 178.696 \cdot 328.0 \\
 &= 1159245.500 \text{ N-mm}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 1159246 / 87226.062 \\
 &= 13.286 \text{ MPa}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($7.2563 < 123.2107$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (7.26)^2 / (2 \cdot 123.21^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (7.26) / (8 \cdot 123.21) - (7.26^3) / (8 \cdot 123.21^3)) \\
 &= 153.7 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba} \\
 &= 3.54 / 153.69 + (1159246 \cdot 110.224 / 9614358) / 173.33 \\
 &= 0.100
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	600.6192 mm ²
Saddle Load Q0 (Weight)	Q0	75763.7 N
Saddle Load QL (Wind/Seismic contribution)	QL	36772.5 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	F1	36772.5 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$\begin{aligned}
 &= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 36772.5 / (600.619 \cdot 2) \\
 &= 30.6 \text{ MPa. Must be less than } 103.2 \text{ MPa.}
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$\begin{aligned}
 &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 0.0 / (600.619 \cdot 2) \\
 &= 0.0 \text{ MPa. Must be less than } 103.2 \text{ MPa.}
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction) }$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned}
 &= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\
 &= 36772.48 / (103.2 \cdot 2.0) \\
 &= 178.1796 \text{ mm}^2
 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned}
 &= B \cdot Ft + \text{Sum of X Moments} \\
 &= 720.0 \cdot 0.0 + 0.0 \\
 &= 0.00 \text{ N-mm}
 \end{aligned}$$



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Eccentricity (e):

= Rmom / QO

= 0.0/75763.73

= 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 178.1796, 0.0]

= 178.1796 mm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

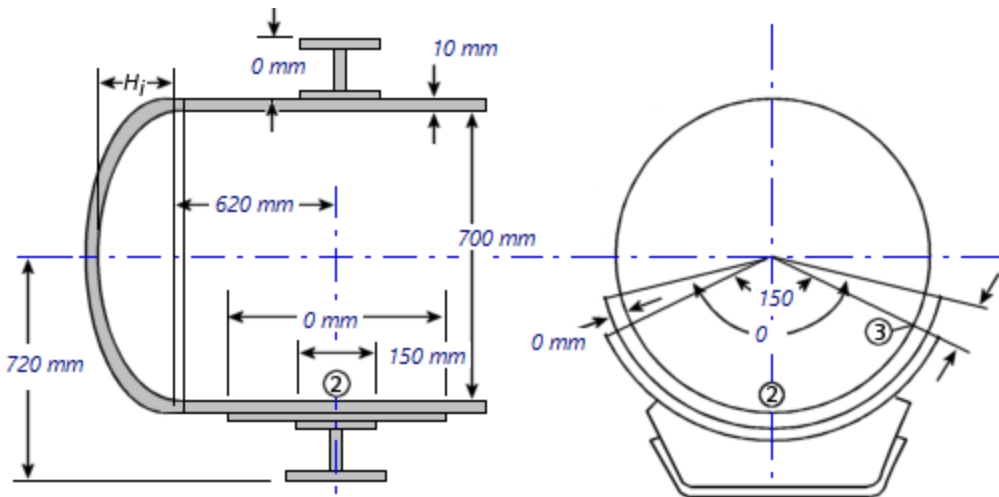
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.9477 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	3626.20	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Shell Allowable Stress used in Calculation		114.50	MPa
Head Allowable Stress used in Calculation		87.41	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	221369.73	N
Pressure used in Analysis	P	1.003 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	0.82	114.50
Long. Stress at Bottom of Midspan	34.81	114.50
Long. Stress at Top of Saddles	48.66	114.50
Long. Stress at Bottom of Saddles	0.06	114.50



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Tangential Shear in Shell	32.78	69.72
Circ. Stress at Horn of Saddle	127.97	143.12
Circ. Compressive Stress in Shell	6.14	114.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(19726.7/2 + 34569) \cdot 720.0/714.7852$$

$$= 134270.4 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(1279.67, 41996) \cdot 720.0/2490.0002$$

$$= 12143.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(10902.13, 41996) \cdot 720.0/2490.0002$$

$$= 12143.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(10902/2 + 34569) \cdot 720.0/714.7852$$

$$= 120937.0 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 87099 + \max(12144, 134270, 12144, 120937)$$

$$= 221369.7 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.72 \cdot 1.2)^2)$$

$$= 1279.740 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	222729.33	N
Transverse Shear Load	44432.64	N
Longitudinal Shear Load	41996.34	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(700.0/2 + 10.0 + 10.0) \sin(150.0/2), 2 \cdot 355.0)$$

$$= 714.785 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -221370 \cdot 620.0 [1 - (1 - 620.0/3626.2 + (355.0^2 - 0.0^2) / (2 \cdot 620.0 \cdot 3626.2)) / (1 + (4 \cdot 0.0) / (3 \cdot 3626.2))]$$

$$= -19627796.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$



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$$= 221370 \cdot 3626 / 4 \left((1 + 2(355^2 - 0^2) / (3626^2)) / (1 + (4 \cdot 0) / (3 \cdot 3626)) \right) - 4 \cdot 620 / 3626$$

$$= 67307520.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.003 \cdot 355.0 / (2 \cdot 10.0) - 67307520 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 0.82 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.003 \cdot 355.0 / (2 \cdot 10.0) + 67307520 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 34.81 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.003 \cdot 355.0 / (2 \cdot 10.0) - 19627796.0 / (0.1607 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 48.66 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.003 \cdot 355.0 / (2 \cdot 10.0) + 19627796.0 / (0.2792 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 0.06 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 221370(3626.2 - 2 \cdot 620.0) / (3626.2 + (4 \cdot 0.0 / 3))$$

$$= 145671.1 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 145671.06 / (355.0 \cdot 10.0)$$

$$= 32.78 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.6733 \cdot 221370 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -6.14 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -221370 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) - 3 \cdot 0.0317 \cdot 221370 / (2 \cdot 10.0^2)$$

$$= -127.97 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(355.0 \cdot 10.0)}, 2 \cdot 620.0)$$

$$= 242.95 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 2490.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000017 \cdot 2490.0(155.0 - 21.1)$$

$$= 5.547 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length BpLen 930.0000 mm



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Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	MPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0) + 200.0})^3) / 12$$

$$= 24357.543 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 24357.543 + 2922.908(171.814 - 5.0)^2$$

$$= 81359896.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	292.3	10.0	5.0	2922.9	14614.5	0.814E+08
Wearplate	200.0	10.0	15.0	2000.0	30000.0	0.492E+08
Web	15.0	328.0	184.0	4920.0	905279.8	0.448E+08
BasePlate	180.0	22.0	359.0	3960.0	1421639.8	0.139E+09
Totals	...	...	...	13802.9	2371534.2	0.314E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2371534 / 13802.907$$

$$= 171.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 221369.734$$

$$= 57417.1914 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 5.2778 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(150.0/2) / 1.309$$

$$= 461.730 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 26522014.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 14.4933 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ MPa}$$



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Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{2 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(221370 + 1360) \cdot 180.0}{2 \cdot 930.0 \cdot 174.663}}$$

$$= 19.242 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (350.0 + 10.0 + 10.0)$$

$$= 714.785 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 714.7851 / (5 - 1)$$

$$= 178.696 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 178.6963 \cdot 180.0 / 2$$

$$= 16082.664 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 222729.312 / (930.0 \cdot 180.0)$$

$$= 1.331 \text{ N/mm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 16082.7 \cdot 1.33$$

$$= 21398.334 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2025.0 + 1340.222$$

$$= 3365.222 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 21398.3 / 3365.2219$$

$$= 6.359 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.488E+07
Web	89.3	15.0	7.5	1340.2	10051.7	0.275E+07
Totals	...	...	...	3365.2	177114.1	0.764E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 15.0$$

$$= 135.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 177114.141 / 3365.222$$

$$= 52.631 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(52.631, 150.0 - 52.631)$$



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= 97.369 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{7636983/3365.222}$

= 47.638 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk

= 720.0 - $\cos(150.0/2)$ (350.0 + 10.0 + 10.0) - 22.0

= 602.237 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199955/262.0}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 602.237/47.638

= 12.642

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$

= 41996.3/(2 • 930.0) • 178.696 • 602.24/2

= 1215422.875 N-mm

Compressive Allowable, $KL/r < C_c$ (12.6419 < 122.7395) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2/(2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r)/(8 \cdot C_c) - (KL/r)^3/(8 \cdot C_c^3))$

= $(1 - (12.64)^2/(2 \cdot 122.74^2)) 262 /$

$(5/3 + 3 \cdot (12.64)/(8 \cdot 122.74) - (12.64^3)/(8 \cdot 122.74^3))$

= 152.8 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$

= 6.36/152.83 + (1215423 • 97.369/7636990)/174.66

= 0.130

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.677E+07
Web	178.7	15.0	7.5	2680.4	20103.3	0.284E+07
Totals	...	...	...	4705.4	187165.8	0.961E+07

Distance to Centroid from Datum [ytot]:

= AY / A

= 187165.828/4705.444

= 39.776 mm

Distance to Centroid [C1]:

= $\max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$

= $\max(39.776, \text{abs}(150.0 - 39.776))$

= 110.224 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 720.0 - $\cos(370.0/2)$ (22.0 + 0.0 + 0.0) - 0.0

= 328.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{9614358/4705.444}$



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= 45.202 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 328.0/45.202

= 7.256

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 41996.34/(2 • 930.0)

= 22.579 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 22.579 • 178.696 • 328.0

= 1323926.375 N-mm

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 1323926/87226.062

= 15.173 MPa

Compressive Allowable, KL/r < Cc (7.2563 < 122.7395) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))

= (1-(7.26)²/(2 • 122.74²))262/

(5/3+3•(7.26)/(8• 122.74)-(7.26³)/(8•122.74³))

= 154.9 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 8.96/154.86 + (1323926 • 110.224/9614358)/174.66

= 0.145

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.29 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	175.00 MPa
Area Available in a Single Bolt	BltArea	600.6192 mm ²
Saddle Load Q0 (Weight)	Q0	88458.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	12143.5 N
Maximum Transverse Force	Ft	44432.6 N
Maximum Longitudinal Force	F1	41996.3 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

= 11305.9 N

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= max(Ff; F1) / (Bolt Area • Number of Bolts)

= 41996.3/(600.619 • 2)

= 35.0 MPa. Must be less than 107.9 MPa.

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)

= 44432.6/(600.619 • 2)

= 37.0 MPa. Must be less than 107.9 MPa.



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Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $41996.34 / (107.87 \cdot 2.0)$
= 194.6729 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $720.0 \cdot 44432.64 + 0.0$
= 32004474.00 N-mm

Eccentricity (e):
= $Rmom / QO$
= $32004474 / 88458.9$
= 361.65 mm > $Bplen / 6$ --> Uplift in Transverse direction

$f = Bplen / 2 - Edgedis$
= $930.0 / 2 - 50.0$
= 415.00 mm

Modular Ratio Of Steel/Concrete (n1):
= ES / EC
= $203402.5 / 21526.32$
= 9.45

$K1 = 3 (e - 0.5 \cdot Bplen)$
= $3 (361.65 - 0.5 \cdot 930.0)$
= -310.04 mm

$K2 = 6 \cdot n1 \cdot At / Bpwid \cdot (f + e)$
= $6 \cdot 9.45 \cdot 600.62 / 180.0 \cdot (415.0 + 361.65)$
= 146923.76 mm²

$K3 = -K2 \cdot (0.5 \cdot Bplen + f)$
= $-146923.77 \cdot (0.5 \cdot 930.0 + 415.0)$
= -129292930.01 mm³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -310.04 \cdot Y^2 + 146923.77 \cdot Y + -0.1E+09 = 0$
 $Y = 513.82 \text{ mm}$

$Num = (Bplen / 2 - Y / 3 - e)$
= $(930.0 / 2 - 513.82 / 3 - 361.65)$
= -67.93

$Denom = (Bplen / 2 - Y / 3 + f)$
= $(930.0 / 2 - 513.82 / 3 + 415.0)$
= 708.73

Total Bolt Tension Force [Tforce]:
= $-QO \cdot Num / Denom$
= $-88458.9 \cdot -67.93 / 708.73$
= 8478.25 N

Bolt Area Required due to Transverse Load [Bltareart]:
= $Tforce / (Stba \cdot Nbt)$
= $8478.25 / (137.29 \cdot 1.0)$
= 61.7582 mm²



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 194.6729, 61.7582]
 = 194.6729 mm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
 = 2(QO + Tforce) / (Y • Bpwid)
 = 2(88458.9 + 8478.25)/(513.82 • 180.0)
 = 2.10 MPa

Distance from Baseplate Edge to the Web [ADIST]:
 = (Bplen - Weblength) / 2
 = (930.0 - 879.2)/2
 = 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
 = Tforce • Adist
 = 8478.25 • 25.4
 = 215434.94 N-mm

Equivalent Bearing Pressure (f1):
 = fc • (Y - Adist) / Y
 = 2.1 • (513.82 - 25.4)/513.82
 = 1.99 MPa

Overturning Moment due to Bearing Pressure [Mc]:
 = (Adist² • Bpwid / 6) • (f1 + 2 • fc)
 = (25.4² • 180.0/6) • (1.99 + 2 • 2.1)
 = 119758.67 N-mm

Baseplate Required Thickness [Treq]:
 = $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
 = $\sqrt{(6 \cdot \max(215434.94, 119758.67) / (180.0 \cdot 262.49))}$
 = 5.2296 mm

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (200.00) is less than 1.56*sqrt(rm*t)
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = min(b + 1.56*sqrt(Rm • t), 2a)
 = min(150.0 + 1.56*sqrt(355.0 • 10.0), 2 • 490.0)
 = 242.9477 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	3626.20	mm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Inside or Mean Depth of Head	Hi or hm	177.55	mm
Shell Allowable Stress used in Calculation		114.50	MPa
Head Allowable Stress used in Calculation		114.50	MPa



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Circumferential Efficiency in Plane of Saddle	1.00
Circumferential Efficiency at Mid-Span	1.00
Distance from Saddle Base to Centerline	B 720.00 mm
Coefficient of Friction	mu 0.30

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 170597.20 N
 Pressure used in Analysis P 1.003 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	1.73	114.50
Long. Stress at Bottom of Midspan	33.90	114.50
Long. Stress at Top of Saddles	39.26	114.50
Long. Stress at Bottom of Saddles	5.47	114.50
Tangential Shear in Shell	26.30	69.72
Circ. Stress at Horn of Saddle	98.62	143.12
Circ. Compressive Stress in Shell	4.73	114.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (19726.7/2 + 34569) \cdot 720.0/714.7852$
 $= 134270.4 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(1279.67, 41996) \cdot 720.0/2490.0002$
 $= 12143.5 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(10902.13, 41996) \cdot 720.0/2490.0002$
 $= 12143.5 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (10902/2 + 34569) \cdot 720.0/714.7852$
 $= 120937.0 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 36327 + \max(12144, 134270, 12144, 120937)$
 $= 170597.2 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.72 \cdot 1.2)^2)$
 $= 1279.740 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	171956.80 N
Transverse Shear Load	44432.64 N
Longitudinal Shear Load	41996.34 N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$



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$$= \min(2(700.0/2 + 10.0 + 10.0) \sin(150.0/2), 2 \cdot 355.0)$$

$$= 714.785 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -170597 \cdot 490.0 [1 - (1 - 490.0/3626.2 + (355.0^2 - 177.55^2) / (2 \cdot 490.0 \cdot 3626.2)) / (1 + (4 \cdot 177.55) / (3 \cdot 3626.2))]$$

$$= -13645099.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 170597 \cdot 3626 / 4 \cdot ((1 + 2(355^2 - 178^2)/(3626^2)) / (1 + (4 \cdot 178) / (3 \cdot 3626)) - 4 \cdot 490 / 3626)$$

$$= 63697028.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 1.003 \cdot 355.0 / (2 \cdot 10.0) - 63697028 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 1.73 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 1.003 \cdot 355.0 / (2 \cdot 10.0) + 63697028 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 33.90 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 1.003 \cdot 355.0 / (2 \cdot 10.0) - 13645099.0 / (0.1607 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 39.26 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 1.003 \cdot 355.0 / (2 \cdot 10.0) + 13645099.0 / (0.2792 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 5.47 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 170597(3626.2 - 2 \cdot 490.0) / (3626.2 + (4 \cdot 177.55 / 3))$$

$$= 116863.1 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.7988 \cdot 116863.09 / (355.0 \cdot 10.0)$$

$$= 26.30 \text{ MPa}$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.6733 \cdot 170597 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -4.73 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -170597 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) -$$



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$$3 \cdot 0.0317 \cdot 170597 / (2 \cdot 10.0^2) \\ = -98.62 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a) \\ = \min(150.0 + 1.56 \cdot \sqrt{ 355.0 \cdot 10.0 }, 2 \cdot 490.0) \\ = 242.95 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	MPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth})t^3)/12 \\ = ((1.56 \cdot \sqrt{ 350.0 \cdot 10.0 } + 200.0)10.0^3)/12 \\ = 24357.543 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\ = 24357.543 + 2922.908(171.814 - 5.0)^2 \\ = 81359896.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	292.3	10.0	5.0	2922.9	14614.5	0.814E+08
Wearplate	200.0	10.0	15.0	2000.0	30000.0	0.492E+08
Web	15.0	328.0	184.0	4920.0	905279.8	0.448E+08
BasePlate	180.0	22.0	359.0	3960.0	1421639.8	0.139E+09
Totals	...	...	...	13802.9	2371534.2	0.314E+09

Distance to Centroid [C1]:

$$= AY / A \\ = 2371534 / 13802.907 \\ = 171.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2 \\ = 180 - 150.0/2 \\ = 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ = (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0)) \\ = 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q \\ = 0.259 \cdot 170597.203 \\ = 44248.1953 \text{ N}$$

Tension Stress, St = (Fh/As) = 4.0673 MPa

Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 MPa



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Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(150.0/2) / 1.309$$

$$= 461.730 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 20439022.0000 \text{ N-mm}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 11.1692 \text{ MPa}$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{2 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(170597 + 1360) \cdot 180.0}{2 \cdot 930.0 \cdot 174.663}}$$

$$= 16.907 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (350.0 + 10.0 + 10.0)$$

$$= 714.785 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 714.785 / (5 - 1)$$

$$= 178.696 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 178.6963 \cdot 180.0 / 2$$

$$= 16082.664 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 171956.781 / (930.0 \cdot 180.0)$$

$$= 1.027 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 16082.7 \cdot 1.03$$

$$= 16520.451 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2025.0 + 1340.222$$

$$= 3365.222 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 16520.5 / 3365.2219$$

$$= 4.910 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.488E+07
Web	89.3	15.0	7.5	1340.2	10051.7	0.275E+07
Totals	...	...	...	3365.2	177114.1	0.764E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$



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= 150.0 - 15.0
= 135.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 177114.141/3365.222
= 52.631 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
= max(52.631, 150.0 - 52.631)
= 97.369 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{7636983/3365.222}$
= 47.638 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 720.0 - cos(150.0/2)(350.0 + 10.0 + 10.0) - 22.0
= 602.237 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199955/262.0}$
= 122.739

Slenderness ratio [KL/r]:

= KL/r
= 1 • 602.237/47.638
= 12.642

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 41996.3 / (2 • 930.0) • 178.696 • 602.24/2
= 1215422.875 N-mm

Compressive Allowable, KL/r < Cc (12.6419 < 122.7395) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
= (1 - (12.64)² / (2 • 122.74²)) 262 /
(5/3 + 3 • (12.64) / (8 • 122.74) - (12.64³) / (8 • 122.74³))
= 152.8 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
= 4.91/152.83 + (1215423 • 97.369/7636990) / 174.66
= 0.121

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.677E+07
Web	178.7	15.0	7.5	2680.4	20103.3	0.284E+07
Totals	...	...	...	4705.4	187165.8	0.961E+07

Distance to Centroid from Datum [ytot]:

= AY / A
= 187165.828/4705.444
= 39.776 mm

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))



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$$= \max(39.776, \text{abs}(150.0 - 39.776))$$

$$= 110.224 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 720.0 - \cos(370.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 328.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{9614358/4705.444}$$

$$= 45.202 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 328.0/45.202$$

$$= 7.256$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 41996.34 / (2 \cdot 930.0)$$

$$= 22.579 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 22.579 \cdot 178.696 \cdot 328.0$$

$$= 1323926.375 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1323926/87226.062$$

$$= 15.173 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($7.2563 < 122.7395$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.26)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (7.26) / (8 \cdot 122.74) - (7.26^3) / (8 \cdot 122.74^3))$$

$$= 154.9 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 6.92/154.86 + (1323926 \cdot 110.224/9614358) / 174.66$$

$$= 0.132$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.29 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	175.00 MPa
Area Available in a Single Bolt	BltArea	600.6192 mm ²
Saddle Load Q0 (Weight)	Q0	37686.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	12143.5 N
Maximum Transverse Force	Ft	44432.6 N
Maximum Longitudinal Force	F1	41996.3 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$



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$$= 41996.3 / (600.619 \cdot 2)$$

$$= 35.0 \text{ MPa. Must be less than } 107.9 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 44432.6 / (600.619 \cdot 2)$$

$$= 37.0 \text{ MPa. Must be less than } 107.9 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 41996.34 / (107.87 \cdot 2.0)$$

$$= 194.6729 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 720.0 \cdot 44432.64 + 0.0$$

$$= 32004474.00 \text{ N-mm}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 32004474 / 37686.38$$

$$= 848.89 \text{ mm} > B_{plen} / 6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 930.0 / 2 - 50.0$$

$$= 415.00 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= E_s / E_c$$

$$= 203402.5 / 21526.32$$

$$= 9.45$$

K1 = 3 (e - 0.5 • Bplen)

$$= 3 (848.89 - 0.5 \cdot 930.0)$$

$$= 1151.66 \text{ mm}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$= 6 \cdot 9.45 \cdot 600.62 / 180.0 \cdot (415.0 + 848.89)$$

$$= 239096.41 \text{ mm}^2$$

K3 = -K2 • (0.5 • Bplen + f)

$$= -239096.41 \cdot (0.5 \cdot 930.0 + 415.0)$$

$$= -210404867.31 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K_1 \cdot Y^2 + K_2 \cdot Y + K_3 = 0$$

$$Y^3 + 1151.66 \cdot Y^2 + 239096.41 \cdot Y + -0.2E+09 = 0$$

$$Y = 306.61 \text{ mm}$$

Num = (Bplen / 2 - Y / 3 - e)

$$= (930.0 / 2 - 306.61 / 3 - 848.89)$$

$$= -486.09$$

Denom = (Bplen / 2 - Y / 3 + f)

$$= (930.0 / 2 - 306.61 / 3 + 415.0)$$

$$= 777.80$$

Total Bolt Tension Force [Tforce]:



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$$\begin{aligned}
 &= - QO \cdot Num / Denom \\
 &= - 37686.38 \cdot -486.09 / 777.8 \\
 &= 23552.52 \text{ N}
 \end{aligned}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned}
 &= Tforce / (Stba \cdot Nbt) \\
 &= 23552.52 / (137.29 \cdot 1.0) \\
 &= 171.5639 \text{ mm}^2
 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned}
 &= \max[Bltarearl, Bltarears, Bltareart] \\
 &= \max[0.0, 194.6729, 171.5639] \\
 &= 194.6729 \text{ mm}^2
 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$\begin{aligned}
 &= 2(QO + Tforce) / (Y \cdot Bpwid) \\
 &= 2(37686.38 + 23552.52) / (306.61 \cdot 180.0) \\
 &= 2.22 \text{ MPa}
 \end{aligned}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$\begin{aligned}
 &= (Bplen - WebIngh) / 2 \\
 &= (930.0 - 879.2) / 2 \\
 &= 25.4000 \text{ mm}
 \end{aligned}$$

Overturning Moment due To Bolt Tension [Mt]:

$$\begin{aligned}
 &= Tforce \cdot Adist \\
 &= 23552.52 \cdot 25.4 \\
 &= 598476.56 \text{ N-mm}
 \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned}
 &= fc \cdot (Y - Adist) / Y \\
 &= 2.22 \cdot (306.61 - 25.4) / 306.61 \\
 &= 2.04 \text{ MPa}
 \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned}
 &= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc) \\
 &= (25.4^2 \cdot 180.0 / 6) \cdot (2.04 + 2 \cdot 2.22) \\
 &= 125348.49 \text{ N-mm}
 \end{aligned}$$

Baseplate Required Thickness [Treq]:

$$\begin{aligned}
 &= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))} \\
 &= \sqrt{(6 \cdot \max(598477, 125348.49) / (180.0 \cdot 262.49))} \\
 &= 8.7163 \text{ mm}
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

Step: 21 9:50am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

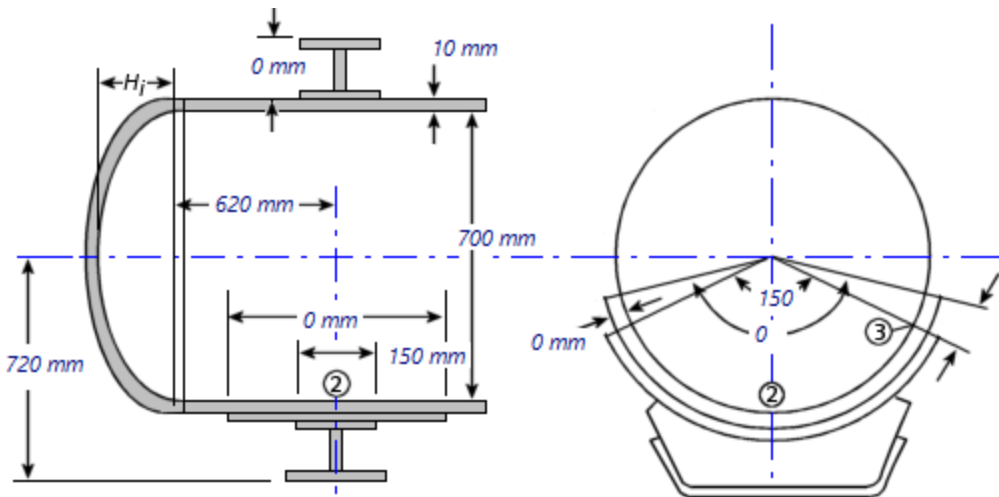
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.9477 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	3626.20	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Shell Allowable Stress used in Calculation		153.00	MPa
Head Allowable Stress used in Calculation		153.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	131191.98	N
Pressure used in Analysis	P	1.304 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	13.07	153.00
Long. Stress at Bottom of Midspan	33.21	153.00
Long. Stress at Top of Saddles	41.42	153.00
Long. Stress at Bottom of Saddles	12.62	153.00



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Saddle Calcs: Test Case:

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Tangential Shear in Shell	19.43	90.61
Circ. Stress at Horn of Saddle	75.84	229.50
Circ. Compressive Stress in Shell	3.64	153.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (6509.8 / 2 + 0) \cdot 720.0 / 714.7852$$

$$= 9836.0 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(422.29, 0) \cdot 720.0 / 2490.0002$$

$$= 122.1 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 121356 + \max(122, 9836, 0, 0)$$

$$= 131192.0 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.72 \cdot 1.2)^2)$$

$$= 1279.740 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 132551.58 N

Transverse Shear Load 3254.90 N

Longitudinal Shear Load 422.29 N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(700.0/2 + 10.0 + 10.0) \sin(150.0/2), 2 \cdot 355.0)$$

$$= 714.785 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607 K2 = 0.7988 K3 = 0.4851 K4 = 0.2952

K5 = 0.6733 K6 = 0.0317 K7 = 0.0317 K8 = 0.3021

K9 = 0.2177 K10 = 0.0355 K1* = 0.2792

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -131192 \cdot 620.0 [1 - (1 - 620.0/3626.2 + (355.0^2 - 0.0^2) /$$

$$(2 \cdot 620.0 \cdot 3626.2)) / (1 + (4 \cdot 0.0) / (3 \cdot 3626.2))]$$

$$= -11632167.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 131192 \cdot 3626 / 4 ((1 + 2(355^2 - 0^2)/(3626^2)) / (1 + (4 \cdot 0) /$$

$$(3 \cdot 3626)) - 4 \cdot 620 / 3626)$$

$$= 39888956.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 1.304 \cdot 355.0 / (2 \cdot 10.0) - 39888956 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 13.07 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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Saddle Calcs: Test Case:

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$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.304 \cdot 355.0 / (2 \cdot 10.0) + 39888956 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 33.21 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.304 \cdot 355.0 / (2 \cdot 10.0) - 11632167.0 / (0.1607 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 41.42 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.304 \cdot 355.0 / (2 \cdot 10.0) + 11632167.0 / (0.2792 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 12.62 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 131192(3626.2 - 2 \cdot 620.0) / (3626.2 + (4 \cdot 0.0 / 3))$$

$$= 86330.1 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 86330.12 / (355.0 \cdot 10.0)$$

$$= 19.43 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.6733 \cdot 131192 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -3.64 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -131192 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) - 3 \cdot 0.0317 \cdot 131192 / (2 \cdot 10.0^2)$$

$$= -75.84 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(355.0 \cdot 10.0)}, 2 \cdot 620.0)$$

$$= 242.95 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0)} + 200.0)^3) / 12$$

$$= 24357.543 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:



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Saddle Calcs: Test Case:

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$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 24357.543 + 2922.908(171.814 - 5.0)^2 \\
 &= 81359896.000 \text{ mm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	292.3	10.0	5.0	2922.9	14614.5	0.814E+08
Wearplate	200.0	10.0	15.0	2000.0	30000.0	0.492E+08
Web	15.0	328.0	184.0	4920.0	905279.8	0.448E+08
BasePlate	180.0	22.0	359.0	3960.0	1421639.8	0.139E+09
Totals	...	...	...	13802.9	2371534.2	0.314E+09

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 2371534/13802.907 \\
 &= 171.814 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 150.0/2 \\
 &= 105.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\
 &= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0)\cos(105.0)) \\
 &= 0.2594
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.259 \cdot 131191.984 \\
 &= 34027.5742 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 3.1278 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 720.0 - 350.0 \cdot \sin(150.0/2) / 1.309 \\
 &= 461.730 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 15717937.0000 \text{ N-mm}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 8.5893 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(131192 + 1360)180.0) / (2 \cdot 930.0 \cdot 173.33)} \\
 &= 14.901 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 150.0/2) (350.0 + 10.0 + 10.0) \\
 &= 714.785 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 714.7851 / (5 - 1) \\
 &= 178.696 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



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Saddle Calcs: Test Case:

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$$= 178.6963 \cdot 180.0/2$$

$$= 16082.664 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 132551.594 / (930.0 \cdot 180.0)$$

$$= 0.792 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 16082.7 \cdot 0.79$$

$$= 12734.664 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2025.0 + 1340.222$$

$$= 3365.222 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 12734.7 / 3365.2219$$

$$= 3.785 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.488E+07
Web	89.3	15.0	7.5	1340.2	10051.7	0.275E+07
Totals	...	...	...	3365.2	177114.1	0.764E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 15.0$$

$$= 135.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 177114.141 / 3365.222$$

$$= 52.631 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(52.631, 150.0 - 52.631)$$

$$= 97.369 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{ (7636983 / 3365.222)}$$

$$= 47.638 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 720.0 - \cos(150.0/2) (350.0 + 10.0 + 10.0) - 22.0$$

$$= 602.237 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199955 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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Saddle Calcs: Test Case:

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$$= 1 \cdot 602.237/47.638$$

$$= 12.642$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 422.3 / (2 \cdot 930.0) \cdot 178.696 \cdot 602.24/2$$

$$= 12221.638 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ (12.6419 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (12.64)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (12.64) / (8 \cdot 123.21) - (12.64^3) / (8 \cdot 123.21^3))$$

$$= 151.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.78/151.69 + (12221.64 \cdot 97.369/7636990) / 173.33$$

$$= 0.026$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.677E+07
Web	178.7	15.0	7.5	2680.4	20103.3	0.284E+07
Totals	...	...	...	4705.4	187165.8	0.961E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 187165.828/4705.444$$

$$= 39.776 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(39.776, \text{abs}(150.0 - 39.776))$$

$$= 110.224 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 720.0 - \cos(370.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 328.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (9614358/4705.444) }$$

$$= 45.202 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 328.0/45.202$$

$$= 7.256$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 422.293 / (2 \cdot 930.0)$$

$$= 0.227 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.227 \cdot 178.696 \cdot 328.0$$

$$= 13312.691 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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Saddle Calcs: Test Case:

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= 13312.691/87226.062
= 0.153 MPa

Compressive Allowable, $KL/r < C_c$ (7.2563 < 123.2107) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r ³)/(8• C_c ³))
= (1-(7.26)²/(2 • 123.21²))260/
(5/3+3•(7.26)/(8• 123.21)-(7.26³)/(8•123.21³))
= 153.7 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$
= 5.33/153.69 + (13312.69 • 110.224/9614358)/173.33
= 0.036

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	600.6192 mm ²
Saddle Load Q0 (Weight)	Q0	122715.6 N
Saddle Load QL (Wind/Seismic contribution)	QL	122.1 N
Maximum Transverse Force	Ft	3254.9 N
Maximum Longitudinal Force	Fl	422.3 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τb,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= 422.3/(600.619 • 2)
= 0.4 MPa. Must be less than 103.2 MPa.

Shear Stress in a Single Bolt, Transverse Direction [τb,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= 3254.9/(600.619 • 2)
= 2.7 MPa. Must be less than 103.2 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
= 422.29/(103.2 • 2.0)
= 2.0462 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$
= 720.0 • 3254.9 + 0.0
= 2344480.75 N-mm

Eccentricity (e):

= R_{mom} / Q_0
= 2344481/122715.62
= 19.10 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:



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Saddle Calcs: Test Case:

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= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 2.0462, 0.0]

= 2.0462 mm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$

and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$

= $\min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 490.0)$

= 242.9477 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	3626.20	mm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Inside or Mean Depth of Head	Hi or hm	177.55	mm
Shell Allowable Stress used in Calculation		153.00	MPa
Head Allowable Stress used in Calculation		153.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		70171.80	N
Pressure used in Analysis	P	1.304	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	16.52	153.00
Long. Stress at Bottom of Midspan	29.75	153.00
Long. Stress at Top of Saddles	31.96	153.00
Long. Stress at Bottom of Saddles	18.06	153.00
Tangential Shear in Shell	10.82	90.61
Circ. Stress at Horn of Saddle	40.56	229.50
Circ. Compressive Stress in Shell	1.94	153.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$

= $3.0 (6509.8/2 + 0) \cdot 720.0/714.7852$

= 9836.0 N

Longitudinal Saddle Reaction Force [Fwl]:



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Saddle Calcs: Test Case: Step: 21 9:50am Aug 12, 2026

$$\begin{aligned}
 &= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s \\
 &= \max(422.29, 0) \cdot 720.0/2490.0002 \\
 &= 122.1 \text{ N}
 \end{aligned}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st}) \\
 &= 60336 + \max(122, 9836, 0, 0) \\
 &= 70171.8 \text{ N}
 \end{aligned}$$

Longitudinal Wind Force [F_l]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.72 \cdot 1.2)^2) \\
 &= 1279.740 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	71531.40	N
Transverse Shear Load	3254.90	N
Longitudinal Shear Load	422.29	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2 (700.0/2 + 10.0 + 10.0) \sin(150.0/2), 2 \cdot 355.0) \\
 &= 714.785 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/3L)] \\
 &= -70172 \cdot 490.0 [1 - (1 - 490.0/3626.2 + (355.0^2 - 177.55^2) / (2 \cdot 490.0 \cdot 3626.2)) / (1 + (4 \cdot 177.55)/(3 \cdot 3626.2))] \\
 &= -5612643.0 \text{ N-mm}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L / 4 ((1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 70172 \cdot 3626 / 4 ((1 + 2(355^2 - 178^2)/(3626^2)) / (1 + (4 \cdot 178)/(3 \cdot 3626)) - 4 \cdot 490/3626) \\
 &= 26200516.0 \text{ N-mm}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 1.304 \cdot 355.0 / (2 \cdot 10.0) - 26200516 / (\pi \cdot 355.0^2 \cdot 10.0) \\
 &= 16.52 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 1.304 \cdot 355.0 / (2 \cdot 10.0) + 26200516 / (\pi \cdot 355.0^2 \cdot 10.0) \\
 &= 29.75 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 1.304 \cdot 355.0 / (2 \cdot 10.0) - 5612643.0 / (0.1607 \cdot \pi \cdot 355.0^2 \cdot 10.0) \\
 &= 31.96 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M1 / (K1* \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 1.304 \cdot 355.0 / (2 \cdot 10.0) + 5612643.0 / (0.2792 \cdot \pi \cdot 355.0^2 \cdot 10.0) \\
 &= 18.06 \text{ MPa}
 \end{aligned}$$



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Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 70172(3626.2 - 2 \cdot 490.0)/(3626.2 + (4 \cdot 177.55/3))$$

$$= 48069.3 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ₂]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.7988 \cdot 48069.34 / (355.0 \cdot 10.0)$$

$$= 10.82 \text{ MPa}$$

Decay Length (4.15.20) [x₁,x₂]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ₆]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.6733 \cdot 70172 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -1.94 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.22) [σ₇]:

$$= -Q/(4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -70172 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) - 3 \cdot 0.0317 \cdot 70172 / (2 \cdot 10.0^2)$$

$$= -40.56 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B₁]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(355.0 \cdot 10.0)}, 2 \cdot 490.0)$$

$$= 242.95 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0)} + 200.0)10.0^3)/12$$

$$= 24357.543 \text{ mm}^4$$

Resolved Inertia for the Shell [I_{shell}]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 24357.543 + 2922.908(171.814 - 5.0)^2$$

$$= 81359896.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	292.3	10.0	5.0	2922.9	14614.5	0.814E+08
Wearplate	200.0	10.0	15.0	2000.0	30000.0	0.492E+08
Web	15.0	328.0	184.0	4920.0	905279.8	0.448E+08
BasePlate	180.0	22.0	359.0	3960.0	1421639.8	0.139E+09
Totals	...	...	...	13802.9	2371534.2	0.314E+09



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Saddle Calcs: Test Case: Step: 21 9:50am Aug 12, 2026

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2371534 / 13802.907$$

$$= 171.814 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 70171.805$$

$$= 18200.6270 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.6730 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta / 2) / (\theta / 2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(150.0 / 2) / 1.309$$

$$= 461.730 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 8407190.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.5942 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3 \cdot (70172 + 1360) \cdot 180.0) / (2 \cdot 930.0 \cdot 173.33)}$$

$$= 10.946 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0 / 2) \cdot (350.0 + 10.0 + 10.0)$$

$$= 714.785 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 714.7851 / (5 - 1)$$

$$= 178.696 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 178.6963 \cdot 180.0 / 2$$

$$= 16082.664 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 71531.398 / (930.0 \cdot 180.0)$$

$$= 0.427 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 16082.7 \cdot 0.43$$

$$= 6872.255 \text{ N}$$



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Saddle Calcs: Test Case:

Step: 21 9:50am Aug 12,2026

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
= 2025.0 + 1340.222
= 3365.222 mm²

Compressive Stress [Sc]:

= P/Ar
= 6872.3/3365.2219
= 2.042 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.488E+07
Web	89.3	15.0	7.5	1340.2	10051.7	0.275E+07
Totals	...	...	...	3365.2	177114.1	0.764E+07

Rib dimension [D]:

= Saddle Width - Web Thickness
= 150.0 - 15.0
= 135.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 177114.141/3365.222
= 52.631 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
= max(52.631, 150.0 - 52.631)
= 97.369 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{7636983/3365.222}$
= 47.638 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 720.0 - cos(150.0/2)(350.0 + 10.0 + 10.0) - 22.0
= 602.237 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 199955/260.0)}$
= 123.211

Slenderness ratio [KL/r]:

= KL/r
= 1 • 602.237/47.638
= 12.642

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 422.3 / (2 • 930.0) • 178.696 • 602.24/2
= 12221.638 N-mm

Compressive Allowable, KL/r < Cc (12.6419 < 123.2107) per AISC E2-1 [Sca]:

= $(1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$
= $(1 - (12.64)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (12.64) / (8 \cdot 123.21) - (12.64^3) / (8 \cdot 123.21^3))$
= 151.7 MPa



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 FileName : 401-E-30001A-Bottom- RP Page 113 of 188
 Saddle Calcs: Test Case: Step: 21 9:50am Aug 12,2026

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.04/151.69 + (12221.64 \cdot 97.369/7636990) / 173.33$$

$$= 0.014$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	2025.0	167062.5	0.677E+07
Web	178.7	15.0	7.5	2680.4	20103.3	0.284E+07
Totals	...	...	...	4705.4	187165.8	0.961E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 187165.828/4705.444$$

$$= 39.776 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(39.776, \text{abs}(150.0 - 39.776))$$

$$= 110.224 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 720.0 - \cos(370.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 328.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (9614358/4705.444) }$$

$$= 45.202 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 328.0/45.202$$

$$= 7.256$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 422.293/(2 \cdot 930.0)$$

$$= 0.227 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.227 \cdot 178.696 \cdot 328.0$$

$$= 13312.691 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 13312.691/87226.062$$

$$= 0.153 \text{ MPa}$$

Compressive Allowable, $KL/r < Cc$ (7.2563 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (7.26)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (7.26) / (8 \cdot 123.21) - (7.26^3) / (8 \cdot 123.21^3))$$

$$= 153.7 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$

$$= 2.88/153.69 + (13312.69 \cdot 110.224/9614358) / 173.33$$

$$= 0.020$$



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Saddle Calcs: Test Case:

Step: 21 9:50am Aug 12,2026

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	600.6192 mm ²
Saddle Load QO (Weight)	QO	61695.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	122.1 N
Maximum Transverse Force	Ft	3254.9 N
Maximum Longitudinal Force	Fl	422.3 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 422.3 / (600.619 \cdot 2)$$

$$= 0.4 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3254.9 / (600.619 \cdot 2)$$

$$= 2.7 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 422.29 / (103.2 \cdot 2.0)$$

$$= 2.0462 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 720.0 \cdot 3254.9 + 0.0$$

$$= 2344480.75 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 2344481 / 61695.45$$

$$= 37.99 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 2.0462, 0.0]$$

$$= 2.0462 \text{ mm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:50am Aug 12,2026

Input, Nozzle Desc: N1- Channel inlet

From: 30

Pressure for Reinforcement Calculations	P	4.6003	MPa
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		735.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	44.0000	mm
Height of Beveled Transition	L'	14.3832	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



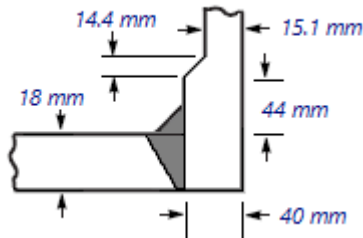
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Nozzle Calcs.: N1- Channel inl

Nozl: 11 9:50am Aug 12,2026



Reinforcement CALCULATION, Description: N1- Channel inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.6 \cdot 351.5 / (138 \cdot 1.0 - 0.6 \cdot 4.6)$
 $= 11.9567 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.6 \cdot 136.525 / (138 \cdot 1.0 + 0.4 \cdot 4.6)$
 $= 4.4913 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.912 / 0.5773$
 $= 43.1496 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	491.7496	mm
Parallel to Vessel Wall, opening length	d	245.8748	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	41.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



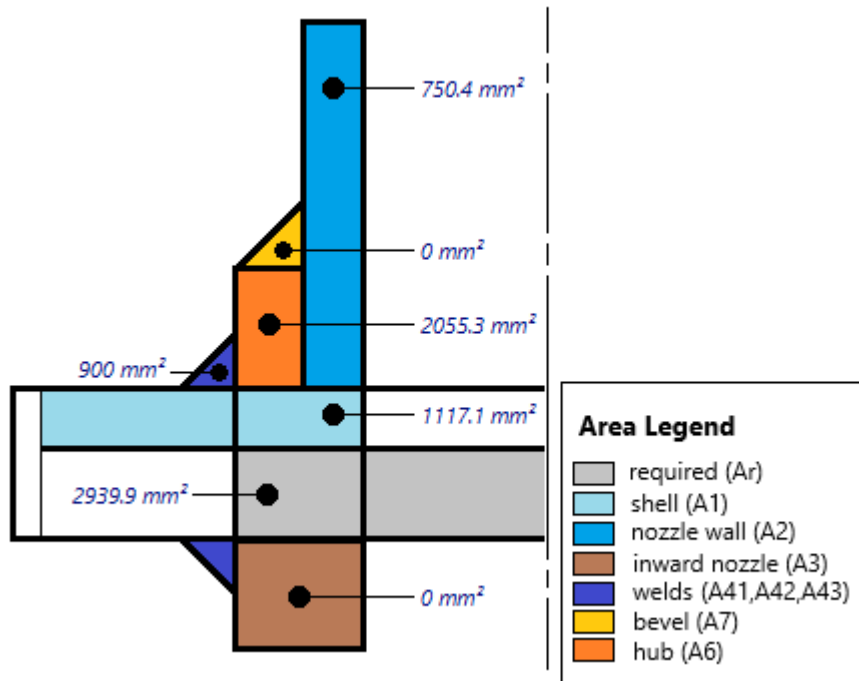
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Nozzle Calcs.: N1- Channel in1

Nozl: 11 9:50am Aug 12,2026



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2939.854	NA	NA
Area in Shell	A1	1117.081	NA	NA
Area in Nozzle Wall	A2	750.445	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	900.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	2055.273	NA	NA
TOTAL AREA AVAILABLE	Atot	4822.799	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (245.8748 \cdot 11.9567 \cdot 1.0 + 2 \cdot 38.5 \cdot 11.9567 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 2939.854 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 245.875(1.0 \cdot 16.5 - 1.0 \cdot 11.957) - 2 \cdot 38.5 \\
 &\quad (1.0 \cdot 16.5 - 1.0 \cdot 11.9567) \cdot (1 - 1.0) \\
 &= 1117.081 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 41.25)(13.59 - 4.49)1.0 \\
 &= 750.445 \text{ mm}^2
 \end{aligned}$$



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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:50am Aug 12,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 30.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 900.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(41.2, 305.0, 44.0)) \cdot (40.0 - 15.1) \cdot 1.0$$

$$= 2055.273 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.9913 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.4567 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.4567 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.6026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.603, \max(13.4567, 3.0)]$$

$$= 9.6026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.9913, 9.6026)$$

$$= 9.6026 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 43.5,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 301.5 MPa	Passed
Occasional	: 19.7,	Allowable	: 183.5 MPa	Passed
Shear	: 15.5,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.202, tr = 4.491, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.384$, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 11.957, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.725$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:50am Aug 12,2026

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.60/10.20 = 0.451$

Weld Size Calculations, Description: N1- Channel inl

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$21.2100 = 0.7 * W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t \cdot t_r)) \cdot S_v)$
 $= \max(0, (2939.8538 - 1117.0808 + 2 \cdot 38.5 \cdot 1.0 \cdot (1.0 \cdot 16.5 - 11.9567)) \cdot 138)$
 $= 299788.81 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (2805.7185 + 900.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 511336.44 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (2805.7185 + 0.0 + 900.0 + (1270.5)) \cdot 138$
 $= 686647.25 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (2805.7185 + 0.0 + 900.0 + 0.0 + (1270.5)) \cdot 138$
 $= 686647.25 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 273.05 \cdot 30.0 \cdot 0.49 \cdot 138$
 $= 869989. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 129.7312) \cdot (40.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 1515610. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 273.05 \cdot (15.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 656930. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (869989 + 1515610) = 2385599 \text{ N}$

Failure Path [Path 2-2]:



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$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (869989 + 0 + 656930 + 0) = 1526919 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (869989 + 656930 + 0) = 1526919 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2385598 N , must exceed W = 299788 N or W1 = 511336 N

Path 2-2 = 1526919 N , must exceed W = 299788 N or W2 = 686647 N

Path 3-3 = 1526919 N , must exceed W = 299788 N or W3 = 686647 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

The Drop for this Nozzle is : 27.7254 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 350.7254 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 98.16 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 245.87 mm

Input Echo, WRC297 Item 1, Description: N1- Channel in:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	322.875	mm
Nozzle Thickness	40.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 4	
Nozzle Cold S.I. Allowable	SNmc 138.00	MPa
Nozzle Hot S.I. Allowable	SNmh 138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	4.60	MPa
Radial Load	P	-22499.99	N
Circumferential Shear	Vc	16250.01	N
Longitudinal Shear	Vl	16250.01	N
Circumferential Moment	Mc	0.125E+08	N-mm
Longitudinal Moment	Ml	0.125E+08	N-mm
Torsional Moment	Mt	0.144E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:50am Aug 12,2026

*Warning - The ratio, Dn/Tn (8.072) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 322.875 + 2 \cdot 1.65 \cdot \sqrt{359.75(18.0 - 1.5)}$$

$$= 577.122 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	719.500 mm
Nozzle Outside Diameter	(d) =	322.875 mm
Vessel Thickness used	(T) =	16.500 mm
Nozzle Thickness used	(t) =	38.500 mm
T / t	=	0.429
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.963$$

Nr/P	=	0.060
Mr/P	=	0.105
M0/P	=	0.034
N0/P	=	0.080
MrD/Mc	=	0.263
NrDL/Mc	=	0.098
M0d/Mc	=	0.079
N0DL/Mc	=	0.086
MrD/M1	=	0.092
NrDL/M1	=	0.051
M0D/M1	=	0.025
N0DL/M1	=	0.089

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	95	100	95	100
Outplane Stress Summary	85	98	153	81

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-78	78	78	-78
Inplane Pressure Stress	47	47	47	47
Inplane Stress Summary	18	72	191	-70

Vessel Stresses (MPa):



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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:50am Aug 12,2026

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			5	5	5	5
Shear Stress Summary			7	7	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			86	100	191	152

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			6	6	6	6
Outplane Bending (P)			16	-16	16	-16
Outplane Membrane (Mc)			-12	-12	12	12
Outplane Bending (Mc)			-67	67	67	-67
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			47	47	47	47
Outplane Stress Summary			-8	92	150	-17

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			4	4	4	4
Inplane Bending (P)			52	-52	52	-52
Inplane Membrane (Mc)			-13	-13	13	13
Inplane Bending (Mc)			-224	224	224	-224
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			95	100	95	100
Inplane Stress Summary			-85	263	391	-157

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			5	5	5	5
Shear Stress Summary			3	3	7	7

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			85	263	391	157



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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:50am Aug 12,2026

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20
Hoop Stress Summary	9	14	34	40

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-10	10	10	-10
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-1	6	31	-3

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	11	15	35	43

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20
Hoop Stress Summary	9	15	34	39

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:50am Aug 12,2026

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-5	-5	5	5
Axial Bending (Mc)	-33	33	33	-33
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-24	29	54	-26
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	33	29	54	66

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	95	100	95	100	95	100	95	100
Circ. Pl (SUS)	-6	-6	18	18	-9	-9	17	17
Circ. Q (SUS)	-5	5	37	-37	-172	172	276	-276
Long. Pm (SUS)	47	47	47	47	47	47	47	47
Long. Pl (SUS)	-3	-3	11	11	-6	-6	18	18
Long. Q (SUS)	-26	26	130	-130	-51	51	83	-83
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	95.0	100.0	95.0	100.0	95.0	100.0	95.0	100.0
Pm+Pl (SUS)	89.0	94.0	113.0	118.0	86.0	91.0	112.0	117.0
Pm+Pl+Q (Total)	84.5	100.2	188.4	153.2	86.2	263.1	388.1	159.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	100.00	138.00	Passed
Pm+Pl (SUS)	118.02	207.00	Passed
Pm+Pl+Q (TOTAL)	388.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	20	15	20	15	20	15	20
Circ. Pl (SUS)	-6	-6	18	18	-6	-6	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-5	-5	5	5	-5	-5	5	5
Long. Q (SUS)	-4	4	16	-16	-27	27	39	-39
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	15.0	20.0	15.0	20.0	15.0	20.0	15.0	20.0
Pm+Pl (SUS)	9.0	14.0	33.0	38.0	9.0	14.0	33.0	38.0
Pm+Pl+Q (Total)	11.7	14.5	33.7	42.2	34.2	29.3	51.2	65.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	138.00	Passed
Pm+Pl (SUS)	38.00	207.00	Passed
Pm+Pl+Q (TOTAL)	65.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:50am Aug 12,2026

Input, Nozzle Desc: N2B- channel outlet

From: 30

Pressure for Reinforcement Calculations	P	4.6004	MPa
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		735.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	44.0000	mm
Height of Beveled Transition	L'	14.3832	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



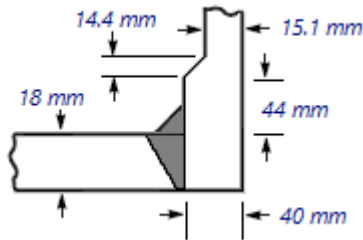
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Nozzle Calcs.: N2B- channel ou

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Reinforcement CALCULATION, Description: N2B- channel outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.6 \cdot 351.5 / (138 \cdot 1.0 - 0.6 \cdot 4.6)$
 $= 11.9570 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.6 \cdot 136.525 / (138 \cdot 1.0 + 0.4 \cdot 4.6)$
 $= 4.4914 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.912 / 0.5773$
 $= 43.1496 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	491.7496	mm
Parallel to Vessel Wall, opening length	d	245.8748	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	41.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



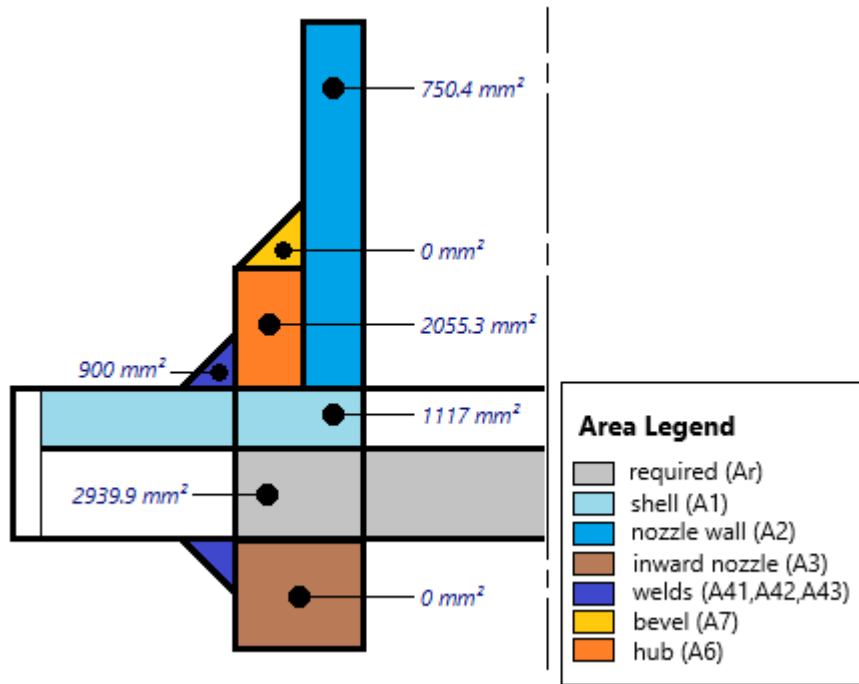
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Nozzle Calcs.: N2B- channel ou

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Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2939.921	NA	NA
Area in Shell	A1	1117.014	NA	NA
Area in Nozzle Wall	A2	750.437	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	900.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	2055.273	NA	NA
TOTAL AREA AVAILABLE	Atot	4822.724	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (245.8748 \cdot 11.957 \cdot 1.0 + 2 \cdot 38.5 \cdot 11.957 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 2939.921 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 245.875(1.0 \cdot 16.5 - 1.0 \cdot 11.957) - 2 \cdot 38.5 \\
 &\quad (1.0 \cdot 16.5 - 1.0 \cdot 11.957) \cdot (1 - 1.0) \\
 &= 1117.014 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 41.25)(13.59 - 4.49)1.0 \\
 &= 750.437 \text{ mm}^2
 \end{aligned}$$



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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:50am Aug 12,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 30.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 900.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(41.2, 305.0, 44.0)) \cdot (40.0 - 15.1) \cdot 1.0$$

$$= 2055.273 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.9914 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.4570 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.4570 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.6026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.603, \max(13.457, 3.0)]$$

$$= 9.6026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.9914, 9.6026)$$

$$= 9.6026 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 43.5,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 301.5 MPa	Passed
Occasional	: 19.7,	Allowable	: 183.5 MPa	Passed
Shear	: 15.5,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.202, tr = 4.491, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.384$, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 11.957, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.725$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
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Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4.60/10.20 = 0.451$

Weld Size Calculations, Description: N2B- channel ou

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$21.2100 = 0.7 * W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t \cdot t_r)) \cdot S_v)$
 $= \max(0, (2939.9209 - 1117.0138 + 2 \cdot 38.5 \cdot 1.0 \cdot (1.0 \cdot 16.5 - 11.957)) \cdot 138)$
 $= 299804.41 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (2805.7102 + 900.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 511335.34 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (2805.7102 + 0.0 + 900.0 + (1270.5)) \cdot 138$
 $= 686646.19 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (2805.7102 + 0.0 + 900.0 + 0.0 + (1270.5)) \cdot 138$
 $= 686646.19 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 273.05 \cdot 30.0 \cdot 0.49 \cdot 138$
 $= 869989. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 129.7312) \cdot (40.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 1515610. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 273.05 \cdot (15.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 656930. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (869989 + 1515610) = 2385599 \text{ N}$

Failure Path [Path 2-2]:



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$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (869989 + 0 + 656930 + 0) = 1526919 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (869989 + 656930 + 0) = 1526919 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2385598 N , must exceed W = 299804 N or W1 = 511335 N

Path 2-2 = 1526919 N , must exceed W = 299804 N or W2 = 686646 N

Path 3-3 = 1526919 N , must exceed W = 299804 N or W3 = 686646 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

The Drop for this Nozzle is : 27.7254 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 350.7254 mm

Input Echo, WRC297 Item 1, Description: N2B- channel o:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	322.875	mm
Nozzle Thickness	40.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 4	
Nozzle Cold S.I. Allowable	SNmc 138.00	MPa
Nozzle Hot S.I. Allowable	SNmh 138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	4.60	MPa
Radial Load	P	-22499.99	N
Circumferential Shear	Vc	16250.01	N
Longitudinal Shear	Vl	16250.01	N
Circumferential Moment	Mc	0.125E+08	N-mm
Longitudinal Moment	Ml	0.125E+08	N-mm
Torsional Moment	Mt	0.144E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (8.072) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:50am Aug 12,2026

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 322.875 + 2 \cdot 1.65 \cdot \sqrt{359.75(18.0 - 1.5)}$$

$$= 577.122 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 719.500 mm
 Nozzle Outside Diameter (d) = 322.875 mm
 Vessel Thickness used (T) = 16.500 mm
 Nozzle Thickness used (t) = 38.500 mm
 T / t = 0.429
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.963$$

Nr/P = 0.060
 Mr/P = 0.105
 M0/P = 0.034
 N0/P = 0.080
 MrD/Mc = 0.263
 NrDL/Mc = 0.098
 M0d/Mc = 0.079
 N0DL/Mc = 0.086
 MrD/Ml = 0.092
 NrDL/Ml = 0.051
 M0D/Ml = 0.025
 N0DL/Ml = 0.089

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	95	100	95	100
Outplane Stress Summary	85	98	153	81

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-78	78	78	-78
Inplane Pressure Stress	47	47	47	47
Inplane Stress Summary	18	72	191	-70

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:50am Aug 12,2026

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	7	7	3	3
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	86	100	191	152
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-67	67	67	-67
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	47	47	47	47

Outplane Stress Summary	-8	92	150	-17
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-224	224	224	-224
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	95	100	95	100

Inplane Stress Summary	-85	263	391	-157
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	7	7
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	85	263	391	157
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:50am Aug 12,2026

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20

Hoop Stress Summary	9	14	34	40
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-10	10	10	-10
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-1	6	31	-3
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	11	15	35	43
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20

Hoop Stress Summary	9	15	34	39
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-5	-5	5	5



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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:50am Aug 12,2026

Axial Bending (Mc)	-33	33	33	-33
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-24	29	54	-26
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	33	29	54	66

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	95	100	95	100	95	100	95	100
Circ. Pl (SUS)	-6	-6	18	18	-9	-9	17	17
Circ. Q (SUS)	-5	5	37	-37	-172	172	276	-276
Long. Pm (SUS)	47	47	47	47	47	47	47	47
Long. Pl (SUS)	-3	-3	11	11	-6	-6	18	18
Long. Q (SUS)	-26	26	130	-130	-51	51	83	-83
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	95.0	100.0	95.0	100.0	95.0	100.0	95.0	100.0
Pm+Pl (SUS)	89.0	94.0	113.0	118.0	86.0	91.0	112.0	117.0
Pm+Pl+Q (Total)	84.5	100.2	188.4	153.2	86.2	263.1	388.1	159.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	100.00	138.00	Passed
Pm+Pl (SUS)	118.02	207.00	Passed
Pm+Pl+Q (TOTAL)	388.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):



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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:50am Aug 12,2026

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	20	15	20	15	20	15	20
Circ. Pl (SUS)	-6	-6	18	18	-6	-6	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-5	-5	5	5	-5	-5	5	5
Long. Q (SUS)	-4	4	16	-16	-27	27	39	-39
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	15.0	20.0	15.0	20.0	15.0	20.0	15.0	20.0
Pm+Pl (SUS)	9.0	14.0	33.0	38.0	9.0	14.0	33.0	38.0
Pm+Pl+Q (Total)	11.7	14.5	33.7	42.2	34.2	29.3	51.2	65.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	138.00	Passed
Pm+Pl (SUS)	38.00	207.00	Passed
Pm+Pl+Q (TOTAL)	65.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4B- Shell outl

Nozl: 13 9:50am Aug 12,2026

Input, Nozzle Desc: N4B- Shell outlet

From: 50

Pressure for Reinforcement Calculations	P	1.0001	MPa
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-240 304L	
Parent Allowable Stress at Temperature	Sv	114.50	MPa
Parent Allowable Stress At Ambient	Sva	115.00	MPa
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1725.50	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304L	
Material UNS Number		S30403	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	87.41	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-182 F304L	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	368.2750	mm
Thickness of Pad	te	8.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



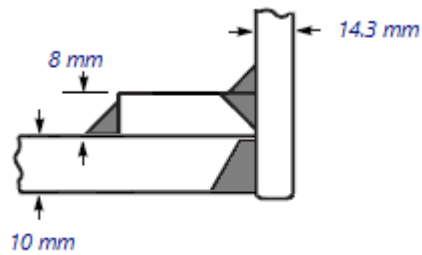
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Nozzle Calcs.: N4B- Shell outl

Nozl: 13 9:50am Aug 12,2026



Reinforcement CALCULATION, Description: N4B- Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.0 \cdot 350.0 / (114 \cdot 1.0 - 0.6 \cdot 1.0)$
 $= 3.0731 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.0 \cdot 84.1375 / (87 \cdot 1.0 + 0.4 \cdot 1.0)$
 $= 0.9582 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	285.4508	mm
Parallel to Vessel Wall, opening length	d	142.7254	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 87.4 / 114.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 87.4 / 114.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 114.5)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.763, 1.0)$
 $= 0.763$



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Nozzle Calcs.: N4B- Shell outl

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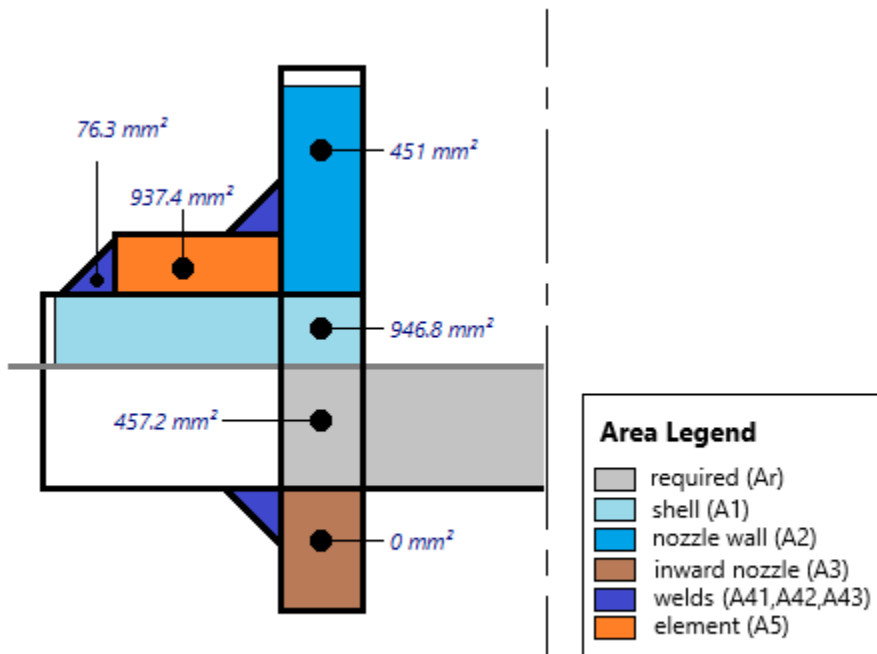


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	457.181	NA	NA
Area in Shell	A1	946.778	NA	NA
Area in Nozzle Wall	A2	451.046	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	76.341	NA	NA
Area in Element	A5	937.406	NA	NA
TOTAL AREA AVAILABLE	Atot	2411.572	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (142.7254 \cdot 3.0731 \cdot 1.0 + 2 \cdot 12.7748 \cdot 3.0731 \cdot 1.0 \cdot (1 - 0.76)) \\
 &= 457.181 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 142.725(1.0 \cdot 10.0 - 1.0 \cdot 3.073) - 2 \cdot 12.775 \\
 &\quad (1.0 \cdot 10.0 - 1.0 \cdot 3.0731) \cdot (1 - 0.763) \\
 &= 946.778 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 25.0) \cdot (12.77 - 0.96) \cdot 0.7634
 \end{aligned}$$



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:50am Aug 12,2026

= 451.046 mm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $10.0^2 \cdot 0.76 + (0.0)^2 \cdot 0.76 + 0.0^2 \cdot 1.0$

= 76.341 mm²

Area Available in Element [A5]:

= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$

= $(285.4508 - 168.275) \cdot 8.0 \cdot 1.0$

= 937.406 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.4582$ mm

Wall Thickness per UG16(b), $tr16b = 3.0000$ mm

Wall Thickness, shell/head, internal pressure $trb1 = 3.0731$ mm

Wall Thickness $tb1 = \max(trb1, tr16b) = 3.0731$ mm

Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000$ mm

Wall Thickness per table UG-45 $tb3 = 7.7200$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb3, \max(tb1, tb2)]$

= $\min[7.72, \max(3.0731, 3.0)]$

= 3.0731 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(ta, tb)$

= $\max(2.4582, 3.0731)$

= 3.0731 mm

Available Nozzle Neck Thickness = $0.875 \cdot 14.275 = 12.490$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	28.6,	Allowable	:	87.4 MPa	Passed
Expansion	:	0.0,	Allowable	:	224.4 MPa	Passed
Occasional	:	2.6,	Allowable	:	116.3 MPa	Passed
Shear	:	17.5,	Allowable	:	61.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govern. thk, $t_g = 8.0$, $c = 0.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govern. thk, $t_g = 8.0$, $c = 0.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C



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 Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:50am Aug 12,2026

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N4B- Shell outl

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	4.0000 = 0.5*TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (457.1807 - 946.7778 + 2 \cdot 12.7748 \cdot 0.7634 \cdot (1.0 \cdot 10.0 - 3.0731)) \cdot 114)$
 $= \max(0, -40585.18) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (451.0464 + 937.4063 + 76.3413 - 0.0 \cdot 0.76) \cdot 114$
 $= 167703.67 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (451.0464 + 0.0 + 76.3413 + (195.0491)) \cdot 114$
 $= 82711.49 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (451.0464 + 0.0 + 76.3413 + 937.4063 + (195.0491)) \cdot 114$
 $= 190034.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 87$
 $= 113204. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 368.275 \cdot 8.0 \cdot 0.49 \cdot 114$
 $= 259624. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 77.7501) \cdot (14.2748 - 1.5) \cdot 0.7 \cdot 87$
 $= 190910. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 168.275 \cdot 8.0 \cdot 0.74 \cdot 138$
 $= 215921. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 168.275 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 114$



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= 134366. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 = (SPEW + SNW) = (259624 + 190910) = 450534 N

Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (113204 + 215921 + 134366 + 0) = 463491 N

Failure Path [Path 3-3]:
 = (Spew + Tngw + Sinw)
 = (259624 + 134366 + 0) = 393989 N

Summary of Failure Path Calculations:

Path 1-1 = 450533 N , must exceed W = 0 N	or W1 = 167703 N
Path 2-2 = 463490 N , must exceed W = 0 N	or W2 = 82711 N
Path 3-3 = 393989 N , must exceed W = 0 N	or W3 = 190034 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

The Drop for this Nozzle is : 10.2635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.2635 mm

Input Echo, WRC297 Item 1, Description: N4B- Shell out:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304L	
Vessel Cold S.I. Allowable	Smc 115.00	MPa
Vessel Hot S.I. Allowable	Smh 114.50	MPa

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	168.275	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304L	
Nozzle Cold S.I. Allowable	SNmc 115.00	MPa
Nozzle Hot S.I. Allowable	SNmh 87.41	MPa

Diameter of Reinforcing Pad	368.275	mm
Thickness of Reinforcing Pad	8.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	1.00	MPa
Radial Load	P	-13499.99	N
Circumferential Shear	Vc	9750.02	N
Longitudinal Shear	Vl	9750.02	N
Circumferential Moment	Mc	0.450E+07	N-mm
Longitudinal Moment	Ml	0.450E+07	N-mm
Torsional Moment	Mt	0.518E+07	N-mm

Include Axial Pressure Thrust	No
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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:50am Aug 12,2026

Include Pressure Stress Indices per Div. 2 No
Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{359.0(10.0 - 0.0)}$$

$$= 366.000 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 718.000 mm
Nozzle Outside Diameter (d) = 168.275 mm
Vessel Thickness used (T) = 18.000 mm
Nozzle Thickness used (t) = 12.775 mm
T / t = 1.409
d / t = 13.172

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.480$$

Nr/P = 0.104
Mr/P = 0.133
M0/P = 0.055
N0/P = 0.130
MrD/Mc = 0.241
NrDL/Mc = 0.150
M0d/Mc = 0.118
N0DL/Mc = 0.069
MrD/Ml = 0.153
NrDL/Ml = 0.105
M0D/Ml = 0.077
N0DL/Ml = 0.166

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-37	37	37	-37
Normal Pressure Stress	18	19	18	19
Outplane Stress Summary	-13	35	89	-12

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-75	75	75	-75
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-37	47	131	-86



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:50am Aug 12,2026

Vessel Stresses (MPa):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			2	2	-2	-2
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			6	6	6	6
Shear Stress Summary			8	8	4	4

Vessel Stresses (MPa):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			40	52	131	86

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			13	-13	13	-13
Outplane Membrane (Mc)			-5	-5	5	5
Outplane Bending (Mc)			-58	58	58	-58
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			9	9	9	9
Outplane Stress Summary			-35	53	92	-51

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			4	4	4	4
Inplane Bending (P)			33	-33	33	-33
Inplane Membrane (Mc)			-12	-12	12	12
Inplane Bending (Mc)			-119	119	119	-119
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			18	19	18	19
Inplane Stress Summary			-75	98	188	-116

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-2	-2	2	2
Torsional Shear (Mt)			6	6	6	6
Shear Stress Summary			4	4	8	8

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			75	98	189	117



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:50am Aug 12,2026

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	-2	0	25	26

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	40	-40	40	-40
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-19	-19	19	19
Axial Bending (ML)	-99	99	99	-99
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-73	44	164	-114

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	11	11	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	75	51	164	141

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	C1	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	5	7	17	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu	C1	Du	D1
	Left	Left	Right	Right



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:50am Aug 12,2026

pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	40	-40	40	-40
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-163	163	163	-163
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-138	109	228	-178
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9

Shear Stress Summary	6	6	11	11
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	144	109	229	198
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{355.0(10.0 - 0.0)}$$

$$= 364.895 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	710.000 mm
Nozzle Outside Diameter	(d) =	368.275 mm
Vessel Thickness used	(T) =	10.000 mm
Nozzle Thickness used	(t) =	112.775 mm
T / t	=	0.089
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.371$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.077
M0/P	=	0.022
N0/P	=	0.031
MrD/Mc	=	0.248
NrDL/Mc	=	0.102
M0d/Mc	=	0.074
N0DL/Mc	=	0.056
MrD/Ml	=	0.056
NrDL/Ml	=	0.055
M0D/Ml	=	0.017
N0DL/Ml	=	0.033

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:50am Aug 12,2026

longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	34	35	34	35
Outplane Stress Summary	39	30	72	13

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	62	-62	62	-62
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-41	41	41	-41
Inplane Pressure Stress	17	17	17	17
Inplane Stress Summary	38	-4	134	-72

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	4	4	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	43	35	134	86

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-54	54	54	-54
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17
Outplane Stress Summary	-21	50	100	-43

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:50am Aug 12,2026

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	62	-62	62	-62
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-181	181	181	-181
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	35	34	35
Inplane Stress Summary	-90	148	297	-189

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	0	0	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	90	148	297	189

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-8	-8	18	18	-8	-8	16	16
Circ. Q (SUS)	-24	24	50	-50	-86	86	152	-152
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-4	-4	12	12	0	0	10	10
Long. Q (SUS)	-42	42	108	-108	-45	45	71	-71
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	10.7	11.6	36.3	37.2	11.6	12.2	34.3	35.2
Pm+Pl+Q (Total)	39.5	51.0	129.4	87.2	76.4	97.4	186.7	118.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	114.50	Passed
Pm+Pl (SUS)	37.25	171.75	Passed
Pm+Pl+Q (TOTAL)	186.66	343.50	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	34	35	34	35	34	35	34	35
Circ. Pl (SUS)	1	1	7	7	-6	-6	18	18
Circ. Q (SUS)	5	-5	29	-29	-119	119	243	-243
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	0	0	12	12	-2	-2	10	10
Long. Q (SUS)	21	-21	103	-103	-37	37	71	-71
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	34.0	35.0	34.0	35.0	34.0	35.0	34.0	35.0
Pm+Pl (SUS)	35.1	36.1	41.1	42.1	28.1	29.1	52.0	53.0
Pm+Pl+Q (Total)	42.2	35.5	132.0	87.0	91.0	148.0	295.0	190.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	35.00	114.50	Passed
Pm+Pl (SUS)	53.04	171.75	Passed
Pm+Pl+Q (TOTAL)	295.05	343.50	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	-8	-8	18	18	0	0	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-17	-17	21	21	-17	-17	21	21
Long. Q (SUS)	-59	59	139	-139	-123	123	203	-203
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	14.3	14.3	26.0	26.6	20.4	21.4	24.5	24.5
Pm+Pl+Q (Total)	74.7	51.0	163.4	140.7	143.7	109.5	227.6	197.2



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:50am Aug 12,2026

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	87.41	Passed
Pm+Pl (SUS)	26.56	131.12	Passed
Pm+Pl+Q (TOTAL)	227.57	262.23	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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 Nozzle Calcs.: K1- Pressure re Nozl: 14 9:50am Aug 12,2026

Input, Nozzle Desc: K1- Pressure relief

From: 50

Pressure for Reinforcement Calculations	P	1.0001	MPa
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-240 304L	
Parent Allowable Stress at Temperature	Sv	114.50	MPa
Parent Allowable Stress At Ambient	Sva	115.00	MPa
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4425.50	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304L	
Material UNS Number		S30403	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	87.41	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-182 F304L	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	368.2750	mm
Thickness of Pad	te	8.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



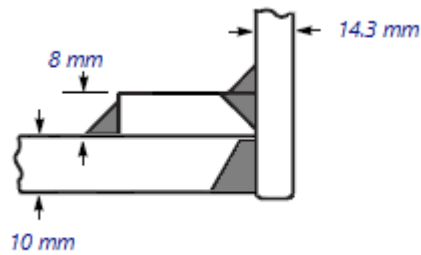
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Nozzle Calcs.: K1- Pressure re

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Reinforcement CALCULATION, Description: K1- Pressure relief

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.0 \cdot 350.0 / (114 \cdot 1.0 - 0.6 \cdot 1.0)$
 $= 3.0731 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.0 \cdot 84.1375 / (87 \cdot 1.0 + 0.4 \cdot 1.0)$
 $= 0.9582 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	285.4508 mm
Parallel to Vessel Wall, opening length	d	142.7254 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 87.4 / 114.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 87.4 / 114.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 114.5)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.763, 1.0)$
 $= 0.763$

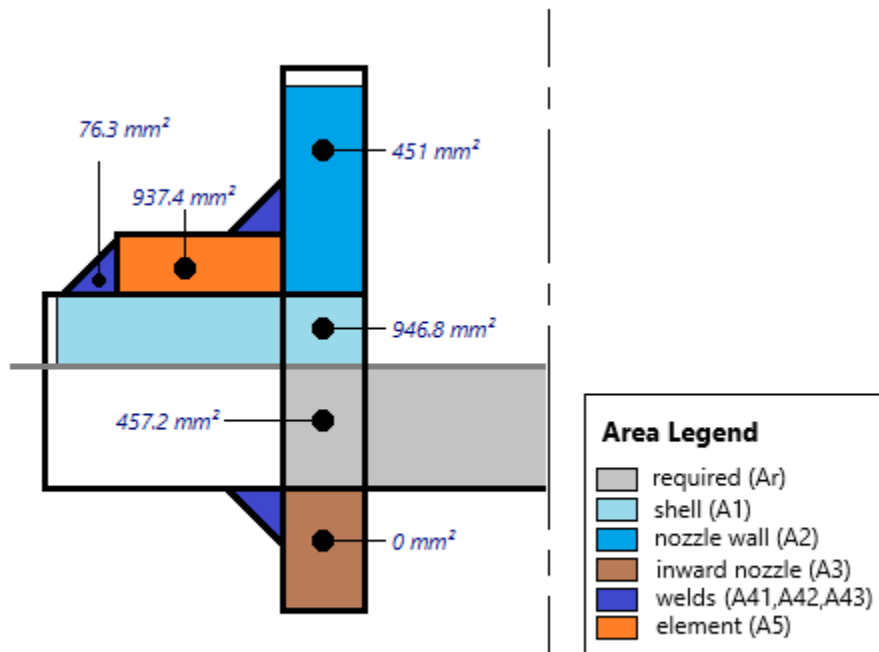


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	457.181	NA	NA
Area in Shell	A1	946.778	NA	NA
Area in Nozzle Wall	A2	451.046	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	76.341	NA	NA
Area in Element	A5	937.406	NA	NA
TOTAL AREA AVAILABLE	Atot	2411.572	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (142.7254 \cdot 3.0731 \cdot 1.0 + 2 \cdot 12.7748 \cdot 3.0731 \cdot 1.0 \cdot (1 - 0.76)) \\
 &= 457.181 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 142.725(1.0 \cdot 10.0 - 1.0 \cdot 3.073) - 2 \cdot 12.775 \\
 &\quad (1.0 \cdot 10.0 - 1.0 \cdot 3.073) \cdot (1 - 0.763) \\
 &= 946.778 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 25.0) \cdot (12.77 - 0.96) \cdot 0.7634
 \end{aligned}$$



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:50am Aug 12,2026

= 451.046 mm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $10.0^2 \cdot 0.76 + (0.0)^2 \cdot 0.76 + 0.0^2 \cdot 1.0$

= 76.341 mm²

Area Available in Element [A5]:

= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$

= $(285.4508 - 168.275) \cdot 8.0 \cdot 1.0$

= 937.406 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.4582$ mm

Wall Thickness per UG16(b), $tr_{16b} = 3.0000$ mm

Wall Thickness, shell/head, internal pressure $tr_{b1} = 3.0731$ mm

Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 3.0731$ mm

Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 3.0000$ mm

Wall Thickness per table UG-45 $tb_3 = 7.7200$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$

= $\min[7.72, \max(3.0731, 3.0)]$

= 3.0731 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$

= $\max(2.4582, 3.0731)$

= 3.0731 mm

Available Nozzle Neck Thickness = $0.875 \cdot 14.275 = 12.490$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	28.6,	Allowable	:	87.4 MPa	Passed
Expansion	:	0.0,	Allowable	:	224.4 MPa	Passed
Occasional	:	2.6,	Allowable	:	116.3 MPa	Passed
Shear	:	17.5,	Allowable	:	61.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govern. thk, $t_g = 8.0$, $c = 0.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govern. thk, $t_g = 8.0$, $c = 0.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C



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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K1- Pressure re

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	4.0000 = 0.5*TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr))Sv)$
 $= \max(0, (457.1807 - 946.7778 + 2 \cdot 12.7748 \cdot 0.7634 \cdot (1.0 \cdot 10.0 - 3.0731))114)$
 $= \max(0, -40585.18) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (451.0464 + 937.4063 + 76.3413 - 0.0 \cdot 0.76) \cdot 114$
 $= 167703.67 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (451.0464 + 0.0 + 76.3413 + (195.0491)) \cdot 114$
 $= 82711.49 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (451.0464 + 0.0 + 76.3413 + 937.4063 + (195.0491)) \cdot 114$
 $= 190034.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 87$
 $= 113204. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 368.275 \cdot 8.0 \cdot 0.49 \cdot 114$
 $= 259624. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 77.7501) \cdot (14.2748 - 1.5) \cdot 0.7 \cdot 87$
 $= 190910. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 168.275 \cdot 8.0 \cdot 0.74 \cdot 138$
 $= 215921. \text{ N}$

Tension, Shell Groove Weld [Tngw]:



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$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 168.275 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 114 \\
 &= 134366. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (259624 + 190910) = 450534 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (113204 + 215921 + 134366 + 0) = 463491 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (259624 + 134366 + 0) = 393989 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 450533 N	, must exceed W = 0 N	or W1 = 167703 N
Path 2-2 = 463490 N	, must exceed W = 0 N	or W2 = 82711 N
Path 3-3 = 393989 N	, must exceed W = 0 N	or W3 = 190034 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

The Drop for this Nozzle is : 10.2635 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.2635 mm

Input Echo, WRC297 Item 1, Description: K1- Pressure r:

Diameter Basis for Cylindrical Shell		ID	
Shell Internal Corrosion Allowance	0.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	700.000	mm	
Shell Thickness	10.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-240 304L		
Vessel Cold S.I. Allowable	Smc	115.00	MPa
Vessel Hot S.I. Allowable	Smh	114.50	MPa
Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance	1.5000	mm	
Nozzle Diameter	168.275	mm	
Nozzle Thickness	14.2748	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-312 TP304L		
Nozzle Cold S.I. Allowable	SNmc	115.00	MPa
Nozzle Hot S.I. Allowable	SNmh	87.41	MPa
Diameter of Reinforcing Pad		368.275	mm
Thickness of Reinforcing Pad	8.0000	mm	

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUS tained loads.

Design Internal Pressure	Dp	1.00	MPa
Radial Load	P	-13499.99	N
Circumferential Shear	Vc	9750.02	N
Longitudinal Shear	Vl	9750.02	N
Circumferential Moment	Mc	0.450E+07	N-mm
Longitudinal Moment	Ml	0.450E+07	N-mm
Torsional Moment	Mt	0.518E+07	N-mm



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Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{359.0(10.0 - 0.0)}$$

$$= 366.000 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 718.000 mm
 Nozzle Outside Diameter (d) = 168.275 mm
 Vessel Thickness used (T) = 18.000 mm
 Nozzle Thickness used (t) = 12.775 mm
 T / t = 1.409
 d / t = 13.172

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.480$$

Nr/P = 0.104
 Mr/P = 0.133
 M0/P = 0.055
 N0/P = 0.130
 MrD/Mc = 0.241
 NrDL/Mc = 0.150
 M0d/Mc = 0.118
 N0DL/Mc = 0.069
 MrD/Ml = 0.153
 NrDL/Ml = 0.105
 M0D/Ml = 0.077
 N0DL/Ml = 0.166

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-37	37	37	-37
Normal Pressure Stress	18	19	18	19
Outplane Stress Summary	-13	35	89	-12

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-75	75	75	-75
Inplane Pressure Stress	9	9	9	9



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:50am Aug 12,2026

Inplane Stress Summary	-37	47	131	-86
------------------------	-----	----	-----	-----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6

Shear Stress Summary	8	8	4	4
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	40	52	131	86
------------------------	----	----	-----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-58	58	58	-58
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9

Outplane Stress Summary	-35	53	92	-51
-------------------------	-----	----	----	-----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-119	119	119	-119
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	18	19	18	19

Inplane Stress Summary	-75	98	188	-116
------------------------	-----	----	-----	------

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	4	4	8	8
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:50am Aug 12,2026

Two * Max Shear Stress	75	98	189	117
------------------------	----	----	-----	-----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	-2	0	25	26

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	40	-40	40	-40
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-19	-19	19	19
Axial Bending (ML)	-99	99	99	-99
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-73	44	164	-114

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	11	11	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	75	51	164	141

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	5	7	17	18

Nozzle Stresses (MPa):



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:50am Aug 12,2026

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	40	-40	40	-40
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-163	163	163	-163
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-138	109	228	-178
----------------------	------	-----	-----	------

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9

Shear Stress Summary	6	6	11	11
----------------------	---	---	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	144	109	229	198

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{355.0(10.0 - 0.0)}$$

$$= 364.895 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	710.000 mm
Nozzle Outside Diameter	(d) =	368.275 mm
Vessel Thickness used	(T) =	10.000 mm
Nozzle Thickness used	(t) =	112.775 mm
T / t	=	0.089
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.371$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.077
M0/P	=	0.022
N0/P	=	0.031
MrD/Mc	=	0.248
NrDL/Mc	=	0.102
M0d/Mc	=	0.074
N0DL/Mc	=	0.056
MrD/Ml	=	0.056
NrDL/Ml	=	0.055
M0D/Ml	=	0.017
N0DL/Ml	=	0.033

Vessel Stresses (MPa):



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:50am Aug 12,2026

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	34	35	34	35
Outplane Stress Summary	39	30	72	13

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	62	-62	62	-62
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-41	41	41	-41
Inplane Pressure Stress	17	17	17	17
Inplane Stress Summary	38	-4	134	-72

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	4	4	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	43	35	134	86

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-54	54	54	-54
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17
Outplane Stress Summary	-21	50	100	-43

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to	Cu Left	Cl Left	Du Right	D1 Right
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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:50am Aug 12,2026

circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	62	-62	62	-62
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (MC)	-181	181	181	-181
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	35	34	35
Inplane Stress Summary	-90	148	297	-189

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		2	2	2	2
Shear Stress Summary		0	0	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		90	148	297	189

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-8	-8	18	18	-8	-8	16	16
Circ. Q (SUS)	-24	24	50	-50	-86	86	152	-152
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-4	-4	12	12	0	0	10	10
Long. Q (SUS)	-42	42	108	-108	-45	45	71	-71
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	10.7	11.6	36.3	37.2	11.6	12.2	34.3	35.2
Pm+Pl+Q (Total)	39.5	51.0	129.4	87.2	76.4	97.4	186.7	118.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	114.50	Passed
Pm+Pl (SUS)	37.25	171.75	Passed
Pm+Pl+Q (TOTAL)	186.66	343.50	Passed



Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 * S_mh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	34	35	34	35	34	35	34	35
Circ. Pl (SUS)	1	1	7	7	-6	-6	18	18
Circ. Q (SUS)	5	-5	29	-29	-119	119	243	-243
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	0	0	12	12	-2	-2	10	10
Long. Q (SUS)	21	-21	103	-103	-37	37	71	-71
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	34.0	35.0	34.0	35.0	34.0	35.0	34.0	35.0
Pm+Pl (SUS)	35.1	36.1	41.1	42.1	28.1	29.1	52.0	53.0
Pm+Pl+Q (Total)	42.2	35.5	132.0	87.0	91.0	148.0	295.0	190.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	35.00	114.50	Passed
Pm+Pl (SUS)	53.04	171.75	Passed
Pm+Pl+Q (TOTAL)	295.05	343.50	Passed

Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 * S_mh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	-8	-8	18	18	0	0	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-17	-17	21	21	-17	-17	21	21
Long. Q (SUS)	-59	59	139	-139	-123	123	203	-203
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	14.3	14.3	26.0	26.6	20.4	21.4	24.5	24.5
Pm+Pl+Q (Total)	74.7	51.0	163.4	140.7	143.7	109.5	227.6	197.2



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:50am Aug 12,2026

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	87.41	Passed
Pm+Pl (SUS)	26.56	131.12	Passed
Pm+Pl+Q (TOTAL)	227.57	262.23	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N3B- Shell inle

Nozl: 15 9:50am Aug 12,2026

Input, Nozzle Desc: N3B- Shell inlet

From: 50

Pressure for Reinforcement Calculations	P	1.0069	MPa
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-240 304L	
Parent Allowable Stress at Temperature	Sv	114.50	MPa
Parent Allowable Stress At Ambient	Sva	115.00	MPa
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1725.50	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304L	
Material UNS Number		S30403	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	87.41	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-182 F304L	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	368.2750	mm
Thickness of Pad	te	8.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



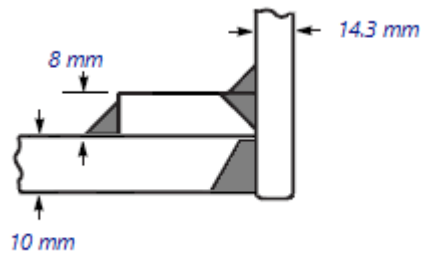
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Nozzle Calcs.: N3B- Shell inlet

Nozl: 15 9:50am Aug 12,2026



Reinforcement CALCULATION, Description: N3B- Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.01 \cdot 350.0 / (114 \cdot 1.0 - 0.6 \cdot 1.01)$
 $= 3.0943 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.01 \cdot 84.1375 / (87 \cdot 1.0 + 0.4 \cdot 1.01)$
 $= 0.9648 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	285.4508	mm
Parallel to Vessel Wall, opening length	d	142.7254	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 87.4 / 114.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 87.4 / 114.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 114.5)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.763, 1.0)$
 $= 0.763$



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Nozzle Calcs.: N3B- Shell inle

Nozl: 15 9:50am Aug 12,2026

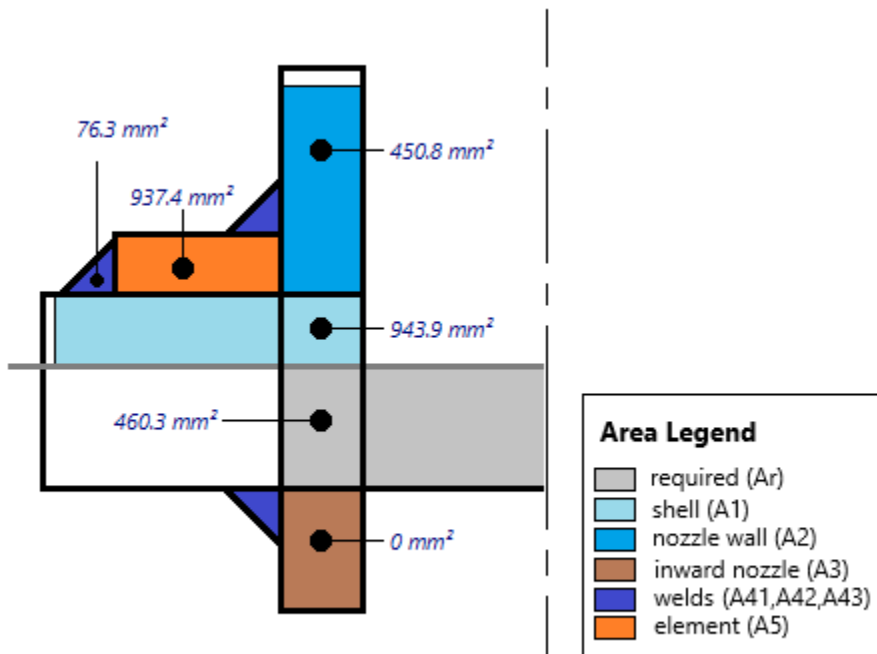


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	460.334	NA	NA
Area in Shell	A1	943.880	NA	NA
Area in Nozzle Wall	A2	450.797	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	76.341	NA	NA
Area in Element	A5	937.406	NA	NA
TOTAL AREA AVAILABLE	Atot	2408.425	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (142.7254 \cdot 3.0943 \cdot 1.0 + 2 \cdot 12.7748 \cdot 3.0943 \cdot 1.0 \cdot (1 - 0.76)) \\
 &= 460.334 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 142.725(1.0 \cdot 10.0 - 1.0 \cdot 3.094) - 2 \cdot 12.775 \\
 &\quad (1.0 \cdot 10.0 - 1.0 \cdot 3.0943) \cdot (1 - 0.763) \\
 &= 943.880 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 25.0) \cdot (12.77 - 0.96) \cdot 0.7634
 \end{aligned}$$



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Nozzle Calcs.: N3B- Shell inle Nozl: 15 9:50am Aug 12,2026

= 450.797 mm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $10.0^2 \cdot 0.76 + (0.0)^2 \cdot 0.76 + 0.0^2 \cdot 1.0$

= 76.341 mm²

Area Available in Element [A5]:

= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr_4$

= $(285.4508 - 168.275) \cdot 8.0 \cdot 1.0$

= 937.406 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.4648$ mm

Wall Thickness per UG16(b), $tr_{16b} = 3.0000$ mm

Wall Thickness, shell/head, internal pressure $tr_{b1} = 3.0943$ mm

Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 3.0943$ mm

Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 3.0000$ mm

Wall Thickness per table UG-45 $tb_3 = 7.7200$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$

= $\min[7.72, \max(3.0943, 3.0)]$

= 3.0943 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$

= $\max(2.4648, 3.0943)$

= 3.0943 mm

Available Nozzle Neck Thickness = $0.875 \cdot 14.275 = 12.490$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	28.6,	Allowable	:	87.4 MPa	Passed
Expansion	:	0.0,	Allowable	:	224.4 MPa	Passed
Occasional	:	2.6,	Allowable	:	116.3 MPa	Passed
Shear	:	17.5,	Allowable	:	61.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govern. thk, $t_g = 8.0$, $c = 0.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govern. thk, $t_g = 8.0$, $c = 0.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C



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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N3B- Shell inle

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm

Intermediate Calc. for pad/shell Welds TminPad 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	4.0000 = 0.5*TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$$

$$= \max(0, (460.3342 - 943.8804 + 2 \cdot 12.7748 \cdot 0.7634 \cdot (1.0 \cdot 10.0 - 3.0943)) \cdot 114)$$

$$= \max(0, -39939.75) \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (450.7966 + 937.4063 + 76.3413 - 0.0 \cdot 0.76) \cdot 114$$

$$= 167675.08 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (450.7966 + 0.0 + 76.3413 + (195.0491)) \cdot 114$$

$$= 82682.89 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (450.7966 + 0.0 + 76.3413 + 937.4063 + (195.0491)) \cdot 114$$

$$= 190006.16 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 87$$

$$= 113204. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 368.275 \cdot 8.0 \cdot 0.49 \cdot 114$$

$$= 259624. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 77.7501) \cdot (14.2748 - 1.5) \cdot 0.7 \cdot 87$$

$$= 190910. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 168.275 \cdot 8.0 \cdot 0.74 \cdot 138$$

$$= 215921. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:



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$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 168.275 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 114 \\
 &= 134366. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (259624 + 190910) = 450534 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (113204 + 215921 + 134366 + 0) = 463491 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\
 &= (259624 + 134366 + 0) = 393989 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 450533 N	, must exceed W = 0 N	or W1 = 167675 N
Path 2-2 = 463490 N	, must exceed W = 0 N	or W2 = 82682 N
Path 3-3 = 393989 N	, must exceed W = 0 N	or W3 = 190006 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

The Drop for this Nozzle is : 10.2635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.2635 mm

Input Echo, WRC297 Item 1, Description: N3B- Shell inl:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304L	
Vessel Cold S.I. Allowable	Smc 115.00	MPa
Vessel Hot S.I. Allowable	Smh 114.50	MPa

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	168.275	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304L	
Nozzle Cold S.I. Allowable	SNmc 115.00	MPa
Nozzle Hot S.I. Allowable	SNmh 87.41	MPa

Diameter of Reinforcing Pad	368.275	mm
Thickness of Reinforcing Pad	8.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp 1.01	MPa
Radial Load	P -13499.99	N
Circumferential Shear	Vc 9750.02	N
Longitudinal Shear	Vl 9750.02	N
Circumferential Moment	Mc 0.450E+07	N-mm
Longitudinal Moment	Ml 0.450E+07	N-mm
Torsional Moment	Mt 0.518E+07	N-mm



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Nozzle Calcs.: N3B- Shell inle Nozl: 15 9:50am Aug 12,2026

Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{359.0(10.0 - 0.0)}$$

$$= 366.000 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 718.000 mm
 Nozzle Outside Diameter (d) = 168.275 mm
 Vessel Thickness used (T) = 18.000 mm
 Nozzle Thickness used (t) = 12.775 mm
 T / t = 1.409
 d / t = 13.172

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.480$$

Nr/P = 0.104
 Mr/P = 0.133
 M0/P = 0.055
 N0/P = 0.130
 MrD/Mc = 0.241
 NrDL/Mc = 0.150
 M0d/Mc = 0.118
 N0DL/Mc = 0.069
 MrD/Ml = 0.153
 NrDL/Ml = 0.105
 M0D/Ml = 0.077
 N0DL/Ml = 0.166

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-37	37	37	-37
Normal Pressure Stress	19	20	19	20
Outplane Stress Summary	-13	36	89	-12

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-75	75	75	-75
Inplane Pressure Stress	9	9	9	9



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Nozzle Calcs.: N3B- Shell inle Nozl: 15 9:50am Aug 12,2026

Inplane Stress Summary	-37	47	131	-86
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6

Shear Stress Summary	8	8	4	4
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	40	52	132	86
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-58	58	58	-58
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9

Outplane Stress Summary	-35	53	92	-51
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-119	119	119	-119
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	19	20	19	20

Inplane Stress Summary	-75	98	188	-115
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	4	4	8	8
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



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Nozzle Calcs.: N3B- Shell inle Nozl: 15 9:50am Aug 12,2026

Two * Max Shear Stress	75	98	189	117
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	-2	0	25	26

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	40	-40	40	-40
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-19	-19	19	19
Axial Bending (ML)	-99	99	99	-99
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-73	44	164	-114

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	11	11	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	75	51	164	141

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	5	7	17	18

Nozzle Stresses (MPa):



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Nozzle Calcs.: N3B- Shell inle Nozl: 15 9:50am Aug 12,2026

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	40	-40	40	-40
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-163	163	163	-163
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-138	109	228	-178

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9
Shear Stress Summary	6	6	11	11

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	144	109	229	198

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{355.0(10.0 - 0.0)}$$

$$= 364.895 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	710.000 mm
Nozzle Outside Diameter	(d) =	368.275 mm
Vessel Thickness used	(T) =	10.000 mm
Nozzle Thickness used	(t) =	112.775 mm
T / t	=	0.089
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.371$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.077
M0/P	=	0.022
N0/P	=	0.031
MrD/Mc	=	0.248
NrDL/Mc	=	0.102
M0d/Mc	=	0.074
N0DL/Mc	=	0.056
MrD/Ml	=	0.056
NrDL/Ml	=	0.055
M0D/Ml	=	0.017
N0DL/Ml	=	0.033

Vessel Stresses (MPa):



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Nozzle Calcs.: N3B- Shell inle Nozl: 15 9:50am Aug 12,2026

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	34	35	34	35
Outplane Stress Summary	40	30	73	13

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	62	-62	62	-62
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-41	41	41	-41
Inplane Pressure Stress	17	17	17	17
Inplane Stress Summary	38	-3	134	-72

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	4	4	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	43	35	134	86

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-54	54	54	-54
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17
Outplane Stress Summary	-21	51	100	-43

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to	Cu Left	Cl Left	Du Right	D1 Right
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circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	62	-62	62	-62
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (MC)	-181	181	181	-181
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	35	34	35
Inplane Stress Summary	-90	148	297	-189

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		2	2	2	2
Shear Stress Summary		0	0	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		90	148	297	189

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-8	-8	18	18	-8	-8	16	16
Circ. Q (SUS)	-24	24	50	-50	-86	86	152	-152
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-4	-4	12	12	0	0	10	10
Long. Q (SUS)	-42	42	108	-108	-45	45	71	-71
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	11.6	12.5	37.2	38.2	12.2	13.0	35.2	36.2
Pm+Pl+Q (Total)	39.4	51.2	129.4	87.2	75.4	98.4	187.7	117.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	114.50	Passed
Pm+Pl (SUS)	38.23	171.75	Passed
Pm+Pl+Q (TOTAL)	187.66	343.50	Passed



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Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	34	35	34	35	34	35	34	35
Circ. Pl (SUS)	1	1	7	7	-6	-6	18	18
Circ. Q (SUS)	5	-5	29	-29	-119	119	243	-243
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	0	0	12	12	-2	-2	10	10
Long. Q (SUS)	21	-21	103	-103	-37	37	71	-71
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	34.0	35.0	34.0	35.0	34.0	35.0	34.0	35.0
Pm+Pl (SUS)	35.1	36.1	41.1	42.1	28.1	29.1	52.0	53.0
Pm+Pl+Q (Total)	42.2	35.5	132.0	87.0	91.0	148.0	295.0	190.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	35.00	114.50	Passed
Pm+Pl (SUS)	53.04	171.75	Passed
Pm+Pl+Q (TOTAL)	295.05	343.50	Passed

Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	-8	-8	18	18	0	0	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-17	-17	21	21	-17	-17	21	21
Long. Q (SUS)	-59	59	139	-139	-123	123	203	-203
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	14.3	14.3	26.0	26.6	20.4	21.4	24.5	24.5
Pm+Pl+Q (Total)	74.7	51.0	163.4	140.7	143.7	109.5	227.6	197.2



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Nozzle Calcs.: N3B- Shell inle

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Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	87.41	Passed
Pm+Pl (SUS)	26.56	131.12	Passed
Pm+Pl+Q (TOTAL)	227.57	262.23	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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ASME TS Calc: DESIGN CASE

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	1.00 MPa
Shell Maximum Operating Pressure	Psox,max	1.00 MPa
Shell Minimum Operating Pressure	Psox,min	0.00 MPa
Shell Thickness	ts	10.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	700.000 mm
Shell Temperature for Internal Pressure	Ts	155.00 °C
Shell Material		SA-240 304L

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30403
Shell Allowable Stress at Temperature	Ss	114.50 MPa
Shell Allowable Stress at Ambient		115.00 MPa

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4.60 MPa
Channel Maximum Operating Pressure	Ptox,max	4.60 MPa
Channel Minimum Operating Pressure	Ptox,min	0.00 MPa
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	700.000 mm
Channel Design Temperature	TEMPC	155.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 MPa
Channel Allowable Stress at Ambient		138.00 MPa

Tube Data:

Number of Tube Holes	Nt	545
Tube Wall Thickness	et	2.0000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	3300.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	3195.00 mm
Design Temperature of the Tubes		155.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		87.41 MPa
Tube Allowable Stress At Ambient		115.00 MPa
Tube Yield Stress At design Temperature	Syt	130.80 MPa
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	330.948 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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ASME TS Calc: DESIGN CASE

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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	312161.0 mm ²
Perimeter of Tube Layout	Cp	2058.4 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	155.00 °C
Tubesheet Material	SA-965 F304L	
Tubesheet Material UNS Number	S30403	
Tubesheet Allowable Stress at Temperature	S	87.41 MPa
Tubesheet Allowable Stress at Ambient	Tt	115.00 MPa
Thickness of Tubesheet	h	105.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	860.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?	Yes
Is the Tubesheet Flat on Tubeside?	Yes
Is the Tubesheet Flat on Shellside?	No
Step 1 Diameter on the Shellside	ds1 763.00 mm
Step 1 Depth on the Shellside	hs1 8.00 mm
Step 2 Diameter on the Shellside	ds2 697.00 mm
Step 2 Depth on the Shellside	hs2 8.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4	860.00 mm

Note: Tubesheet diameter is now: 860.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 mm ²
Junction Stress Reduction option	Perform Plastic Calculation	
Perform Differential Pressure Design	No	
Run Multiple Load Cases	Yes	

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side	Shell
Flange Outside Diameter	A 860.000 mm
Flange Inside Diameter	B 700.000 mm
Flange Face Outside Diameter	Fod 766.000 mm
Flange Face Inside Diameter	Fid 700.000 mm
Gasket Outside Diameter	Go 755.000 mm
Gasket Inside Diameter	Gi 729.000 mm
Small end Hub thk.	g0 6.0000 mm
Large end Hub thk.	g1 8.0000 mm
Gasket Factor,	m 3.00
Gasket Design Seating Stress	y 68.95 MPa
Flange Facing Sketch	Code Sketch 1a
Gasket Thickness	tg 4.5000 mm
Full face Gasket Flange Option	Program Selects

Bolting Information:

Diameter of Bolt Circle	C	800.000 mm
Nominal Bolt Diameter	dB	25.4000 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	20
Bolt Material	SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00 MPa
Bolt Allowable Stress At Ambient	Sa	172.00 MPa
Weld between Flange and Shell/Channel		0.0000 mm

Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes



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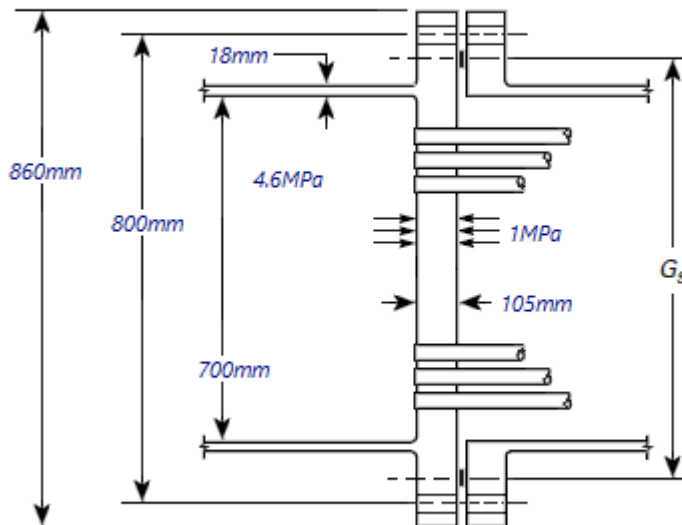
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ASME TS Calc: DESIGN CASE

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Thickness of Extended Portion of Tubesheet Tf 89.0000 mm
Is Bolt Load Transferred to the Tubesheet No

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Bolt load is not transferred to tubesheet which is extended for bolting. Req. Tubesheet extension thk. can't be computed. Please check input.

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group G

Elastic Mod. at Design Temperature	155.0 °C	0.18571E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 °C	0.19513E+06 MPa

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	155.0 °C	0.19483E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 MPa

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesheet Design Temp.	155.0 °C	0.18571E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 °C	0.19513E+06 MPa

TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature	155.0 °C	0.18571E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 °C	0.19513E+06 MPa

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:



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ASME TS Calc: DESIGN CASE

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$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (4.6 \cdot (19.0/2 - 0)) / (87.41 \cdot 1.0 + 0.4 \cdot 4.6)$$

$$= 0.4897 + 0.0000 = 0.4897 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 155.00 °C
Elastic Modulus for Material 182877.28 MPa

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.0000	19.00	3525.95	9.50	50.0000	0.0121884	68.63

EMAWP = (2.167 / (D/T) - 0.0833) * B = 9.9377 MPa

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3812	19.00	3525.95	49.84	50.0000	0.0004428	37.39

EMAWP = (4 * B) / (3 * (D/T)) = (4 * 37.3887) / (3 * 49.8432) = 1.0002 MPa

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4897 mm
Allowable Internal Pressure at Corroded thickness	20.09 MPa
Required Internal Design Pressure	4.60 MPa
Allowable External Pressure at Corroded thickness	9.94 MPa
Required External Design Pressure	1.00 MPa
Required Thickness due to Shell Side pressure	0.3812 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = (Goc - Gic) / 2	13.000 mm
Basic Gasket Width,	b0 = N / 2	6.500 mm
Effective Gasket Width,	b = sqrt(b0) * 2.5	6.422 mm
Gasket Reaction Diameter,	G = Go - 2.0 * b	742.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 330.948 + 19.0$$

$$= 680.896 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (25.0 - 19.0) / 25.0$$

$$= 0.240$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= Gs / Do \text{ (Configurations d, e, f)}$$

$$= 742.1556 / 680.896$$

$$= 1.0900$$

Ratio [pc]:



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ASME TS Calc: DESIGN CASE

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= Dc / Do (Configurations a, e, f)
= 700.0/680.896
= 1.0281

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

= ltx / h
= 0/105.0
= 0.000 (must be $0 \leq \rho \leq 1$)

Effective tube hole Diameter [d*]:

= max(dt - 2tt(EtT/E) • (StT/S) • (ρ); dt - 2tt)
= max(19.0 - 2 • 2.0 (185710/185710) •
(87/87) • (0); 19.0 - 2 • 2.0)
= 19.0000 mm

Effective tube pitch [p*]:

= $p / \sqrt{(1 - 4 \cdot \min(AI; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$
= $25.0 / \sqrt{(1 - 4 \cdot \min(0; 4 \cdot 680.896 \cdot 25.0) / (\pi \cdot 680.896^2))}$
= 25.0000 mm

Effective ligament efficiency for bending [μ*]:

= (p* - d*) / p*
= (25.0 - 19.0) / 25.0
= 0.2400

h/p = 4.200000 ; μ* = 0.240000 ; E* = 37890 MPa

E*/E = 0.204029 ; ν* = 0.407183

Note: As h/p (4.200) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

βs, ks, λs, δs and ws are all = 0

Channel coefficient [βc]:

= $((12(1 - \nu_c^2))^{1/4}) / \sqrt{(Dc + tc)tc}$
= $((12(1 - 0.3^2))^{1/4}) / \sqrt{(700.0 + 18.0)18.0}$
= 0.0160 1/mm

Channel coefficient [kc]:

= $\beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_s^2))$
= $0.016 \cdot 194825.812 \cdot 18.0^3 / (6(1 - 0.3^2))$
= 3327579.7500 MPa•mm²

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

CqDelta = CqTheta = CmTheta = CmDelta = 1

Channel coefficient [λc]:

= $6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (CmTheta +$
 $h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$
= $6(700.0 + 18.0/\cos(0)) \cdot 3327579/105.0^3 \cdot (1.0 +$
 $105.0 \cdot 0.016 \cdot (1.0 + 1.0)/2 + (105.0 \cdot 0.016)^2/2 \cdot 1.0)$
= 50628.8 MPa

Channel coefficient [δc]:

= $Dc^2 / (4 \cdot E_c \cdot tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc)))$
= $700.0^2 / (4 \cdot 194825 \cdot 18.0) \cdot (1 - 700.0 \cdot 0.3 / (2(700.0 + 18.0)))$
= 0.29823E-01 mm/MPa

Channel coefficient [zetac]:

zetac = 0



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ASME TS Calc: DESIGN CASE

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Parameter [wc]:

$$\begin{aligned}
 &= p_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\
 &\quad \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)) \\
 &= 1.028 \cdot 3327579 [0.029823 (1.0 + 0.016 \cdot 1.0 \cdot 105.0) - \\
 &\quad 0 (1.0/105.0 + 1.0 \cdot 0.016/2)) \\
 &= 4370.4482 \text{ mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / D_o \\
 &= 860.0/680.896 \\
 &= 1.2630
 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned}
 &= (1 - v^*) / E^* \cdot (\lambda c + E \cdot \ln(K)) \\
 &= (1 - 0.41) / 37890 \cdot (50628.82 + 185710 \cdot \ln(1.26)) \\
 &= 1.4706
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t] \\
 &= 680.896^2 / 16 \cdot [(1.09 - 1) \cdot (1.09^2 + 1) \cdot 1.0 - \\
 &\quad (1.028 - 1) \cdot (1.028^2 + 1) \cdot 4.6] \\
 &= -1988.2886 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= M_{ts} + (w_c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o) \\
 &= -1988 + (20105) + 522421 \cdot (800.0 - 742.156) / (2 \cdot \pi \cdot 680.896) \\
 &= 25181.0723 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= ((M^*) - D_o^2 / 32 \cdot F (P_s - P_t)) / (1 + F) \\
 &= ((25181.07) - 680.896^2 / 32 \cdot 1.471 (1.0 - 4.6)) / (1 + 1.47) \\
 &= 41240.2617 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t) \\
 &= 41240.26 + 680.896^2 / 64 \cdot (3 + 0.407) \cdot (1.0 - 4.6) \\
 &= -47619.4258 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(| M_p | ; | M_o |) \\
 &= \max(| 41240.262 | ; | -47619.426 |) \\
 &= 47619.4258 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - h_g')^2) \\
 &= 6 \cdot 47619.426 / (0.24 (105.0 - 0)^2) \\
 &= 107.9806 \text{ MPa}
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 87.41 \\
 &= 174.82 \text{ MPa}
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - h_g')^2) \\
 &= 6 \cdot 46702.41 / (0.24 (81.7674 - 0)^2) \\
 &= 174.6299 \text{ MPa}
 \end{aligned}$$



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ASME TS Calc: DESIGN CASE

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Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + \text{Cats} + \text{Catc} \\
 &= 81.7674 + 0 + 0 \\
 &= 81.7674 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(\text{HreqB}; \text{HreqS}) \\
 &= \max(81.7674; 32.6316) \\
 &= 81.7674 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= | \text{Ps} - \text{Pt} | \\
 &= | 1.0 - 4.6 | \\
 &= 3.600 \text{ MPa}
 \end{aligned}$$

Shear Stress check [τlimit]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / \text{Do} \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\
 &= 4 \cdot 0.24 \cdot 105.0 / 680.9 \cdot \min(0.8 \cdot 87.41; 0.533 \cdot 130.801) \\
 &= 10.321 \text{ MPa}
 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot \text{Ap} / \text{Cp}) \cdot | \text{Ps} - \text{Pt} | + \text{Cats} + \text{Catc} \\
 &= 1 / (4 \cdot 0.24 \cdot 69.717) \cdot (4 \cdot 312161.0 / 2058.35) \cdot | 1.0 - 4.6 | + 0 \\
 &= 32.6316 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 18.000 mm

Min. Channel len. adjacent to the tubesheet is: 202.049mm

Axial Channel Membrane Stress [σc,m]:

$$\begin{aligned}
 &= \text{Dc}^2 / (4 \cdot \text{tc} \cdot (\text{Dc} + \text{tc}) \cdot \cos(\theta)) \cdot \text{Pt} \\
 &= 700.0^2 / (4 \cdot 18.0 \cdot (700.0 + 18.0) \cdot \cos(0)) \cdot 4.6 \\
 &= 43.604 \text{ MPa}
 \end{aligned}$$

Axial Channel Bending Stress [σc,b]:

$$\begin{aligned}
 &= 6 \cdot \text{kc} / \text{tc}^2 \cdot [(\delta_c \cdot \text{Cm} \Delta \cdot \beta_c - \zeta_c \cdot \text{Cm} \Theta) \cdot \text{Pt} - \\
 &\quad 6((1 - \nu^*) / E^*) \cdot \text{Do} / h^3 \cdot (\text{Cm} \Theta + \\
 &\quad h \cdot \beta_c / 2 \cdot \text{Cm} \Delta) \cdot (\text{Mp} + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt}))] \\
 &= 6 \cdot 748043 / 18.0^2 \cdot [(0.029823 \cdot 1.0 \cdot 0.016 - 0 \cdot 1.0) \cdot 4.6 - \\
 &\quad 6((1 - 0.407) / 37890) \cdot 680.896 / 105.0^3 \cdot (1.0 + \\
 &\quad 0.016 \cdot 105.0 / 2 \cdot 1.0) \cdot (41240.262 + 680.896^2 / 32 \cdot (1.0 - 4.6))] \\
 &= 203.530 \text{ MPa}
 \end{aligned}$$

Channel Membrane + Bending Stress [σc]:

$$\begin{aligned}
 &= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c) \\
 &= 247.134 \text{ MPa (Should be } \leq 207.00)
 \end{aligned}$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Ec]:

$$\begin{aligned}
 &= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)} \\
 &= 194825 \cdot \sqrt{(1.5 \cdot 138.0 / 247.13)} \\
 &= 178304.3 \text{ MPa}
 \end{aligned}$$



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Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 47924.629 / (0.24 \cdot (105.0 - 0)^2)$$

$$= 108.7 \text{ MPa (Should be } \leq 174.8 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + Cats + Catc = 82.1151 + 0 + 0 = 82.1151 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(HreqB; HreqS) = \max(82.1151; 32.6316) = 82.1151 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (19.0 - 2.0) 2.0$$

$$= 106.814 \text{ mm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.0 (19.0 - 2.0) 87.41$$

$$= 9335.865 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2.0 (19.0 + 0.67 \cdot 2.0) 87.41$$

$$= 9335.8652 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 87.41 / 87.41$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.0 (19.0 + 0.67 \cdot 2.0) 87.41$$

$$= 6143.5483 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19.0)^2 + 1.07 \cdot 2.0 \cdot (19.0 - 2.0) 1.0 \cdot 1.0} - 0.75 \cdot 19.0 \right)$$

$$= 2.448 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2239 mm

Minimum Required Groove Weld Leg [agr]: 1.2239 mm

Tube-Tubesheet Jt allowable, 9335.87 is $\geq$ tube strength 9335.87 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
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FileName : 401-E-30001A-Bottom- RP

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ASME TS Calc: DESIGN CASE

Case: 1 9:50a Aug 12,2026

Tubesheet bend. stress	108.7	<=	174.8 MPa	Ok
Tubesheet shear stress	21.7	<=	69.7 MPa	Ok
Stress in Channel at Tubesheet	247.1	<=	462.4 MPa	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	82.115	105.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.224	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.224	2.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	100.249	...	159	175	28	70 Ps+Pt D1 Ok
D2uc	45.457	...	48	175	6	70 Ps+Pt D2 Ok
D3uc	82.115	...	109	175	22	70 Ps+Pt D3 Ok
D1c	103.348	...	169	175	28	70 Ps+Pt-c D1 Ok
D2c	48.402	...	51	175	6	70 Ps+Pt-c D2 Ok
D3c	85.216	...	116	175	22	70 Ps+Pt-c D3 Ok
Max:	103.3482	... mm	0.968	0.403	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	316	462	Ok
D2uc	...	...	...	69	207	Ok
D3uc	...	...	...	247	462	Ok
D1c	...	...	...	348	462	Ok
D2c	...	...	...	73	207	Ok
D3c	...	...	...	275	462	Ok
Max	...	...	...	0.752		

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:



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FileName : 401-E-30001A-Bottom- RP

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ASME TS Calc: DESIGN CASE

Case: 1 9:50a Aug 12, 2026

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	103.348	105.000	Ok
Tubesheet thickness flanged extension :	0.000	89.000	Ok
Tube thickness :	0.490	2.000	Ok
Tube-tubesheet fillet weld leg :	1.224	2.000	Ok
Tube-tubesheet groove weld leg :	1.224	2.000	Ok

Given Shell Thickness: 10.0000 mm

Given Channel Thickness: 18.0000 mm

Min Channel length of thk, (18.000) adj. to tubesheet: 202.049 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	4.739	0.999	3.403	1.000
Tubesheet Shear Stress	11.419	1.000	11.419	1.000
Tubesheet-Channel Junction	4.739	0.999	3.403	1.000
Tube Pressure Stress	20.094	1.000	9.938	1.000
Minimum MAWP	4.739		3.403	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6.529	0.999	4.615	1.000
Tubesheet Shear Stress	15.056	1.000	15.056	1.000
Tubesheet-Channel Junction	6.529	0.999	4.615	1.000
Tube Pressure Stress	26.436	1.000	13.438	1.000
Minimum MAPnc	6.529		4.615	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: